Robust Inference for Conditional Z-Estimation via Distributional Nearest Neighbors

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Abstract

We study pointwise inference for parameters defined by conditional estimating equations using a localized Z-estimator built from distributional nearest-neighbor (DNN) weights. A single tuning parameter controls localization (an inverse bandwidth), yielding a transparent bias-variance trade-off. We combine orthogonal (locally robust) scores with a leave-one-out stability condition on the nuisance estimates and the weighted Jacobian, which replaces entropy control and cross-fitting while preserving scarce local information near the target covariate value. Under mild regularity conditions, the estimator is asymptotically linear and normal, with normalization governed by weight concentration, that is, by the effective local sample size. The analysis is constructive and exploits the L-statistic structure of DNN weights and their connection to infinite-order U-statistics.

We implement two workhorse applications, conditional average treatment effects and partially linear quantile regression at a point, showing that the procedure is simple to deploy with modern first stages and is supported by basic diagnostics for localization and overlap. We view the DNN approach as a stable baseline that clarifies the roles of locality and orthogonality and as a platform for dimension-adaptive extensions based on data-driven proximity weights.

Keywords: conditional Z-estimation, Neyman orthogonality, distributional nearest neighbors, infinite-order U-statistics, generalized U-statistics, Jackknife, CATE, localized inference

1. Introduction

We study pointwise inference for parameters defined by conditional estimating equations at a fixed covariate value. Such targets arise when the empirical question is inherently local: treatment effects for a subgroup, conditional policy effects at a benchmark covariate profile, or distributional parameters in regions with weak overlap or sparse support. In these problems, precision is determined not by the nominal sample size alone but by how much weight can be placed on observations genuinely near the target point. Standard global procedures are not tailored to this local geometry, especially when nuisance components are high dimensional or functional. Our goal is a general and practical approach that localizes the moment condition while retaining valid inference with modern first stages.

We propose a localized Z-estimator that aggregates scores using distributional nearest-neighbor (DNN) weights. These weights are convex, depend only on covariate proximity, and admit a simple L-statistic representation that makes computation straightforward. A single tuning parameter s controls localization: larger s concentrates weight on the closest observations and therefore acts like an inverse bandwidth, inducing the familiar bias-variance trade-off. This gives a rank-based localization device that is simple to compute and analytically tractable.

A central design choice is to avoid sample splitting. In conditional problems, only a small share of observations receives non-negligible weight near the target. Splitting partitions that already sparse neighborhood and reduces the effective number of informative observations used in each fit, which can inflate first-order variance and introduce instability as the local neighborhood changes across folds. We instead combine orthogonal (locally robust) scores with leave-one-out stability conditions on both the nuisance estimates and the weighted Jacobian. In our setting, these stability requirements replace mandatory cross-fitting in the main argument while preserving scarce local information. Under these conditions, same-sample nuisance estimation is valid: the effect of first-stage errors is second order, whereas the local precision loss from splitting is first order. This yields confidence intervals that better reflect the local geometry of the problem and can be tighter in settings with limited local support.

Our main theoretical results establish local asymptotic linearity, asymptotic normality, and feasible Wald inference for the localized Z-estimator, with variance scaling governed by DNN weight concentration, equivalently by the effective local sample size. The analysis is constructive: it leverages the L-statistic structure of the DNN weights and their connection to infinite-order U-statistics, together with orthogonal scores, to deliver a transparent inference recipe. We provide feasible estimators of the variance components that are easy to compute and require no additional resampling, along with diagnostics for localization, overlap, and local identification strength. We also make explicit the bias-centering distinction familiar from kernel and nearest-neighbor methods: baseline Gaussian limits arise at the local scale, while point-target centering requires undersmoothing (or a debiasing/multi-scale construction).

Although formulated for general conditional Z-problems, we specialize the procedure to two workhorse applications. First, we implement conditional average treatment effects under standard overlap and mild smoothness near the target covariate value, including an empirical illustration based on the National Job Training Partnership Act data. Second, we show how the same localization device yields pointwise inference for partially linear quantile regression. These examples underscore that the method is simple to deploy, naturally compatible with modern first stages, and well suited to problems where the information that matters is intrinsically local.

The paper proceeds as follows. Sections 2.1 and 2.2 formulate the conditional Z-estimation problem with functional nuisances and introduce the DNN localization scheme. Sections 3.1, 3.2, 3.3 and 3.4 develop the main asymptotic linearity, normality, variance estimation, and Wald inference results. Section 4 discusses tuning, bias, and centering, including the undersmoothing implications for pointwise inference. Sections 5 and 6 specialize the framework to CATE and partially linear quantile regression. Section 7 concludes.

1.1. Notation

We collect notational conventions used throughout. Sets of integers are $[n] \coloneqq \{1, \dots, n\}$ and, more generally, $[p] \coloneqq \{1, \dots, p\}$. For any finite index set $\mathcal{I} \subset \mathbb{N}$ and order $s \in \mathbb{N}$, define

$$L_s(\mathcal{I}) \coloneqq \{ (l_1, \dots, l_s) \in \mathcal{I}^s : l_1 < \cdots < l_s \}, \qquad \mathcal{L}_{n,s} \coloneqq L_s([n]), \tag{1.1}$$

with cardinality $|L_s(\mathcal{I})| = \binom{|\mathcal{I}|}{s}$. We also use the falling factorial $(n)_s \coloneqq \prod_{i=0}^{s-1}(n-i)$ so that $\binom{n}{s} = (n)_s/s!$.

Data, distributions, expectations. We write W_i for a generic observation and let $D_{[n]} \coloneqq (W_1, \dots, W_n)$ denote the sample of size n, drawn i.i.d. from a distribution P_n. We write $D_\ell \coloneqq (W_{l_1}, \dots, W_{l_s})$ for the sub-sample indexed by $\ell \in \mathcal{L}_{n,s}$ and $D_{[n],-\ell}$ for the data with indices in ℓ removed; in the special case of a single index i, we abbreviate $D_{[n],-i}$. Realizations are denoted by lowercase letters (e.g., $d_{[n]}$).

Index concatenation and permutations. For index vectors $\ell = (l_1, \dots, l_s)$ and $\iota = (i_1, \dots, i_t)$ with disjoint entries, we denote ordered concatenation by

$$\ell \parallel \iota \coloneqq (l_1, \dots, l_s, i_1, \dots, i_t). \tag{1.2}$$

We reserve $\ell \cup \iota$ for set-union of index sets. The set of permutations on $\mathcal{I}$ is $\Pi(\mathcal{I}) \coloneqq \{\pi \colon \mathcal{I} \to \mathcal{I} \text{bijective}\}$.

Norms and balls. For vectors v, $\| v \|_2$ denotes the Euclidean norm. For nuisance elements η in a linear space $\mathbb{H}$, we use a problem-specific norm $\| \cdot \|_\eta$ and the product norm $\left\| (\Delta_\alpha, \Delta_\eta) \right\| \coloneqq |\Delta_\alpha| + \left\| \Delta_\eta \right\|_\eta$ on $\mathbb{R} \times \mathbb{H}$.

Asymptotics and order symbols. We write $a_n \to 0$ in probability as $a_n \longrightarrow_p 0$, and $a_n \rightsquigarrow A$ for weak convergence. The symbols $O_{\mathbb{P}}(\cdot)$ and $o_{\mathbb{P}}(\cdot)$ have their standard meanings. For nonnegative sequences/functions a, b, the relation $a \lesssim b$ means $a \leqslant C\, b$ for a constant C independent of n, s, and $P \in \mathcal{P}_n$; we write $a \gtrsim b$ for the reverse and $a \asymp b$ if both hold. Unless stated otherwise, asymptotics are taken along regimes with $n \to \infty$ and $s = s_n \to \infty$ satisfying $s = o(n)$.

Other Notation. We use $\mathbb{1}\{\cdot\}$ for indicator functions, $a \wedge b \coloneqq \min\{a, b\}$ and $a \vee b \coloneqq \max\{a, b\}$. For a class $\mathcal{F}$ of measurable functions and a probability measure Q, $\mathcal{N}(\varepsilon, \mathcal{F}, \| \cdot \|_{L_2(Q)})$ denotes the covering number and

$$\operatorname{ent}(\varepsilon, \mathcal{F}, \| \cdot \|_{L_2(Q)}) \coloneqq \log \mathcal{N}(\varepsilon, \mathcal{F}, \| \cdot \|_{L_2(Q)}) \tag{1.3}$$

the (metric) entropy. When $Q = P$ is the data-generating law, we abbreviate $\| \cdot \|_{L_2(P)}$ by $\| \cdot \|_{P,2}$.

1.2. Related Literature

This paper contributes to three broad conversations:

(i) Classical Z-estimation and semiparametric efficiency for conditional moments

(ii) Inference with complex, high-dimensional or nonparametric nuisance components via orthogonal/locally robust moments

(iii) Asymptotics for (generalized) U- and V-statistics underpinning distributional nearest-neighbor (DNN) and random-forest style weights

(iv) Nearest-neighbor and random-forest views of localization

We also connect to the nearest-neighbor and random-forest literatures that interpret tree- or rank-based procedures as localized weighting schemes.
LITERATURE REVIEW PLACEHOLDER:

Inference for Z-Estimation Problems.

- Newey and McFadden (1994)
- Van der Vaart (1998)
- Ai and Chen (2003)

Inference in the presence of Functional Nuisance Parameters.

- Belloni et al. (2015)
- Chernozhukov et al. (2018)
- Semenova and Chernozhukov (2021)
- Chernozhukov et al. (2023)
- Chernozhukov et al. (2022)
- Chernozhukov et al. (2024)
- Chen et al. (2022)

Classical U-Statistics.

- Hoeffding (1948)
- Hoeffding (1963)
- Serfling (1980)
- Koroljuk and Borovskich (1994)
- Lee (2019)
- Arvesen (1969)
- Arcones and Gine (1992)

Generalized U-Statistics.

- Janson (1984)
- Peng et al. (2022)
- Peng et al. (2026)
- DiCiccio and Romano (2022)
- Song et al. (2019)
- Chen and Kato (2019)
- Juergens (2025)

DNN and Nearest-Neighbors.

- Lin and Jeon (2006)
- Biau et al. (2008)
- Steele (2009)
- Biau and Devroye (2010)
- Biau (2012)
- Demirkaya et al. (2024)

Random Forests.

- Breiman (1996)
- Breiman (2001)
- Meinshausen (2006)
- Scornet et al. (2015)
- Mentch and Hooker (2016)
- Breiman et al. (2017)
- Wager et al. (2014)
- Wager and Athey (2018)
- Athey et al. (2019)

Relative to the above, our contribution is to (i) formulate conditional Z-estimation with rank-localized weights that concentrate as the neighborhood parameter grows, (ii) deliver asymptotic linearity and normality without sample splitting despite functional nuisances, by leveraging orthogonal scores and generalized U-statistic theory, and (iii) characterize the bias-variance tradeoff induced by DNN localization and provide feasible inference using estimable variance components. This yields a simple, scalable alternative to bandwidth-based local methods and to honesty-based forest procedures for conditional targets.

2. Setup

The goal of this section is to formalize the conditional Z-estimation problem at a fixed covariate value $\mathbf{x}$ and to introduce the distributional nearest-neighbor (DNN) weighting scheme that turns global moments into local ones. Many economic targets can be written as the solution to a conditional moment $m(\mathbf{x}; \alpha(\mathbf{x}), \boldsymbol{\eta}_0) = 0$: heterogeneous treatment effects and marginal treatment effects in program evaluation, pointwise elasticities in demand and labor-supply models, conditional Euler-equation residuals in consumption, or structural first-order conditions in auctions and entry games indexed by market characteristics $X = \mathbf{x}$. In each case the estimand $\boldsymbol{\alpha}_0(\mathbf{x})$ is a scalar or low-dimensional function evaluated at $\mathbf{x}$, while $\boldsymbol{\eta}_0$ collects potentially high-dimensional or functional nuisances learned from data.

2.1. Conditional Z-Estimation with Functional Nuisance Parameters

Fix a point $\mathbf{x} \in \mathcal{X}$ and a dimension $p_\alpha \in \mathbb{N}$. Let $\boldsymbol{\alpha}_0 \colon \mathcal{X} \to \mathbb{R}^{p_\alpha}$ be the target function and write $\boldsymbol{\alpha}_0(x) = (\alpha_{0,1}(x), \ldots, \alpha_{0,p_\alpha}(x))'$; the object of interest is $\boldsymbol{\alpha}_0(\mathbf{x})$. The nuisance parameter is a measurable map $\eta_0 = (\eta_{0,1}, \ldots, \eta_{0,p_\alpha})$ with components $\eta_{0,j} \colon \mathcal{Z}_{j,n} \to \mathcal{E}_j$, where each $\mathcal{E}_j \subset \mathbb{R}^{o_j}$ is Borel and o_j does not depend on n.
For each $n \in \mathbb{N}$, we observe i.i.d. data $(W_{i,n})_{i=1}^n$ with $W_{i,n} = (X_i, Z_{i,n}, Y_{i,n})$ taking values in

$$\mathcal{W}_n = \mathcal{X} \times \mathcal{Z}_n \times \mathcal{Y}_n, \qquad \mathcal{Z}_n = \bigtimes_{j=1}^{p_\alpha} \mathcal{Z}_{j,n}, \tag{2.1}$$

where $\mathcal{X} \subset \mathbb{R}^{k_x}$ is a fixed Borel set, while $\mathcal{Z}_{j,n} \subset \mathbb{R}^{q_{j,n}}$ and $\mathcal{Y}_n \subset \mathbb{R}^{k_{y,n}}$ are Borel sets whose dimensions $q_{j,n}$ and $k_{y,n}$ may grow with n. We allow coordinates to be shared across the subvectors and with X: there exist measurable maps $\Pi_{j,n} \colon \mathcal{X} \times \mathcal{Z}_n \to \mathcal{Z}_{j,n}$ such that $Z_{i,n}^j = \Pi_{j,n}(X_i, Z_{i,n})$; components may repeat across j and with X. For notational convenience, we include shared coordinates multiple times inside $W_{i,n}$. Let P_n denote the law of $W_{i,n}$ on $\mathcal{W}_n$, with marginals P_X on $\mathcal{X}$ (constant in n by assumption), $P_{Z,n}$ on $\mathcal{Z}_n$, and $P_{Y,n}$ on $\mathcal{Y}_n$. We require marginal consistency of the triangular array: for any $m \leqslant n$, let $\pi_{m \leftarrow n} \colon \mathcal{W}_n \to \mathcal{W}_m$ be the projection that retains X, the first $q_{j,m}$ coordinates of each Z^j, and the first $k_{y,m}$ coordinates of Y; then

$$P_n \circ \pi_{m \leftarrow n}^{-1} = P_m. \tag{2.2}$$

Equivalently, the probability measure is identical on the marginals corresponding to variables already present in earlier rows of the array. We write $P_n^{\otimes k}$ for the k-fold product measure on $\mathcal{W}_n^k$ and suppress the index n when harmless. Throughout, probabilities, expectations and variances are with respect to P_n, which we suppress when harmless.
For each $x \in \mathcal{X}$ (in particular for $x = \mathbf{x}$), the parameter space is

$$\mathcal{A}(x) = \bigtimes_{j=1}^{p_\alpha} \mathcal{A}_j(x), \tag{2.3}$$

where each $\mathcal{A}_j(x) \subset \mathbb{R}$ is a nonstochastic closed interval. Identification is by conditional moments that are invariant along the array under the above marginal consistency: there exist measurable score functions

$$\psi_j \colon \mathcal{W}_n \times \mathcal{A}_j(x) \times \mathcal{E}_j \to \mathbb{R}, \qquad j \in [p_\alpha], \tag{2.4}$$

such that $\boldsymbol{\alpha}_0(x)$ is the unique element of $\mathcal{A}(x)$ satisfying, for all $j \in [p_\alpha]$,

$$\mathbb{E}\Big[\psi_j\Big(W_{i,n},\ \alpha_{0,j}(x),\ \eta_{0,j}\Big(Z_{i,n}^j\Big)\Big) \,\Big|\, X_i = x\Big] = 0. \tag{2.5}$$

Here, $(w, a, t) \mapsto \psi_j(w, a, t)$ is measurable on an open neighborhood of $\mathcal{W}_n \times \mathcal{A}_j(x) \times \mathcal{E}_j$, and uniqueness is understood within $\mathcal{A}(x)$. We view this as a localized (conditional) analogue of manytarget Zproblems in Belloni et al. (2015), with localization later induced by Distributional Nearest Neighbor (DNN) weights centered at $\mathbf{x}$.

Applying Belloni et al. (2015) literally to our conditional moments at $x = \mathbf{x}$, one would estimate $\boldsymbol{\alpha}_0(\mathbf{x})$ by (approximately) solving the empirical moment equations using only observations with $X_i = \mathbf{x}$:

$$\forall j \in [p_\alpha]: \quad \left| \sum_{i=1}^{n} \mathbb{1}\{X_i = \mathbf{x}\} \psi_j\left(W_i,\ \widehat{\alpha}_j(\mathbf{x}),\ \widehat{\eta}_j\left(Z_i^j\right)\right) \right| \leqslant \inf_{a \in \widehat{\mathcal{A}}_j(\mathbf{x})} \left| \sum_{i=1}^{n} \mathbb{1}\{X_i = \mathbf{x}\} \psi_j\left(W_i,\ a,\ \widehat{\eta}_j\left(Z_i^j\right)\right) \right| + \epsilon_n, \tag{2.6}$$

where $\widehat{\eta}_j$ is a firststage estimator, $\widehat{\mathcal{A}}(\mathbf{x}) = \times_{j=1}^{p_\alpha} \widehat{\mathcal{A}}_j(\mathbf{x})$ satisfies $\boldsymbol{\alpha}_0(\mathbf{x}) \in \widehat{\mathcal{A}}(\mathbf{x})$ with high probability, and $\epsilon_n \geqslant 0$ is a tolerance. When X is continuously distributed, however, the event $\{X_i = \mathbf{x}\}$ has probability zero so the literal sample analogue uses (at most) a vanishing number of observations.

We will deviate from this setup due to the localized nature of our parameter and the fact that no such observations might be available in many scenarios of interest. Instead, we will use all observations and introduce data-dependent weights

$$\boldsymbol{w}_{s,n}(\mathbf{x}) = \boldsymbol{w}(\mathbf{x}) = (w_{i,s,n}(\mathbf{x}))_{i\in[n]} = (w_{s,i}(\mathbf{x}))_{i\in[n]} \in \Delta_n \coloneqq \left\{ (w_i)_{i\in[n]} :\ w_i \in [0,1], \sum_{i=1}^{n} w_i = 1 \right\} \tag{2.7}$$

that reflect proximity to $\mathbf{x}$. When necessary, we will include the data set with respect to which the weights are formed as an additional argument, that is $w_{i,s,n}(D;\mathbf{x}) = w_{i,s}(D;\mathbf{x})$, where we suppress the additional n in the subscript as it is implied by the data set. The resulting estimator $\breve{\boldsymbol{\alpha}}(\mathbf{x})$ solves, for each $j \in [p_\alpha]$,

$$\left| \sum_{i=1}^{n} w_{s,i}\, \psi_j\left(W_i,\ \breve{\alpha}_j(\mathbf{x}),\ \widehat{\eta}_j\left(Z_i^j\right)\right) \right| \leqslant \inf_{a \in \widehat{\mathcal{A}}_j(\mathbf{x})} \left| \sum_{i=1}^{n} w_{s,i}\, \psi_j\left(W_i,\ a,\ \widehat{\eta}_j\left(Z_i^j\right)\right) \right| + \epsilon_n. \tag{2.8}$$

Given the weights w at the fixed point $\mathbf{x}$, we define the effective sample size

$$n_{\text{eff}}(\mathbf{x}) \coloneqq \left(\sum_{i=1}^{n} w_i^2(\mathbf{x}) \right)^{-1} = \| w \|_2^{-2}. \tag{2.9}$$

Intuitively, $n_{\text{eff}}(\mathbf{x})$ is the size of an equally weighted subsample that would deliver the same variance as the localized average at $\mathbf{x}$. As it will turn out, the leading term in our linear representation will be a weighted average of centered scores, $\sum_{i=1}^{n} w_{s,i}(\mathbf{x})\ \psi_j\left(W_i, \alpha_{0,j}(\mathbf{x}), \eta_{0,j}(Z_i^j)\right)$. Writing $\sigma_x^2 \coloneqq \operatorname{Var}(\psi_j(W, \alpha_{0,j}(\mathbf{x}), \eta_{0,j}) \mid X \approx x)$, its variance is

$$\operatorname{Var}\left(\sum_i w_{s,i}(\mathbf{x})\ \psi_j(\cdot) \right) \approx \sigma_x^2 \sum_i w_i^2(\mathbf{x}) = \sigma_x^2 / n_{\text{eff}}(\mathbf{x}) \tag{2.10}$$

by the definition in Equation (2.9). By contrast, an equally weighted average of m observations drawn from the same local distribution has variance σ_x^2/m. Equating the two variances shows that $m = n_{\text{eff}}(\mathbf{x})$, that is, $n_{\text{eff}}(\mathbf{x})$ is precisely the size of an equally weighted subsample that delivers the same first-order variance as the localized, weighted fit at x. This is why all stochastic orders and normalizations in this paper are written in terms of $n_{\text{eff}}(\mathbf{x})$ rather than n.

2.2. Distributional Nearest Neighbor Weighting Scheme

In this paper, the weights $w_{s,i}$ are induced by the distributional nearest neighbor (DNN) scheme, a generalized Ustatistic in the sense of Peng et al. (2022). Define the rank of a covariate vector relative to $\mathbf{x}$ within a sample D by

$$\operatorname{rk}(\mathbf{x}; X, D) \coloneqq \#\left\{ X' \in D :\ \left\| X' - \mathbf{x} \right\|_2 < \| X - \mathbf{x} \|_2 \right\} + 1, \tag{2.11}$$

and let $\mathcal{L}_{n,s} \coloneqq \{\ell \subset [n] :\ |\ell| = s\}$ denote the collection of size-s subsamples with $D_\ell \coloneqq \{W_i :\ i \in \ell\}$. The DNN kernel and weights are

$$\kappa(\mathbf{x}; W_i, D_\ell) = \mathbb{1}\{\operatorname{rk}(\mathbf{x}; X_i, D_\ell) = 1\}, \tag{2.12}$$

$$w_{s,i} = \binom{n}{s}^{-1} \sum_{\ell \in \mathcal{L}_{n,s}} \mathbb{1}\{i \in \ell\}\, \kappa(\mathbf{x}; W_i, D_\ell), \tag{2.13}$$

where $\mathbb{1}\{\cdot\}$ denotes the indicator function.

Remark 2.1 (Choice of Distance and Standardization)**.**
Our results use distances only through ranks (for the L-form) and through small-ball geometry (for the mass-radius and drift bounds). In finite dimension, any norm on $\mathbb{R}^{k_x}$ is equivalent to ℓ_2:

$$c \,\|\, v \,\|_a \leqslant \|\, v \,\|_b \leqslant C \,\|\, v \,\|_a \qquad (\forall v), \tag{2.14}$$

for some $0 < c \leqslant C < \infty$ depending only on the pair of norms. Hence replacing $\|\cdot\|_2$ by any other norm changes only fixed constants (unit-ball volume, Lipschitz moduli), not rates. In particular, all arguments and rates go through verbatim with any of the following choices:

- *ℓ_p norms for $1 \leqslant p \leqslant \infty$ (including ℓ_1 and ℓ_∞)*
- *weighted Euclidean norms $\|\, v \,\|_M := \|\, Mv \,\|_2$ with M invertible*
- *block-separable norms $\left\| (v^{(1)}, v^{(2)}) \right\| := \left\| v^{(1)} \right\|_{p_1} + \left\| v^{(2)} \right\|_{p_2}$ after standardizing each block*

For these alternatives, the two important Propositions 3.1 and 3.2 remain valid up to constants: $\sum_{i=1}^{n} w_{s,i}^2 \asymp s/n$ is unaffected (rank-only), and the mass-radius bound keeps the same exponent k_x with a different unit-ball volume.
Because nearest-neighbor geometry is scale-dependent, the choice is primarily practical. We recommend standardizing (or whitening) the continuous coordinates of X so that proximity reflects economically meaningful similarity rather than raw units. This corresponds to the weighted Euclidean instance above with $M = \widehat{\Sigma}_X^{-1/2}$, i.e.

$$\|\, u \,\|_{\text{Mah}} := \left\| \widehat{\Sigma}_X^{-1/2} u \right\|_2, \tag{2.15}$$

which is Euclidean distance after an affine rescaling; all statements below are unchanged under such reparameterizations.
Distances based on angles (e.g., cosine similarity) alter the relevant local dimension to that of an embedded manifold and would require restating the design conditions for that geometry. We therefore keep ℓ_2 as the default and regard non-Euclidean, non-norm similarities as application-specific extensions.

Letting $w_{s,(i)}(\mathbf{x})$ denote the weight associated with the i-th closest observation to the point of interest in the sense of Eq. Equation (2.16), Steele (2009) shows that there is a simple closed-form representation for the DNN weights that results in an equivalent L-statistic with a useful asymptotic approximation for the $s = o(n)$ regime. Order the sample by distance to $\mathbf{x}$,

$$\left\| X_{(1)} - \mathbf{x} \right\|_2 < \left\| X_{(2)} - \mathbf{x} \right\|_2 < \cdots < \left\| X_{(n)} - \mathbf{x} \right\|_2, \tag{2.16}$$

which holds almost surely when P_X is continuous; if numerical ties occur, we break them deterministically (e.g., by index), and results are tiebreaking invariant. The weights then take the following, often more convenient, form.

$$w_{s,(i)}(\mathbf{x}) = \begin{cases} \binom{n}{s}^{-1} \binom{n-i}{s-1} \approx \frac{s}{n} \left(1 - \frac{s}{n}\right)^{i-1}, & \text{if} \quad 1 \leqslant i \leqslant n - s + 1, \\ 0, & \text{otherwise} \end{cases} \tag{2.17}$$

While we define the weights in Equation (2.8) using the U-statistics definition of DNN for intuition and mathematical convenience; all computations should only ever use the equivalent L-statistic weights in Equation (2.17), avoiding enumeration of $\mathcal{L}_{n,s}$. Recent results on the properties of the DNN weights motivate an assumption on the choice of kernel order as the sample size grows. At this stage we therefore introduce the following condition.

Condition 2.1 (DNN Kernel Order)**.**
The kernel-order associated with the DNN weights obeys the following conditions.

(i) **Growth**: The kernel order diverges as the sample size grows, that is $s = s_n \to \infty$ as $n \to \infty$.

(ii) **Growth Rate**: The kernel order grows slower than the sample size, that is $s = s_n = o(n)$.

Unlike the conventional k-NN scheme, where increasing k spreads weight over a larger random neighborhood, distributional nearest neighbors (DNN) become more local as the kernel order s grows. Intuitively, each DNN term is a 1-NN taken within a random subsample of size s, and the typical nearest-neighbor distance inside such subsamples shrinks with s. This is also visible in the L-statistic representation of DNN, where the weight on the i-th nearest observation is $w_{s,(i)}(\mathbf{x}) \propto \binom{n-i}{s-1}$. The decay across ranks steepens with s because

$$\frac{w_{s,(i)}(\mathbf{x})}{w_{s,(i+1)}(\mathbf{x})} = \frac{\binom{n-i}{s-1}}{\binom{n-i-1}{s-1}} = \frac{n-i}{n-i-s+1}, \tag{2.18}$$

which is increasing in s. Hence larger s concentrates mass on the very closest points and reduces the effective neighborhood around $\mathbf{x}$. At the extremes, $s = 1$ yields the global average (maximally dispersed), while $s = n$ reduces to the usual 1-NN based on the full sample (maximally concentrated). In short, under DNN, s plays the role of an inverse bandwidth: larger s means tighter localization. This creates a familiar tradeoff. If s is too small, the weights are spread over a wider region and we average across observations that do not closely resemble $\mathbf{x}$, which introduces bias from insufficient localization. If s is too large, the weights concentrate on very few points and the estimate fluctuates more from sample to sample, which increases variance. A useful heuristic is that the effective number of non-negligible neighbors behaves like n/s, so larger s means a smaller effective sample size. This once more contextualizes our motivation to forgo sample splitting. Because only a small fraction of observations receive meaningful weight near $\mathbf{x}$, splitting the data could cut the already scarce local information and inflate first-order variance. The L-statistic form in Equation (2.17) makes the scale of $n_{\text{eff}}(\mathbf{x})$ transparent. When $s = o(n)$, the geometric approximation $w_{s,(i)}(\mathbf{x}) \approx (s/n)(1-s/n)^{i-1}$ implies that only the first $O(n/s)$ nearest neighbors carry nonnegligible mass, with the remaining weights decaying at a geometric rate. A direct calculation gives

$$\sum_{i=1}^{\infty} w_{s,(i)}(\mathbf{x})^2 \;\approx\; \left(\tfrac{s}{n}\right)^2 \sum_{i=0}^{\infty} \left(1-\tfrac{s}{n}\right)^{2i} = \frac{(s/n)^2}{1-(1-s/n)^2} = \frac{s/n}{2-(s/n)}, \tag{2.19}$$

hence $\sum_i w_i^2(\mathbf{x})$ is of order s/n uniformly over sequences with $s = o(n)$, and therefore $n_{\text{eff}}(\mathbf{x}) = (\sum_i w_i^2(\mathbf{x}))^{-1}$ is of order n/s. Proposition 3.1 records this fact without appealing to the approximation and will serve as our pointwise bookkeeping device for stochastic rates at $\mathbf{x}$. Using the L-statistic representation of the weights, the estimator at the center of this paper, $\breve{\boldsymbol{\alpha}}(\mathbf{x}) = (\breve{\alpha}_1(\mathbf{x}), \cdots, \breve{\alpha}_{p_\alpha}(\mathbf{x}))^\intercal$ is thus implicitly defined by the following set of equations.

$$\begin{aligned} &\left| \sum_{i=1}^{n-s+1} \binom{n}{s}^{-1} \binom{n-i}{s-1} \psi_j\left(W_{(i)},\, \breve{\alpha}_j(\mathbf{x}),\, \widehat{\eta}_j\left(Z_{(i)}^j\right)\right) \right| \\ &\qquad \leqslant \inf_{a \in \widehat{\mathcal{A}}_j(\mathbf{x})} \left| \sum_{i=1}^{n-s+1} \binom{n}{s}^{-1} \binom{n-i}{s-1} \psi_j\left(W_{(i)},\, a,\, \widehat{\eta}_j\left(Z_{(i)}^j\right)\right) \right| + \epsilon_n. \end{aligned} \tag{2.20}$$

2.3. Conditions for the Main Results

The conditions below are grouped by role. They are ordered from primitive assumptions to derived ingredients used in the proofs. We first state regularity conditions on the target, design, and score objects; next we record implied properties of the conditional moment map; and finally we add localization-, stability-, and rate conditions specific to DNN weighting, including the undersmoothing condition that makes localization bias negligible at the inference scale. This organization is meant to separate what needs to be assumed from what can be verified as a consequence of earlier conditions.

In the conditions below, ζ, n_0, c_1 and C_1 are given positive constants, M is a fixed positive integer, and $\delta_n \downarrow 0$ and $\rho_n \downarrow 0$ are given sequences of constants. Let $a_n = \max(p_\alpha, p, n, \mathrm{e})$ and $b_n = \{\log(ep_\alpha)\}^{1/2}$.

Each condition below requires one or more open ℓ_2-balls centered at $\mathbf{x}$ and contained in $\mathcal{X}$. Rather than naming a separate ball for each purpose, we fix a single neighborhood $\mathcal{B}_\mathbf{x} \subset \mathcal{X}$, an open ℓ_2-ball centered at $\mathbf{x}$, that is small enough to satisfy all conditions simultaneously. Such a ball exists as the intersection of finitely many open balls (one per condition and per $j \in [p_\alpha]$), which is again open, nonempty, and centered at $\mathbf{x}$, provided $\mathbf{x}$ is an interior point of $\mathcal{X}$ as required in Condition 2.3. We write $\mathfrak{r}_\mathbf{x}$ for its radius, $\pi_\mathbf{x} \coloneqq P_X(\mathcal{B}_\mathbf{x})$ for the local probability mass it carries, and use $\mathcal{B}(\mathbf{x}, r) \coloneqq \{u : \|u - \mathbf{x}\|_2 < r\}$ for any open ℓ_2-ball centered at $\mathbf{x}$ with explicit radius r. We also write $G \coloneqq \{X \in \mathcal{B}_\mathbf{x}\}$ and $G_i \coloneqq \{X_i \in \mathcal{B}_\mathbf{x}\}$ for the events that a generic or i-th observation falls in $\mathcal{B}_\mathbf{x}$.

Condition 2.2 (Target Parameter Conditions)**.**
Uniformly in $n \geqslant n_0$ and $j \in [p_\alpha]$, the parameter function of interest $\boldsymbol{\alpha}_0 \colon \mathcal{X} \to \mathbb{R}^{p_\alpha}$ fulfills the following conditions.

(i) **Local Lipschitz Continuity in** x: There is a constant $K_{\alpha,j} \geqslant 0$ such that for all $x, x' \in \mathcal{B}_{\mathbf{x}}$ we have

$$\left| \alpha_{0,j}(x) - \alpha_{0,j}(x') \right| \leqslant K_{\alpha,j} \left\| x - x' \right\|_2. \tag{2.21}$$

We define $K_\alpha = \max_{j \in [p_\alpha]} K_{\alpha,j}$.

(ii) **Interior Point:** For all $u \in \mathcal{B}_{\mathbf{x}}$ we have

$$\alpha_{0,j}(\mathbf{x}), \alpha_{0,j}(u) \in \operatorname{int}\left(\mathcal{A}_j(u)\right). \tag{2.22}$$

The first part of this condition only ensures the target map is well behaved close to the point of interest. It is used when we compare the conditional moments that are naturally involved in conditional Z-estimation problems and when we argue that small perturbations in α stay in the parameter space. In practice, the check is: does the econometric target you care about (e.g., a CATE, a conditional quantile, a policy value) vary smoothly in x near the chosen point? If yes, you are covered. The second part of this condition will not be invoked explicitly in our proofs. However, it will allow us to ignore unintuitive boundary cases that do not serve the exposition of the theory developed in this paper throughout.

Condition 2.3 (Distribution Conditions)**.**
Uniformly in $n \geqslant n_0$, the following conditions hold:

(i) **Interior Point**: The point of interest $\mathbf{x}$ is an interior point of $\mathcal{X}$.

(ii) **Continuous Distribution**: The marginal density of X, say f_X exists and is continuous on $\mathcal{B}_{\mathbf{x}}$. Furthermore, there are constants f_- and f_+ such that $\forall x \in \mathcal{B}_{\mathbf{x}}: \quad 0 < f_- \leqslant f_X(x) \leqslant f_+ < \infty$.

(iii) **i.i.d. Observations**: The observations $\{ W_{i,n} \}_{i \in [n]}$ are (row-wise) i.i.d. draws from a distribution P_n on $\mathcal{W}_n$ as outlined in Section 2.1.

This assumption helps us avoid boundary corrections and ensures that nearest-neighbor ranks are almost surely unique so the DNN order statistics and weights $\kappa(u; W_i, D_\ell)$ are well-defined without tie-breaking.
However, if some coordinates of X are discrete (or numerically coarsened), equal distances to $\mathbf{x}$ can occur. Our results only require that ties be resolved by a fixed, measurable rule that depends on $X_{[n]}$ (but not on Y or auxiliary randomness). Any such rule leaves all first-order statements unchanged because the L-form weights depend on the ranks only, and within a tie block the set of occupied ranks is a contiguous range whose total weight is invariant to how the block is ordered. One convenient choice is to employ a lexicographic tie-breaking rule: Order observations by the pair $(\| X_i - \mathbf{x} \|_2,\ i)$; that is, if $\| X_i - \mathbf{x} \|_2 = \left\| X_j - \mathbf{x} \right\|_2$ we declare $i \prec j$ when $i < j$. This "distance-then-index" total order is measurable, stable, and makes the ranking (hence $w_{s,(i)}$) unique without introducing extra randomness.

Condition 2.4 (Score Function Conditions)**.**
Uniformly in $n \geqslant n_0$ and $j \in [p_\alpha]$, the following conditions hold:

(i) **Measurability**: the class of functions $\mathcal{F}_j = \{ w \mapsto \psi_j(w, a, \tilde{\eta}) : a \in \mathcal{A}_j(\mathbf{x}), \tilde{\eta} \in \mathcal{H}_j \cup \{ \eta_{0,j} \} \}$ is pointwise measurable.

(ii) **Envelope**: the class $\mathcal{F}_j$ has measurable envelope $F_j \geqslant \sup_{f \in \mathcal{F}_j} |f|$, such that $F = \max_{j=1,\ldots,p_\alpha} F_j$ satisfies

$$\sup_{u \in \mathcal{X}} \mathbb{E}\left[\left| F(W) \right|^{\mathfrak{q}} \,\middle|\, X = u \right] \leqslant C_1 \tag{2.23}$$

for some $\mathfrak{q} \geqslant 4$. I think I can weaken this to $\mathfrak{q} = 2 + \delta$ for $\delta > 0$

(iii) **Neyman-Orthogonality**: ψ_j is Neyman-orthogonal at $\mathbf{x}$ with respect to η_j.

The entropy/envelope parts are standard: they give uniform laws of large numbers and keep moments under control. As a brief reminder for readers that are new to the literature on robust inference, Neyman-orthogonality takes the following form in our scenario at hand.

Definition 2.1 (Neyman-Orthogonality)**.**
We say a function ψ_j as considered above is Neyman-orthogonal at $x \in \mathcal{X}$ with respect to nuisance parameter η_j if for any admissible h

$$\frac{\partial}{\partial \eta}\mathbb{E}\big[\psi_j\left(W, \alpha_{0,j}\left(x\right), \eta_{0,j}\left(Z^j\right)\right) \mid X = x\big][h] = \frac{\partial}{\partial t}\mathbb{E}\big[\psi_j\left(W, \alpha_{0,j}\left(x\right), \left[\eta_{0,j} + th\right]\left(Z^j\right)\right) \mid X = x\big]\Big|_{t=0} = 0, \tag{2.24}$$

that is, the Gâteaux derivative for any admissible function h is zero. We say it is Neyman-orthogonal with respect to η if it is Neyman-orthogonal at x for all $x \in \mathcal{X}$.

The orthogonality part is the main workhorse: It makes first-order sensitivity to η vanish at $\mathbf{x}$. Most off-the-shelf doubly-robust or locally robust scores satisfy this by construction.
In addition to these conditions we impose directly on the score functions, we add a number of conditions on the conditional moment function that derives from it. For this purpose we define the following conditional moment function that we will use throughout the paper.

$$m_j\left(u; a, \tilde{\eta}\right) = \mathbb{E}\big[\psi_j\left(W, a, \tilde{\eta}\left(Z^j\right)\right) \mid X = u\big] \tag{2.25}$$

Before imposing additional conditions, however, we make a number of observations that follow from prior assumptions on more primitive objects. Instead of showing these results in the appendix we present them here to outline how some of the assumptions we impose on the score function translate over to the conditional moment to relate them to other branches of the literature that impose assumptions on the latter.

Lemma 2.1 (Conditional Moment: Well-Definedness & Measurability)**.** (→ Proof, p. 53)
Let m_j be defined as in Equation (2.25). Suppose Condition 2.3 (i)(iii) hold and Condition 2.4 (i) and (ii) hold. Then for every fixed $a \in \mathcal{A}_j\left(\mathbf{x}\right)$ and $\tilde{\eta} \in \mathcal{H}_j \cup \{\, \eta_{0,j} \,\}$,

(i) The conditional moment in Equation (2.25) exists and is finite for P_X-almost every u.

(ii) There exists a version of $m_j\left(u, a, \tilde{\eta}\right)$ that is Borel-measurable in u on $\mathcal{B}_{\mathbf{x}}$.

(iii) $\left(a, \tilde{\eta}\right) \mapsto m_j\left(u, a, \tilde{\eta}\right)$ is Borel-measurable for P_X-almost every $u \in \mathcal{B}_{\mathbf{x}}$.

Remark 2.2 (Conditional Moment: Orthogonality)**.**
Under Condition 2.4 (iii) the Gâteaux derivative of the conditional moment with respect to η at $\left(u, a, \tilde{\eta}\right) = \left(\mathbf{x}, \alpha_{0,j}\left(\mathbf{x}\right), \eta_{0,j}\right)$ vanishes in every admissible direction h.

$$\frac{\partial}{\partial \eta} m_j\left(\mathbf{x}; \alpha_{0,j}\left(\mathbf{x}\right), \eta_{0,j}\left(Z^j\right)\right)[h] = \frac{\partial}{\partial t} m_j\left(\mathbf{x}; \alpha_{0,j}\left(\mathbf{x}\right), \left[\eta_{0,j} + th\right]\left(Z^j\right)\right)\Big|_{t=0} = 0 \tag{2.26}$$

In addition to these implied properties, we impose a number of actual assumptions on the conditional moment function. This puts assumptions exactly where they are used: in Taylor expansions at $\left(x; \boldsymbol{\alpha}_0\left(\mathbf{x}\right), \boldsymbol{\eta}_0\right)$ and in controlling the small drift from evaluating at X_i instead of $\mathbf{x}$. Lipschitz statements are chosen for readability; Remark 2.3 explains the one-for-one Hölder swap if weaker smoothness is preferred. These continue to be rather mild in the context of nonparametric inference and allow us to avoid stronger assumptions on the score function itself.

Condition 2.5 (Conditional Moment Conditions)**.**
Uniformly in $n \geqslant n_0$ and $j \in [p_\alpha]$, the following conditions hold:

(i) **Local Lipschitz Continuity in** α: There exists a constant $L_{\alpha,j} \geqslant 0$ and an open interval centered at $\alpha_{0,j}\left(\mathbf{x}\right)$, say $\mathcal{B}_{\alpha,j}$, such that for any $a, a' \in \mathcal{B}_{\alpha,j}$, we have

$$\sup_{u \in \mathcal{B}_{\mathbf{x}}} \big|\, m_j\left(u; a, \eta_{0,j}\right) - m_j\left(u; a', \eta_{0,j}\right) \big| \leqslant L_{\alpha,j} \big|\, a - a' \,\big|. \tag{2.27}$$

We define $L_\alpha = \max_{j \in [p_\alpha]} L_{\alpha,j}$.

(ii) **Local Lipschitz Continuity in η**: There is a seminorm $\|\cdot\|_\eta$ on $\mathcal{H}_j$, an open $\|\cdot\|_\eta$ ball $\mathcal{B}_{\eta,j}$ centered at $\eta_{0,j}$, and a constant $L_{\eta,j} \geqslant 0$ such that for all $\tilde{\eta}, \tilde{\eta}' \in \mathcal{B}_{\eta,j}$ we have

$$\sup_{u \in \mathcal{B}_{\mathbf{x}}} \left| m_j\left(u; \alpha_{0,j}(\mathbf{x}), \tilde{\eta}\right) - m_j\left(u; \alpha_{0,j}(\mathbf{x}), \tilde{\eta}'\right) \right| \leqslant L_{\eta,j} \left\| \tilde{\eta} - \tilde{\eta}' \right\|_\eta. \tag{2.28}$$

We define $L_\eta = \max_{j \in [p_\alpha]} L_{\eta,j}$.

(iii) **Local Lipschitz Continuity in u**: There exists a constant $L_{u,j} \geqslant 0$ such that for any $u, u' \in \mathcal{B}_{\mathbf{x}}$

$$\left| m_j\left(u; \alpha_{0,j}(\mathbf{x}), \eta_{0,j}\right) - m_j\left(u'; \alpha_{0,j}(\mathbf{x}), \eta_{0,j}\right) \right| \leqslant L_{u,j} \left\| u - u' \right\|_2. \tag{2.29}$$

We define $L_u = \max_{j \in [p_\alpha]} L_{u,j}$.

(iv) **Local Smoothness (uniform near $\mathbf{x}$)**: There exists an open $\left(\|\cdot\|_2 \times \|\cdot\|_\eta\right)$-ball $\mathcal{O}_j$ centered at $\left(\alpha_{0,j}(\mathbf{x}), \eta_{0,j}\right)$ such that for all $u \in \mathcal{B}_{\mathbf{x}}$:

(a) $m_j(u; \cdot, \cdot)$ is Hadamard directionally differentiable at every $(a, \tilde{\eta}) \in \mathcal{O}_j$ with a bounded linear derivative

$$Dm_j(u; a, \tilde{\eta}) \in \mathcal{L}\left(\mathbb{R} \times \mathcal{H}_j,\ \mathbb{R}\right), \qquad Dm_j(u; a, \tilde{\eta})\left[\Delta_\alpha, \Delta_\eta\right] = \partial_\alpha m_j(u; a, \tilde{\eta})\ \Delta_\alpha + \partial_\eta m_j(u; a, \tilde{\eta})\left[\Delta_\eta\right] \tag{2.30}$$

(b) There exists $L_{v,j} \geqslant 0$ such that for all $u \in \mathcal{B}_{\mathbf{x}}$ and all $(a, \tilde{\eta}), (a', \tilde{\eta}') \in \mathcal{O}_j$,

$$\left\| Dm_j(u; a, \tilde{\eta}) - Dm_j(u; a', \tilde{\eta}') \right\|_{\text{op}} \leqslant L_{v,j} \left(\left| a - a' \right| + \left\| \tilde{\eta} - \tilde{\eta}' \right\|_\eta \right), \tag{2.31}$$

where $\|\cdot\|_{\text{op}}$ is the operator norm induced by the product norm $\left\|(\Delta_\alpha, \Delta_\eta)\right\| \coloneqq |\Delta_\alpha| + \left\|\Delta_\eta\right\|_\eta$ on $\mathbb{R} \times \mathcal{H}_j$. We define $L_v = \max_{j \in [p_\alpha]} L_{v,j}$.

(c) For all $u \in \mathcal{B}_{\mathbf{x}}$ and $(a, \tilde{\eta}) \in \mathcal{O}_j$, the map $a \mapsto m_j(u; a, \tilde{\eta})$ is twice directionally differentiable with $\partial_{\alpha\alpha} m_j(u; a, \tilde{\eta})$ uniformly bounded on $\mathcal{O}_j$.

(d) There exists $L_{\alpha\alpha,j} \geqslant 0$ such that for all $u \in \mathcal{B}_{\mathbf{x}}$ and $a, a' \in \mathcal{B}_{\alpha,j}$,

$$\sup_{(a,\tilde{\eta}) \in \mathcal{O}_j} \left| \partial_{\alpha\alpha} m_j(u; a, \tilde{\eta}) - \partial_{\alpha\alpha} m_j(u; a', \tilde{\eta}) \right| \leqslant L_{\alpha\alpha,j} |a - a'|. \tag{2.32}$$

We define $L_{\alpha\alpha} = \max_{j \in [p_\alpha]} L_{\alpha\alpha,j}$.

(v) **Local Invertibility**: The derivative $\Gamma_j(\mathbf{x}) = \frac{\partial}{\partial a} m_j\left(\mathbf{x}; a, \eta_{0,j}\right)\Big|_{a = \alpha_{0,j}(\mathbf{x})}$ exists. Furthermore, $\inf_{j \in [p_\alpha]} \Gamma_j(\mathbf{x}) \geqslant c > 0$ for some $c \in \mathbb{R}_+$.
Define the diagonal Jacobian $\mathbf{\Gamma}_{\mathbf{x}} \coloneqq \operatorname{diag}\left(\Gamma_1(\mathbf{x}), \ldots, \Gamma_{p_\alpha}(\mathbf{x})\right)$.

(vi) **Identification**: The equation $m_j\left(\mathbf{x}; \alpha_{0,j}(\mathbf{x}), \eta_{0,j}\right) = 0$ has a unique solution in $\mathcal{A}_j(\mathbf{x})$.

Remark 2.3 (Hölder Continuity)**.**
For simplicity we impose local Lipschitz continuity in several places (e.g., moment-level derivatives in (a, η), local dependence on u near $\mathbf{x}$). All such Lipschitz requirements can be weakened to uniform Hölder continuity at essentially no conceptual cost. Concretely, whenever a bound of the form $\left\| G(\theta) - G(\theta') \right\| \leqslant L \left\| \theta - \theta' \right\|$ is used, one may instead assume $\left\| G(\theta) - G(\theta') \right\| \leqslant L_\beta \left\| \theta - \theta' \right\|^\beta$ with exponent $\beta \in (0, 1]$ uniformly over the relevant neighborhood. The proofs go through with the following mechanical changes:

(i) Taylor remainders become $O\left(\left\|(\Delta_\alpha, \Delta_\eta)\right\|^{1+\beta}\right)$ (hence rate conditions such as $\sqrt{n}(\rho_n + r_n)^{1+\beta} \to 0$ replace the quadratic case)

(ii) Localization bounds and incremental empirical-process controls in u acquire a $\|u - \mathbf{x}\|^\beta$ factor

(iii) Any display that previously featured a Lipschitz constant L is replaced by its Hölder analogue L_β with the obvious exponent adjustment

Entropy and weight-negligibility conditions are unaffected. We present the Lipschitz version in the main text for readability and to keep exponents uniform; the Hölder variant is a straight substitution throughout.
Consequence for undersmoothing. *The localization bias of the conditional moment in u drives the bias of the pseudo-target $\boldsymbol{\alpha}_s(\mathbf{x})$ relative to the point target $\boldsymbol{\alpha}_0(\mathbf{x})$. Under Lipschitz continuity ($\beta = 1$), this bias is $O(s^{-1/k_x})$ and Condition 2.7 requires $s_n \gg n^{k_x/(k_x+2)}$. Under β-Hölder continuity, the bias becomes $O(s^{-\beta/k_x})$ and the undersmoothing condition tightens to $s_n \gg n^{k_x/(k_x+2\beta)}$; see Section Appendix B.1 for the derivation.*

Next, we record a local mean-square continuity (MSC) result for the moment m_j and its Jacobian kernel. The role of MSC in our analysis is two-fold. First, it converts the pointwise Lipschitz and local smoothness statements in Conditions 2.4 and 2.5 into the L_2-moduli that drive the stability-based empirical process step at $\mathbf{x}$. Second, MSC furnishes uniform L_2 controls needed downstream for variance ratio consistency. Importantly, MSC is not an extra assumption: it is an immediate consequence of our local Lipschitz and local smoothness conditions, which we package below for later reference.

Lemma 2.2 (Local Mean-Square Continuity)**.** (→ Proof, p. 53)
Assume that Conditions 2.4 and 2.5 hold. Then, for any $j \in [p_\alpha]$ and u in a neighborhood of $\mathbf{x}$, the following L_2 bounds hold.

(i) ***Local Mean-Square Continuity of m_j in $\tilde{\eta}$:*** *For all $\tilde{\eta}, \tilde{\eta}' \in \mathcal{B}_{\eta,j}$ and all $u \in \mathcal{B}_{\mathbf{x}}$,*

$$\left| m_j\left(u; \alpha_{0,j}(\mathbf{x}), \tilde{\eta}\right) - m_j\left(u; \alpha_{0,j}(\mathbf{x}), \tilde{\eta}'\right) \right|^2 \leqslant L_{\eta,j}^2 \left\| \tilde{\eta} - \tilde{\eta}' \right\|_\eta^2. \tag{2.33}$$

(ii) ***Local Mean-Square Continuity in $\tilde{\eta}$ for the Jacobian Kernel:*** *For all $\tilde{\eta}, \tilde{\eta}' \in \mathcal{B}_{\eta,j}$ and all $u \in \mathcal{B}_{\mathbf{x}}$,*

$$\left| \partial_\alpha m_j\left(u; \alpha_{0,j}(\mathbf{x}), \tilde{\eta}\right) - \partial_\alpha m_j\left(u; \alpha_{0,j}(\mathbf{x}), \tilde{\eta}'\right) \right|^2 \leqslant L_{v,j}^2 \left\| \tilde{\eta} - \tilde{\eta}' \right\|_\eta^2. \tag{2.34}$$

(iii) ***Local Mean-Square Continuity m_j in a:*** *For all $a, a' \in \mathcal{B}_{\alpha,j}$ and all $u \in \mathcal{B}_{\mathbf{x}}$,*

$$\left| m_j\left(u; a, \eta_{0,j}\right) - m_j\left(u; a', \eta_{0,j}\right) \right|^2 \leqslant L_{\alpha,j}^2 \left| a - a' \right|^2. \tag{2.35}$$

(iv) ***Local Joint Mean-Square Continuity of m_j in $(a, \tilde{\eta})$:*** *For $(a, \tilde{\eta}), (a', \tilde{\eta}')$ in the neighborhood O_j and for all $u \in \mathcal{B}_{\mathbf{x}}$,*

$$\left| m_j\left(u; a, \tilde{\eta}\right) - m_j\left(u; a', \tilde{\eta}'\right) \right|^2 \leqslant \left(L_{\alpha,j} \left| a - a' \right| + L_{\eta,j} \left\| \tilde{\eta} - \tilde{\eta}' \right\|_\eta \right)^2. \tag{2.36}$$

(v) ***Local Continuity of m_j in u:*** *For all $u, u' \in \mathcal{B}_{\mathbf{x}}$,*

$$\left| m_j\left(u; \alpha_{0,j}(\mathbf{x}), \eta_{0,j}\right) - m_j\left(u'; \alpha_{0,j}(\mathbf{x}), \eta_{0,j}\right) \right|^2 \leqslant L_{u,j}^2 \left\| u - u' \right\|_2^2. \tag{2.37}$$

All displays remain valid after multiplying the integrands by $\mathbb{1}\{X \in U\}$ and then integrating over any measurable $U \subseteq \mathcal{B}_{\mathbf{x}}$ (the right-hand sides pick up a factor $\mathbb{P}(X \in U)$).

Conditions 2.2, 2.4 and 2.5 are target-side regularity assumptions. They do not restrict the first stage. Intuitively, they ask for three things near the point of interest $\mathbf{x}$. First, that the target $x \mapsto \alpha_0(x)$ varies smoothly in x at a local scale. Second, that the target is locally identified with a well-conditioned bread (the score reacts to small changes in the parameter). And finally, that the relevant variables have enough finite moments to control stochastic fluctuations.
We now introduce a localized leave-one-out (LOO) stability requirement that replaces global entropy/Donsker controls with a direct, algorithmic sensitivity bound tailored to our weighted, bagged estimators at the target point $\mathbf{x}$. This follows the stability-based route of Chen et al. (2022), who show that $o(n^{-1/2})$ replace-one stability of the score/Jacobian together with mean-square continuity (MSC) yields stochastic equicontinuity without entropy or cross-fitting. Our formulation adapts their idea to the localized setting:

(i) it constrains perturbations only on the good region where the weighting scheme places mass

(ii) it scales stability by the effective local sample size $n_{\text{eff}}(\mathbf{x}) = (\sum_{i=1}^{n} w_i^2(\mathbf{x}))^{-1}$

(iii) it adds a finite-difference clause for the Jacobian estimator $\widehat{\Gamma}_{h,j}(\mathbf{x})$

In concert with the MSC bounds in Lemma 2.2, this localized stability delivers the stochastic equicontinuity used later to control empirical-process remainders and to establish variance ratio consistency; the rates $\sqrt{n_{\text{eff}}(\mathbf{x})}\,\beta_n^{\text{loc}}(\mathbf{x})/h \to 0$ and $\sqrt{n_{\text{eff}}(\mathbf{x})}\,h^2 \to 0$ ensure that finite-difference noise and bias are negligible at the $n_{\text{eff}}(\mathbf{x})^{-1/2}$ scale.

Condition 2.6 (Local Leave-One-Out Stability)**.**
The following properties hold uniformly in $j \in [p_\alpha]$, where $G := \{X \in \mathcal{B}_{\mathbf{x}}\}$ is the local good event defined above. Let $D = (W_1, \ldots, W_n)$, let $D^{(i)}$ be D with W_i replaced by an i.i.d. copy W_i', and let $w(D;\mathbf{x}) = (w_1, \ldots, w_n)$ denote the data-dependent localization weights at $\mathbf{x}$. Define the localized replace-one modulus

$$\beta_n^{\text{loc}}(\mathbf{x}) := \sup_D \max_{l\in[n]} \sum_{i=1}^{n} \Big| w_i(D;\mathbf{x}) - w_i\left(D^{(l)};\mathbf{x}\right) \Big| \, \mathbb{1}\{X_i \in \mathcal{B}_{\mathbf{x}}\}. \tag{2.38}$$

(i) **Weight Negligibility and Localized Weight Stability:**

$$\max_{i\in[n]} \big| w_{s,i}(\mathbf{x}) \big| = o_{\mathbb{P}}\Big(n_{\text{eff}}(\mathbf{x})^{-1/2} \Big), \qquad \sqrt{n_{\text{eff}}(\mathbf{x})}\,\beta_n^{\text{loc}}(\mathbf{x}) \longrightarrow 0. \tag{2.39}$$

(ii) **Local Nuisance LOO Stability:** Let $\widehat{\eta}_j$ be the full-sample nuisance fit and $\widehat{\eta}_j^{(-l)}$ its leave-one-out version. Then

$$\max_{l\in[n]} \Big\| \left[m_j\left(W;\alpha_{0,j}(\mathbf{x}),\widehat{\eta}_j\right) - m_j\left(W;\alpha_{0,j}(\mathbf{x}),\widehat{\eta}_j^{(-l)}\right) \right] \mathbb{1}\{G\} \Big\|_{L_2} = o\left(n_{\text{eff}}(\mathbf{x})^{-1/2} \right), \tag{2.40}$$

$$\max_{l\in[n]} \Big\| \left[\partial_\alpha m_j\left(W;\alpha_{0,j}(\mathbf{x}),\widehat{\eta}_j\right) - \partial_\alpha m_j\left(W;\alpha_{0,j}(\mathbf{x}),\widehat{\eta}_j^{(-l)}\right) \right] \mathbb{1}\{G\} \Big\|_{L_2} = o\left(n_{\text{eff}}(\mathbf{x})^{-1/2} \right). \tag{2.41}$$

(iii) **Local finite-difference Jacobian Stability:** Suppose $\Gamma_j(\mathbf{x})$ is estimated by the central finite difference

$$\widehat{\Gamma}_{h,j}(\mathbf{x}) := \frac{\sum_{i=1}^{n} w_{s,i}(\mathbf{x}) \left[\psi_j\left(W_i, \alpha_{0,j}(\mathbf{x}) + h, \widehat{\eta}_j\right) - \psi_j\left(W_i, \alpha_{0,j}(\mathbf{x}) - h, \widehat{\eta}_j\right) \right]}{2h}, \tag{2.42}$$

Then

$$\sup_D \max_{l\in[n]} \Big| \left[\widehat{\Gamma}_{h,j}(D) - \widehat{\Gamma}_{h,j}(D^{(l)}) \right] \mathbb{1}\{G\} \Big| = O\left(\frac{\beta_n^{\text{loc}}(\mathbf{x})}{h} \right), \qquad \sqrt{n_{\text{eff}}(\mathbf{x})}\,\frac{\beta_n^{\text{loc}}(\mathbf{x})}{h} \to 0, \qquad \sqrt{n_{\text{eff}}(\mathbf{x})}\,h^2 \to 0. \tag{2.43}$$

At a high level, leave-one-out stability says that no single observation can move the fitted objects in a meaningful way. If removing any one data point barely changes the nuisance estimate or the (local) Jacobian, then two things follow. First, the usual worry about "using the same data twice" becomes harmless. Each score term is being evaluated using a model that cannot be influenced too much by its corresponding observation. Second, Statistics built from these fitted objects do not change much when the data are perturbed one point at a time. A standard variance argument (Efron-Stein) formalizes this intuition: if one-point perturbations barely matter, then the overall sampling fluctuation is small in exactly the way needed for the local stochastic equicontinuity result (Lemma 2.4) that is to follow. We do not aim to provide an exhaustive list of LOO stable first stage estimators in this paper as the preferred choice of learner can be highly domain-specific. LOO stability is a first-stage requirement: deleting one observation should not materially change the nuisance fit in the local window around $\mathbf{x}$, so the weighted score/Jacobian for $\alpha_0(\mathbf{x})$ is insensitive to one-point perturbations. Chen et al. (2022) show that $o(n^{-1/2})$ stability (globally) can replace sample splitting under orthogonality; in our localized setting, interpret their n as $n_{\text{eff}}(\mathbf{x})$. For applied work, the following choices are usually the safest:

(i) **Bagging/subsampling:** Use subsample size $m \ll n$ and many bags B so each unit has low leverage. Bagged 1-NN, trees with minimum leaf size, and random-forest style aggregation are common stable defaults in practice. **References:** (Elisseeff et al., 2005; Wang et al., 2023; Chen et al., 2022).

(ii) **Regularized ERM / GLMs:** Prefer explicit ℓ_2 regularization (ridge/weight decay), small step sizes, and early stopping. In local windows, standardize covariates and avoid near-separation designs; unregularized logistic and very deep trees are more LOO-sensitive. **References:** (Bousquet and Elisseeff, 2002, Ex. 2); (Hardt et al., 2016; Kuzborskij and Lampert, 2018; Arsov et al., 2019).

(iii) **Series/splines with mild ridge:** Use basis dimension well below $n_{\text{eff}}(\mathbf{x})$ and a small Tikhonov penalty; this keeps fits smooth under one-point deletions. **Reference:** (Vito et al., 2006).

(iv) **Use with caution:** Very small-k nearest neighbors, deep boosting, and unregularized deep nets can be unstable in small local windows unless aggressively regularized. For AdaBoost, available guarantees are approximation-stability results under fixed rounds and bounded weak-learner edges. **References:** (Bousquet and Elisseeff, 2002, Ex. 1); (Hardt et al., 2016; Kuzborskij and Lampert, 2018; Gao and Zhou, 2010).

Lemma 2.3 (DNN Weights: Negligibility and Stability)**.** (→ Proof, p. 54)
Let $\boldsymbol{w}(D;\mathbf{x}) = (w_{s,i}(\mathbf{x}))_{i\in[n]}$ be the DNN weights at $\mathbf{x}$, with kernel order $s = s_n$. Suppose Condition 2.1 holds and let the localized replace-one modulus for the DNN weights be

$$\beta_n^{\text{loc}}(\mathbf{x}) := \sup_D \max_{l\in[n]} \sum_{i=1}^{n} \Big| w_i(D;\mathbf{x}) - w_i(D^{(l)};\mathbf{x}) \Big| \, \mathbb{1}\{ X_i \in \mathcal{B}_{\mathbf{x}} \}. \tag{2.44}$$

Then:

(i) ***Weight negligibility:***

$$\max_{i\in[n]} |\, w_i(D;\mathbf{x}) \,| \leqslant \frac{s}{n} \quad \textit{and} \quad \max_{i\in[n]} |\, w_i(D;\mathbf{x}) \,| = o_{\mathbb{P}}\Big(n_{\text{eff}}(\mathbf{x})^{-1/2} \Big). \tag{2.45}$$

(ii) ***Localized Weight Stability:***

$$\beta_n^{\text{loc}}(\mathbf{x}) \leqslant \frac{2s}{n} \quad \textit{and} \quad \sqrt{n_{\text{eff}}(\mathbf{x})}\, \beta_n^{\text{loc}}(\mathbf{x}) \longrightarrow 0. \tag{2.46}$$

Lemma 2.3 shows that, for DNN weights, both weight negligibility and the localized replace-one stability bound are automatic consequences of the Lform representation, $s \to \infty$ with $s/n \to 0$, and the geometry summarized by $n_{\text{eff}}(\mathbf{x}) \asymp n/s$. Thus, strictly speaking, we do not need to assume these properties as separate conditions when working

with DNN. We nonetheless keep them stated explicitly because they isolate the minimal properties used throughout the analysis and become the natural verification targets when replacing DNN by other localization schemes.

The next result is the only empirical-process input we need. It packages the mean-square continuity (MSC) bounds from Lemma 2.2 with the localized LOO stability in Condition 2.6 to deliver local stochastic equicontinuity (LSE) at the natural scale $\sqrt{n_{\text{eff}}(\mathbf{x})}$. Intuitively, replacing one observation perturbs only a small amount of the local weight mass near $\mathbf{x}$, while MSC guarantees that such perturbations translate into proportionally small changes in the moment and its Jacobian; together, these imply that empirical and population quantities move in lockstep along the estimated path. This stability-plus-MSC route (cf. Chen et al. (2022)) lets us avoid global entropy/Donsker assumptions and keeps the argument focused on the actual localized procedure we implement.

Lemma 2.4 (Local Stochastic Equicontinuity)**.** (→ Proof, p. 55)
Assume Conditions 2.4, 2.5 and 2.6 and Lemma 2.2 (MSC) hold. Fix $j \in [p_\alpha]$ and define the normalized weighted process

$$\mathbb{G}_{n,w}[f] \coloneqq \sqrt{n_{\text{eff}}(\mathbf{x})}\left(\sum_{i=1}^{n} w_{s,i}(\mathbf{x})\ f(W_i)\ -\ \mathbb{E}[f(W) \mid X=\mathbf{x}]\right). \tag{2.47}$$

Let $\theta = (a,\tilde{\eta})$ and $\theta_0 = (\alpha_{0,j}(\mathbf{x}), \eta_{0,j})$, and let $\delta_n \downarrow 0$ be any deterministic sequence. Then

(i) ***Moment Block:***

$$\sup_{\|\theta-\theta_0\|\leqslant\delta_n} \left|\mathbb{G}_{n,w}\left(\{\psi_j(\cdot;a,\tilde{\eta}) - \psi_j(\cdot;a,\eta_{0,j})\}\ \mathbb{1}\{G\}\right)\right| = o_{\mathbb{P}}(1). \tag{2.48}$$

In particular, taking $a = \alpha_{0,j}(\mathbf{x})$ and $\tilde{\eta} = \widehat{\eta}_j$,

$$\begin{aligned}\sqrt{n_{\text{eff}}(\mathbf{x})}\Bigg|&\left(\sum_{i=1}^{n} w_{s,i}(\mathbf{x})\left\{m_j(\mathbf{x};\alpha_{0,j}(\mathbf{x}),\widehat{\eta}_j) - \psi_j(W_i;\alpha_{0,j}(\mathbf{x}),\widehat{\eta}_j)\right\}\mathbb{1}\{G_i\}\right)\\ &-\left(\sum_{i=1}^{n} w_{s,i}(\mathbf{x})\left\{m_j(\mathbf{x};\alpha_{0,j}(\mathbf{x}),\eta_{0,j}) - \psi_j(W_i;\alpha_{0,j}(\mathbf{x}),\eta_{0,j})\right\}\mathbb{1}\{G_i\}\right)\Bigg| = o_{\mathbb{P}}(1).\end{aligned} \tag{2.49}$$

(ii) ***Jacobian Block (Finite Difference):*** *Suppose $\Gamma_j(\mathbf{x})$ is estimated by the central finite difference*

$$\widehat{\Gamma}_{h,j}(\mathbf{x}) \coloneqq \frac{\sum_{i=1}^{n} w_{s,i}(\mathbf{x})\left[\psi_j(W_i,\alpha_{0,j}(\mathbf{x})+h,\widehat{\eta}_j) - \psi_j(W_i,\alpha_{0,j}(\mathbf{x})-h,\widehat{\eta}_j)\right]}{2h}, \tag{2.50}$$

If the rates in Equation (2.43) hold, then

$$\sqrt{n_{\text{eff}}(\mathbf{x})}\left|\widehat{\Gamma}_{h,j}(\mathbf{x}) - \Gamma_j(\mathbf{x}) - \left(\widehat{\Gamma}^{(0)}_{h,j}(\mathbf{x}) - \Gamma_j(\mathbf{x})\right)\right| = o_{\mathbb{P}}(1), \tag{2.51}$$

where $\widehat{\Gamma}^{(0)}_{h,j}(\mathbf{x})$ denotes the same finite-difference estimator with η fixed at $\eta_{0,j}$ in both calls.

Local stochastic equicontinuity (LSE) says that, after we localize near $\mathbf{x}$, the empirical and population versions of any score in a small neighborhood of the truth move in lockstep: their difference is $o_{\mathbb{P}}(1)$ at the $\sqrt{n_{\text{eff}}(\mathbf{x})}$ scale. Two ingredients deliver this: stability of the learned objects (replacing one point barely changes weights or the fitted nuisance) and mean-square continuity (small perturbations of $(a,\tilde{\eta})$ produce proportionally small changes in the moment). Together, a single-observation perturbation can only shift a small fraction of the local weight mass near $\mathbf{x}$, and continuity converts this into a small change in the localized score. Classical empirical-process approaches would impose global entropy or Donsker conditions on the whole score class. Our route leverages how the estimator is built: LOO stability plus MSC give a localized equicontinuity bound tailored to the DNN neighborhood. This keeps the proof targeted to the actual algorithm and avoids strong global complexity controls (see also the stability-based SE discussion of Chen et al. (2022)).

The conditions stated so far are sufficient for local asymptotic linearity and Gaussian approximation at the natural $\sqrt{n_{\text{eff}}(\mathbf{x})}$ scale, namely for fluctuations around the localized population target induced by DNN weighting. For inference centered at the point target $\boldsymbol{\alpha}_0(\mathbf{x})$, however, one must additionally control localization bias relative to that same scale. This introduces the usual undersmoothing requirement familiar from kernel and nearest-neighbor inference: Condition 2.1 governs localization and variance, while point-target centering requires an extra lower growth condition on s_n.

Condition 2.7 (Undersmoothing)**.**
Uniformly in $n \geqslant n_0$, assume Condition 2.1. The kernel order satisfies

$$\sqrt{\frac{n}{s_n}}\, s_n^{-1/k_x} \to 0 \qquad \Longleftrightarrow \qquad s_n \gg n^{k_x/(k_x+2)}. \tag{2.52}$$

Condition 2.7 is a standing assumption throughout the main text. It ensures that the localization bias $\sqrt{n_{\text{eff}}(\mathbf{x})}\,|\alpha_{s,j}(\mathbf{x}) - \alpha_{0,j}(\mathbf{x})| \to 0$, so all main-text results are stated and proved centered at the point target $\boldsymbol{\alpha}_0(\mathbf{x})$. The pseudo-target $\boldsymbol{\alpha}_s(\mathbf{x})$ and the analysis under Condition 2.1 alone (without undersmoothing) are treated in Section Appendix B.1.

3. Main Results

3.1. Main Theorems - Local Asymptotic Linearity

The goal of this subsection is to pass from the DNN-weighted U-statistic representation of the localized score to a first-order, sample-mean representation. We proceed in three steps:

(i) Quantify the geometry of the DNN localizer (effective sample size and neighborhood radius)

(ii) Establish Hájek dominance for the resulting oracle kernels (uniform over linear combinations)

(iii) Transfer this linearity to the estimated score via our local SE lemmas

Hájek dominance hinges on two geometric facts delivered by the DNN localizer that we state as Lemmas. Neyman-orthogonality and the local Lipschitz conditions then ensure that plug-in nuisance effects enter only at $o_{\mathbb{P}}\Big(\sqrt{s/n}\Big)$ (made precise via our LSE lemmas).

Proposition 3.1 (Effective sample size with DNN weights)**.** (→ Proof, p. 58)
Assume Condition 2.3 holds. If $s \to \infty$ and $s/n \to 0$, then there exist constants $0 < c < C < \infty$ depending only on (k_x, f_-, f_+) such that, with probability $\to 1$,

$$c\,\frac{s}{n} \leqslant \sum_{i=1}^{n} w_{s,i}^2 \leqslant C\,\frac{s}{n}. \tag{3.1}$$

Consequently, the effective sample size $n_{\text{eff}}(\mathbf{x}) \coloneqq \big(\sum_{i=1}^{n} w_{s,i}^2\big)^{-1}$ satisfies $c'\,\frac{n}{s} \leqslant n_{\text{eff}}(\mathbf{x}) \leqslant C'\,\frac{n}{s}$ with probability $\to 1$, for constants $0 < c' < C' < \infty$ depending only on (k_x, f_-, f_+).

Proposition 3.2 (Radius of the DNN neighborhood)**.** (→ Proof, p. 59)
Let Condition 2.3 hold. For the DNN weights $w_{s,i}$ of order s at $\mathbf{x}$, define

$$R_{s,\delta}(\mathbf{x}) \coloneqq \inf\left\{ r > 0 : \sum_{i=1}^{n} w_{s,i}\ \mathbb{1}\{\,\|X_i - \mathbf{x}\|_2 \leqslant r\,\} \geqslant 1-\delta \right\}, \qquad \delta \in (0,1). \tag{3.2}$$

If $s \to \infty$ and $s/n \to 0$, then for each fixed $\delta \in (0,1)$ there exist constants $0 < c_\delta < C_\delta < \infty$ depending only on (k_x, δ, f_-, f_+) such that, with probability $\to 1$,

$$c_\delta\Big(\tfrac{\log(1/\delta)}{s}\Big)^{1/k_x} \leqslant R_{s,\delta}(\mathbf{x}) \leqslant C_\delta\Big(\tfrac{\log(1/\delta)}{s}\Big)^{1/k_x}. \tag{3.3}$$

In particular, $R_{s,\delta}(\mathbf{x}) = \Theta_{\mathbb{P}}(s^{-1/k_x})$ for fixed $\delta \in (0,1)$.

3.1.1. Oracle Kernels and their Hájek projection

For $j \in [p_\alpha]$ define the localized DNN score kernel of order s by

$$h_{j,s}(D_{[s]}) \coloneqq \sum_{i=1}^{s} \kappa\left(\mathbf{x}; W_i, D_{[s]}\right)\ \psi_j\big(W_i; \alpha_{0,j}(\mathbf{x}), \eta_{0,j}(Z_i^j)\big), \tag{3.4}$$

and write $\theta_{j,s} \coloneqq \mathbb{E}\big[h_{j,s}(D_{[s]})\big]$, $g_{j,s} \coloneqq h_{j,s} - \theta_{j,s}$. The first Hájek projection kernel is

$$h_{j,s}^{\langle 1\rangle}(w) \coloneqq \mathbb{E}[\ g_{j,s}(w, D_{2:s})\ ]. \tag{3.5}$$

The localized statistic admits the U-statistic representation

$$\overline{m}_j(\mathbf{x}) = \binom{n}{s}^{-1} \sum_{\ell \in \mathcal{L}_{n,s}} h_{j,s}(D_\ell) = \theta_{j,s} + \frac{s}{n}\sum_{i=1}^{n} h_{j,s}^{\langle 1\rangle}(W_i) + \mathsf{R}_{j,n}, \tag{3.6}$$

where $\mathsf{R}_{j,n}$ collects the higher-order Hoeffding projections. For any $\gamma \in \mathbb{R}^{p_\alpha}$, define the linear combination kernels

$$h_{\gamma,s} \coloneqq \sum_{j=1}^{p_\alpha} \gamma_j h_{j,s}, \qquad g_{\gamma,s} \coloneqq h_{\gamma,s} - \mathbb{E}[h_{\gamma,s}], \qquad h_{\gamma,s}^{\langle 1\rangle} \coloneqq \mathbb{E}[\, g_{\gamma,s}(\,\cdot\,, D_{2:s})\,], \tag{3.7}$$

and the variance components

$$\zeta_{ss}(\gamma) \coloneqq \mathrm{Var}\big(h_{\gamma,s}(D_{[s]})\big), \qquad \zeta_{1s}(\gamma) \coloneqq \mathrm{Var}\Big(h_{\gamma,s}^{\langle 1\rangle}(W_1)\Big). \tag{3.8}$$

Definition 3.1 (Hájek kernel of the Weight / DNN Localizer)**.**
For fixed $\mathbf{x} \in \mathcal{X}$ and subset size s, define the (single-observation) Hájek kernel of the weight

$$\tilde{\tau}_{s,\mathbf{x}}(u) \coloneqq s\, \mathbb{E}\big[\kappa\big(\mathbf{x}; W_1, D_{2:s}\big) \,\big|\, X_1 = u\big], \qquad \tilde{\tau}^{\circ}_{s,\mathbf{x}}(u) \coloneqq \tilde{\tau}_{s,\mathbf{x}}(u) - \mathbb{E}[\tilde{\tau}_{s,\mathbf{x}}(X)]. \tag{3.9}$$

We refer to $\tilde{\tau}_{s,\mathbf{x}}$ as the DNN localizer.

Under Condition 2.7, the localized pseudo-target $\boldsymbol{\alpha}_s(\mathbf{x})$ satisfies $\sqrt{n_{\mathrm{eff}}(\mathbf{x})}\,|\alpha_{s,j}(\mathbf{x}) - \alpha_{0,j}(\mathbf{x})| \to 0$ for each j, so all main-text results are centered at $\boldsymbol{\alpha}_0(\mathbf{x})$. The pseudo-target and the analysis without undersmoothing are treated in Section Appendix B.1.

Lemma 3.3 (Product Form of the Hájek Kernel)**.** (→ Proof, p. 56)
Let Condition 2.4 hold and let $h_{j,s}^{\langle 1\rangle}$ be as in Equation (3.5). Then, for each $j \in [p_\alpha]$,

$$h_{j,s}^{\langle 1\rangle}(W) = \tilde{\tau}^{\circ}_{s,\mathbf{x}}(X)\, \psi_j\big(W; \alpha_{0,j}(\mathbf{x}), \eta_{0,j}\big). \tag{3.10}$$

Proposition 3.4 (Hájek Dominance for DNN kernels - Uniform in γ)**.** (→ Proof, p. 38)
Let Conditions 2.3 and 2.4 hold and assume $s \to \infty$ with $s = o(n)$. Then, uniformly over $\|\gamma\|_2 = 1$,

$$\frac{s}{n}\left(\frac{\zeta_{ss}(\gamma)}{s\,\zeta_{1s}(\gamma)} - 1\right) \longrightarrow 0. \tag{3.11}$$

In particular, $\overline{m}_{\gamma,s}(\mathbf{x}) \coloneqq \binom{n}{s}^{-1} \sum_{\ell \in \mathcal{L}_{n,s}} h_{\gamma,s}(D_\ell)$ is asymptotically linear with influence function $h_{\gamma,s}^{\langle 1\rangle}(W)$ at the natural $\sqrt{n_{\mathrm{eff}}(\mathbf{x})}$ scale, simultaneously for all $\|\gamma\|_2 = 1$.

Using the arguments from Juergens (2025), this result follows nearly immediately from the structure of the DNN weights and Condition 2.4 (ii). Hájek dominance here means the U-statistic is, to first order, a simple weighted mean of single-observation influence terms $h_{\gamma,s}^{\langle 1\rangle}(W_i)$. Intuitively, the higher-order "interaction" components wash out because the DNN neighborhood is small (radius $\asymp s^{-1/k_x}$) and only about $n_{\mathrm{eff}}(\mathbf{x}) \asymp n/s$ observations carry nonnegligible weight. The variance is therefore governed by the first projection, and the CLT reduces to a Lindeberg condition for a triangular array of localized weighted terms.

3.1.2. Existence of an Approximate Root

Our next task is to guarantee that the localized estimating equation has a well-defined empirical solution at the right scale and with a solver tolerance that is asymptotically negligible for first-order inference. Placing this existence result here lets us exploit the already-established control of the empirical-oracle gap at the $\sqrt{n_{\mathrm{eff}}(\mathbf{x})}$ scale, while postponing an explicit linearization identity to where it naturally drives the Bahadur expansion.

Proposition 3.5 (Existence, Bracketing, and High-Probability Uniqueness of an Approximate Empirical Root)**.**
(→ Proof, p. 39)
Fix $j \in [p_\alpha]$ and for a in a neighborhood $\mathcal{A}_j(\mathbf{x})$ of $\alpha_{0,j}(\mathbf{x})$ define

$$F_{n,j}(a) \coloneqq \frac{1}{\sqrt{n_{\mathrm{eff}}(\mathbf{x})}} \sum_{i=1}^{n} w_i(\mathbf{x})\, \psi_j\big(W_i; a, \widehat{\eta}_j\big)\, \mathbb{1}\{\, X_i \in \mathcal{B}_{\mathbf{x}}\,\}, \tag{3.12}$$

and the centered increment

$$\Delta F_{n,j}(a) \coloneqq F_{n,j}(a) - F_{n,j}\big(\alpha_{0,j}(\mathbf{x})\big). \tag{3.13}$$

Suppose Conditions 2.4, 2.5, 2.6 and 2.7 hold (so that Lemmas 2.2 and 2.4 apply), and let $\Gamma_j(\mathbf{x}) \coloneqq \partial_\alpha m_j\big(\mathbf{x}; \alpha_{0,j}(\mathbf{x}), \eta_{0,j}\big)$ *with* $\inf_j |\,\Gamma_j(\mathbf{x})\,| \geqslant c > 0$*. Then, for any fixed* $M > 0$*:*

(i) ***Sign-Change Bracket at the*** $n_{\text{eff}}\,(\mathbf{x})^{-1/2}$ ***Scale:*** *For any* $\Delta_{n,j}$ *with* $\Delta_{n,j} = o_{\mathbb{P}}\Big(\, b_n^{-1} / \sqrt{n_{\text{eff}}\,(\mathbf{x})}\,\Big)$*, with probability tending to one,*

$$\operatorname{sgn}\Big(F_{n,j}\big(\alpha_{0,j}(\mathbf{x}) - M/\sqrt{n_{\text{eff}}\,(\mathbf{x})}\big) - \Delta_{n,j}\Big) = -\operatorname{sgn}\big(\Gamma_j(\mathbf{x})\big), \tag{3.14}$$

$$\operatorname{sgn}\Big(F_{n,j}\big(\alpha_{0,j}(\mathbf{x}) + M/\sqrt{n_{\text{eff}}\,(\mathbf{x})}\big) - \Delta_{n,j}\Big) = \operatorname{sgn}\big(\Gamma_j(\mathbf{x})\big). \tag{3.15}$$

(ii) ***Uniform Increment Approximation:*** *Uniformly over* $|\,a - \alpha_{0,j}(\mathbf{x})\,| \leqslant M/\sqrt{n_{\text{eff}}\,(\mathbf{x})}$*,*

$$\Delta F_{n,j}(a) = \sqrt{n_{\text{eff}}\,(\mathbf{x})}\,\Gamma_j(\mathbf{x})\,\big(a - \alpha_{0,j}(\mathbf{x})\big) + o_{\mathbb{P}}(\,1\,). \tag{3.16}$$

(iii) ***High-Probability Local Monotonicity:*** *If Condition 2.5 (iv) and (v) hold, then with probability tending to one the map* $a \mapsto F_{n,j}(a)$ *is strictly monotone on* $\big[\alpha_{0,j}(\mathbf{x}) - M/\sqrt{n_{\text{eff}}\,(\mathbf{x})},\ \alpha_{0,j}(\mathbf{x}) + M/\sqrt{n_{\text{eff}}\,(\mathbf{x})}\big]$ *(increasing if* $\Gamma_j(\mathbf{x}) > 0$*, decreasing if* $\Gamma_j(\mathbf{x}) < 0$*).*

(iv) ***Existence of an Approximate Root:*** *With probability tending to one there exists* $\tilde{\alpha}_j(\mathbf{x}) \in \big[\alpha_{0,j}(\mathbf{x}) - M/\sqrt{n_{\text{eff}}\,(\mathbf{x})},\ \alpha_{0,j}(\mathbf{x}) + M/\sqrt{n_{\text{eff}}\,(\mathbf{x})}\big]$ *such that*

$$F_{n,j}\big(\tilde{\alpha}_j(\mathbf{x})\big) = \Delta_{n,j}, \tag{3.17}$$

and $|\,\tilde{\alpha}_j(\mathbf{x}) - \alpha_{0,j}(\mathbf{x})\,| = O_{\mathbb{P}}\Big(\, n_{\text{eff}}\,(\mathbf{x})^{-1/2}\,\Big)$*.*

(v) ***Uniqueness and Exact Root under Continuity:*** *On the high-probability monotonicity event in (iii), the solution to Equation (3.17) is unique on the bracket. If, in addition,* $a \mapsto \psi_j(W; a, \widehat{\eta}_j)$ *is almost surely continuous (so that* $F_{n,j}$ *is continuous), then taking* $\Delta_{n,j} \equiv 0$ *yields an exact empirical root in the bracket, unique with probability tending to one.*

Proposition 3.5 is an existence and well-posedness result at the natural $n_{\text{eff}}\,(\mathbf{x})^{-1/2}$ scale. It says that, in a small $O(n_{\text{eff}}\,(\mathbf{x})^{-1/2})$ neighborhood around the point target $\alpha_{0,j}(\mathbf{x})$, the localized sample score behaves just like its population counterpart: it drifts up or down with a clear sign, and the influence of randomness becomes too small to overturn the drift. Under Condition 2.7, the localized moment map $m^{\text{loc}}_{j,s}(\cdot;\mathbf{x})$ has its zero within $o(n_{\text{eff}}\,(\mathbf{x})^{-1/2})$ of $\alpha_{0,j}(\mathbf{x})$, so the point target lies inside the bracket and local identification is inherited. Because our localizer concentrates information near $\mathbf{x}$ and our weights are stable, the empirical curve inherits the same sign change across that window. A crossing therefore exists. Moreover, the population curve is smooth enough that the crossing is unique with high probability. We do not need the score itself to be differentiable; only that the average moment is well-behaved near the point target. Finally, we allow a tiny solver tolerance $\Delta_{n,j}$: think of it like a numerical tolerance that is strictly smaller than the statistical noise driving first-order uncertainty. That tolerance cannot affect standard errors or Wald tests. With a unique, well-localized solution in hand, the next subsection can safely "linearize" the score and derive the usual Bahadur expansion that powers inference.

3.1.3. Bahadur Representation in Localizer Form

Combining Lemma 3.3 and Proposition 3.4 with Equation (3.6) yields a linear statistic driven by the DNN localizer:

Definition 3.2 (Localized oracle linear statistic)**.**
For $j \in [p_\alpha]$, define

$$L_{n,s,j}(\mathbf{x}) \coloneqq \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \tilde{\tau}^{\circ}_{s,\mathbf{x}}(X_i)\ \psi_j\big(W_i; \alpha_{0,j}(\mathbf{x}), \eta_{0,j}\big). \tag{3.18}$$

The oracle influence function in Proposition 3.4 matches the first-order term in our estimators score expansion, up to two adjustments: namely, substituting $(\boldsymbol{\alpha}_0(\mathbf{x}), \boldsymbol{\eta}_0)$ by plug-ins, and evaluating scores on the localized sample (rather than exactly at $X = \mathbf{x}$). Both adjustments can be shown to be $o_{\mathbb{P}}(1)$ at the $\sqrt{n_{\text{eff}}(\mathbf{x})}$ scale thanks to Lemma 2.4 and Condition 2.6. Intuitively, LSE ensures that small L_2 perturbations of the score within the DNN neighborhood produce only small changes in the localized centered average, while stability guarantees that data-dependent localization and nuisance fitting do not inflate the variance. The following theorem records the resulting first-order expansion we will use in the next subsection to show asymptotic normality.

Theorem 3.6 (Local Bahadur Representation)**.** (→ Proof, p. 40)
Assume Conditions 2.4, 2.5, 2.6 and 2.7 hold, so that Lemmas 2.2, 2.4 and Appendix E.4 apply. Fix $j \in [p_\alpha]$, write $G \coloneqq \{ X \in \mathcal{B}_{\mathbf{x}} \}$, and let $\widehat{\alpha}_j(\mathbf{x})$ be any solution to

$$\sum_{i=1}^{n} w_{s,i}(\mathbf{x})\ \psi_j(W_i; \alpha, \widehat{\eta}_j)\ \mathbb{1}\{ X_i \in G \} = 0, \qquad \big|\alpha - \alpha_{0,j}(\mathbf{x})\big| \leqslant \delta_n, \delta_n \downarrow 0, \tag{3.19}$$

with $\mathbb{P}\big(\big|\widehat{\alpha}_j(\mathbf{x}) - \alpha_{0,j}(\mathbf{x})\big| \leqslant \delta_n\big) \to 1$. If $\Gamma_j(\mathbf{x}) \geqslant c > 0$ as in Condition 2.5, then

$$\sqrt{n_{\text{eff}}(\mathbf{x})}\ \big(\widehat{\alpha}_j(\mathbf{x}) - \alpha_{0,j}(\mathbf{x})\big) = \Gamma_j(\mathbf{x})^{-1}\ \mathbb{G}_{n,w}\Big[\psi_j\big(\cdot; \alpha_{0,j}(\mathbf{x}), \eta_{0,j}\big)\ \mathbb{1}\{ X \in G \}\Big] + o_{\mathbb{P}}(1). \tag{3.20}$$

Corollary 3.7 (Bahadur representation with the DNN localizer)**.**
Under the assumptions of Theorem 3.6,

$$\sqrt{n_{\text{eff}}(\mathbf{x})}\ \big(\widehat{\alpha}_j(\mathbf{x}) - \alpha_{0,j}(\mathbf{x})\big) = \Gamma_j(\mathbf{x})^{-1}\ L_{n,s,j}(\mathbf{x}) + o_{\mathbb{P}}(1), \tag{3.21}$$

where $L_{n,s,j}(\mathbf{x})$ is defined in Equation (3.18) and evaluates the oracle score at $\alpha_{0,j}(\mathbf{x})$; under Condition 2.7 the centering at $\alpha_{0,j}(\mathbf{x})$ is exact to leading order.

The next subsection states the Gaussian limit. Once Equation (3.20) is in place, asymptotic normality at rate $\sqrt{n_{\text{eff}}(\mathbf{x})}$ follows from a localized Lindeberg condition (or, equivalently, from a CLT for the first Hájek projection $h_{j,s}^{\langle 1 \rangle}$), yielding variance $\Gamma(\mathbf{x})^{-1}\Sigma(\mathbf{x})\Gamma(\mathbf{x})^{-1}$.

3.2. Main Theorems - Asymptotic Normality

We now state the Gaussian limit for the DNN Z-estimator at a fixed interior point $\mathbf{x}$. Write $\widehat{\boldsymbol{\alpha}}(\mathbf{x}) \coloneqq \big(\widehat{\alpha}_1(\mathbf{x}), \ldots, \widehat{\alpha}_{p_\alpha}(\mathbf{x})\big)^\top$ and recall the localized process Equation (2.47). Let $\tilde{\tau}_{s,\mathbf{x}}$ denote the DNN localizer at $\mathbf{x}$ associated with the weights $\{w_{s,i}(\mathbf{x})\}$. Throughout this subsection we set $G \coloneqq \{ X \in \mathcal{B}_{\mathbf{x}} \}$. By Theorem 3.6 (combined with Condition 2.7), each coordinate admits the local Bahadur expansion centered at the point target $\alpha_{0,j}(\mathbf{x})$:

$$\sqrt{n_{\text{eff}}(\mathbf{x})}\ \big(\widehat{\alpha}_j(\mathbf{x}) - \alpha_{0,j}(\mathbf{x})\big) = \Gamma_j(\mathbf{x})^{-1}\ \mathbb{G}_{n,w}\Big[\psi_j\big(\cdot; \alpha_{0,j}(\mathbf{x}), \eta_{0,j}\big)\ \mathbb{1}\{ X \in G \}\Big] + o_{\mathbb{P}}(1), \qquad j \in [p_\alpha]. \tag{3.22}$$

Hence asymptotic normality centered at $\boldsymbol{\alpha}_0(\mathbf{x})$ follows from a CLT for the leading (vector) empirical process term for which we employ the CLT machinery developed in Peng et al. (2022).

Theorem 3.8 (Asymptotic Normality (Vector Form))**.** (→ Proof, p. 46)
Assume Conditions 2.2, 2.3, 2.4, 2.5, 2.6 and 2.7 hold. Then

$$\sqrt{n_{\text{eff}}(\mathbf{x})}\ \big(\widehat{\boldsymbol{\alpha}}(\mathbf{x}) - \boldsymbol{\alpha}_0(\mathbf{x})\big) = \Gamma(\mathbf{x})^{-1}\ \mathbb{G}_{n,w}\big[\boldsymbol{\psi}(\,\cdot\,; \boldsymbol{\alpha}_0(\mathbf{x}), \boldsymbol{\eta}_0)\ \mathbb{1}\{ X \in G \}\big] + o_{\mathbb{P}}(1), \tag{3.23}$$

and, consequently,

$$\sqrt{n_{\text{eff}}(\mathbf{x})}\ \big(\widehat{\boldsymbol{\alpha}}(\mathbf{x}) - \boldsymbol{\alpha}_0(\mathbf{x})\big) \xrightarrow{d} \mathcal{N}\big(0, \Omega(\mathbf{x})\big), \qquad \Omega(\mathbf{x}) \coloneqq \Gamma(\mathbf{x})^{-1}\ \Sigma(\mathbf{x})\ \Gamma(\mathbf{x})^{-1}, \tag{3.24}$$

where $\Sigma(\mathbf{x})$ is given in Equation (3.29).

In our separable-score setup, $\Gamma(x)$ is diagonal with $\Gamma_{jj}(x) = \Gamma_j(x)$, so $\omega_j^2(x) = \Gamma_j(x)^{-2}\ \Sigma_{jj}(x)$ and $\sigma_j^2(x) = \Sigma_{jj}(x)$, making coordinate-wise asymptotic normality an immediate consequence.

3.3. Main Theorems - Variance Estimation

The Gaussian limit in Section 3.2 shows that, at a fixed interior point $\mathbf{x}$,

$$\sqrt{n_{\text{eff}}(\mathbf{x})}\big(\widehat{\boldsymbol{\alpha}}(\mathbf{x}) - \boldsymbol{\alpha}_0(\mathbf{x})\big) \xrightarrow{d} \mathcal{N}\big(0,\ \Omega(\mathbf{x})\big), \tag{3.25}$$

with

$$\Omega(\mathbf{x}) \coloneqq \boldsymbol{\Gamma}_{\mathbf{x}}^{-1}\ \Sigma(\mathbf{x})\ \boldsymbol{\Gamma}_{\mathbf{x}}^{-1}, \tag{3.26}$$

where $\boldsymbol{\Gamma}_{\mathbf{x}}$ is the Jacobian ("bread") of the conditional moment map and $\Sigma(\mathbf{x})$ is the covariance ("meat") of the localized estimating equation at the truth (cf. Equation (3.29)). Thus inference reduces to constructing consistent, numerically stable estimators of these two objects. Two features of our localization matter for estimation. First, the meat $\Sigma(\mathbf{x})$ arises from a triangular array: the localizer $\tilde{\tau}_{s,\mathbf{x}}$ and effective sample size $n_{\text{eff}}(\mathbf{x})$ depend on n. Second, the bread $\boldsymbol{\Gamma}_{\mathbf{x}}$ can be obtained either by localized finite differences or, when the score is differentiable, by a derivative-based plug-in. To accommodate the triangular-array nature of $\Sigma(\mathbf{x})$, we adopt matrix ratio-consistency (Definition 3.3) as our criterion; this is the natural notion when the target covariance itself varies with n. We proceed in two steps: we estimate the meat via a delete-1 jackknife scaled by $n_{\text{eff}}(\mathbf{x})$ (yielding $\widehat{\Sigma}_{\mathbf{x}}$), and estimate the bread via a symmetric localized finite difference (yielding $\widehat{\boldsymbol{\Gamma}}_{\mathbf{x}}$), with a derivative-based alternative when available. The resulting sandwich $\widehat{\Omega}(\mathbf{x}) \coloneqq \widehat{\boldsymbol{\Gamma}}_{\mathbf{x}}^{-1}\widehat{\Sigma}_{\mathbf{x}}\widehat{\boldsymbol{\Gamma}}_{\mathbf{x}}^{-T}$ is then shown to be ratio-consistent for $\Omega(\mathbf{x})$.

Definition 3.3 (Matrix Ratio-Consistency)**.**
Let $\Sigma_n \in \mathbb{S}_+^p$ be a (population) covariance and $\widehat{\Sigma}_n$ an estimator. We say $\widehat{\Sigma}_n$ is ratio-consistent for Σ_n if

$$\rho_n \coloneqq \sup_{\substack{v\in\mathbb{R}^p \\ ;\|v\|_2=1,\ v^\top\Sigma_n v>0}} \left| \frac{v^\top\widehat{\Sigma}_n v}{v^\top\Sigma_n v} - 1 \right| \xrightarrow{\mathbb{P}} 0. \tag{3.27}$$

If $\Sigma_n \succ 0$, this is equivalent to $\left\| \Sigma_n^{-1/2}\ \widehat{\Sigma}_n\ \Sigma_n^{-1/2} - I_p \right\|_{\text{op}} \xrightarrow{\mathbb{P}} 0$.

3.3.1. Jackknifing the "Meat"

For each $j \in [p_\alpha]$, define the localized empirical moment at $a \in \mathcal{A}_j(\mathbf{x})$ by

$$\widehat{m}_j(a;\mathbf{x}) \coloneqq \sum_{i=1}^n w_{s,i}(\mathbf{x})\ \psi_j\left(W_i;\ a,\ \widehat{\eta}_j(Z_i^j)\right) \tag{3.28}$$

and let $\widehat{\boldsymbol{m}}(a;\mathbf{x})$ be the p_α-vector stacking these components. The (population) "meat" at the truth coincides with the covariance of the leading process in Theorem 3.8:

$$\Sigma(\mathbf{x}) \coloneqq \text{Var}((\tilde{\tau}_{s,\mathbf{x}} - \mathbb{E}[\tilde{\tau}_{s,\mathbf{x}}])\ \boldsymbol{\psi}(W;\boldsymbol{\alpha}_0(\mathbf{x}),\boldsymbol{\eta}_0)) = \lim_{n\to\infty} n_{\text{eff}}(\mathbf{x})\ \text{Var}(\widehat{\boldsymbol{m}}(\boldsymbol{\alpha}_0(\mathbf{x});\mathbf{x})). \tag{3.29}$$

Under Hájek dominance (Proposition 3.4), $\text{Var}(\widehat{\boldsymbol{m}}(\boldsymbol{\alpha}_0;\mathbf{x})) \asymp n_{\text{eff}}(\mathbf{x})^{-1}$. We impose Condition 2.6 so that keeping $\widehat{\eta}_j$ fixed in jackknife deletes is valid up to $o_{\mathbb{P}}\Big(n_{\text{eff}}(\mathbf{x})^{-1/2}\Big)$. For $a \in \mathcal{A}_j(\mathbf{x})$, define delete-1 localized scores by dropping i only from the second stage:

$$\widehat{m}_j^{(-i)}(a;\mathbf{x}) \coloneqq \sum_{r\neq i} w_r^{(-i)}(\mathbf{x})\ \psi_j\left(W_r;\ a,\ \widehat{\eta}_j(Z_r^j)\right), \tag{3.30}$$

where $w_r^{(-i)}(\mathbf{x})$ are the DNN L-weights of order s recomputed on the $n-1$ sample with observation i removed (same s and tie-break). Stack $\widehat{\boldsymbol{m}}^{(-i)}(a;\mathbf{x})$ across j and define jackknife pseudo-values

$$\boldsymbol{P}_i(a) \ \coloneqq\ n\,\widehat{\boldsymbol{m}}(a;\mathbf{x}) \ -\ (n-1)\,\widehat{\boldsymbol{m}}^{(-i)}(a;\mathbf{x}), \qquad \overline{\boldsymbol{P}}(a) \ \coloneqq\ \frac{1}{n}\sum_{i=1}^n \boldsymbol{P}_i(a). \tag{3.31}$$

Let

$$\widehat{\Omega}_{\mathbf{x}}^{\mathrm{JK}}(a) \coloneqq \frac{n-1}{n}\sum_{i=1}^{n}\big(\boldsymbol{P}_i(a)-\overline{\boldsymbol{P}}(a)\big)\big(\boldsymbol{P}_i(a)-\overline{\boldsymbol{P}}(a)\big)^{\top}, \qquad \widehat{\Sigma}_{\mathbf{x}}(a) \;\coloneqq\; n_{\mathrm{eff}}(\mathbf{x})\;\widehat{\Omega}_{\mathbf{x}}^{\mathrm{JK}}(a). \tag{3.32}$$

We begin by showing ratio-consistency of an infeasible α-oracle Jackknife, i.e. when $a = \boldsymbol{\alpha}_0(\mathbf{x})$. This α-oracle version, will however already use the plug-in estimator for the nuisance as outlined in Equation (3.28).

Proposition 3.9 (Partial-Oracle Jackknife for the Meat)**.** (→ Proof, p. 47)

Assume Conditions 2.3, 2.4 and 2.6 and Hájek dominance (Proposition 3.4). Assume further that $\Sigma(\mathbf{x}) \succ 0$ *as defined in Equation (3.29). Then the oracle jackknife at the truth is ratio-consistent:*

$$\Big\| \;\widehat{\Sigma}_{\mathbf{x}}\big(\boldsymbol{\alpha}_0(\mathbf{x})\big)\;\Sigma(\mathbf{x})^{-1} - I\;\Big\|_{\mathrm{op}} \xrightarrow{\;\mathbb{P}\;} 0. \tag{3.33}$$

As pointed out, this oracle version is infeasible, we thus proceed to show ratio-consistency of the corresponding plug-in Jackknife.

Theorem 3.10 (Feasible Jackknife via Plug-In)**.** (→ Proof, p. 48)

Assume Conditions 2.3, 2.4, 2.5, 2.6 and 2.7 and Hájek dominance (Proposition 3.4). Let $\Sigma(\mathbf{x})$ *be as in Equation (3.29), and define* $\widehat{\Sigma}_x(a)$ *as in Equation (3.32) with scores evaluated at a. Then*

$$\Big\|\widehat{\Sigma}_x\big(\widehat{\boldsymbol{\alpha}}(\mathbf{x})\big) - \widehat{\Sigma}_x\big(\boldsymbol{\alpha}_0(\mathbf{x})\big)\Big\|_{\mathrm{op}} = o_{\mathbb{P}}(1), \qquad \Big\|\widehat{\Sigma}_x\big(\widehat{\boldsymbol{\alpha}}(\mathbf{x})\big)\,\Sigma(\mathbf{x})^{-1} - I\Big\|_{\mathrm{op}} \xrightarrow{\;\mathbb{P}\;} 0. \tag{3.34}$$

Using the Steele L-form (Equation (2.17)), Equation (3.30) is computed in $O(n)$ per delete without enumerating subsamples; only the sorted ranks by $\|X_i - \mathbf{x}\|_2$ and their closed-form L-weights update deterministically when i is removed. Thus $\widehat{\Sigma}_{\mathbf{x}}$ is obtained with essentially the same complexity as the original DNN estimator. This highlights that the computation of standard errors for our estimator is computationally uncharacteristically light for a U-statistic-based estimator, making implementations surprisingly lightweight and fast.

3.3.2. Estimating the "Bread"

To estimate the Jacobian without score derivatives, we use a symmetric localized finite difference evaluated at a generic center $a \in \mathcal{A}(\mathbf{x})$:

$$\widehat{\Gamma}_{\cdot k}(a;\mathbf{x}) \coloneqq \frac{\widehat{\boldsymbol{m}}(a + h\,e_k;\mathbf{x}) \;-\; \widehat{\boldsymbol{m}}(a - h\,e_k;\mathbf{x})}{2h}, \qquad k \in [p_\alpha], \tag{3.35}$$

and stack columns into $\widehat{\boldsymbol{\Gamma}}_{\mathbf{x}}(a)$. The population Jacobian (bread) is

$$\boldsymbol{\Gamma}_{\mathbf{x}} \coloneqq \frac{\partial}{\partial a^{\top}}\,\overline{\boldsymbol{m}}(a;\mathbf{x})\Big|_{a=\boldsymbol{\alpha}_0(\mathbf{x})}. \tag{3.36}$$

$$\overline{\boldsymbol{m}}(a;\mathbf{x}) \coloneqq \mathbb{E}[\widehat{\boldsymbol{m}}(a;\mathbf{x})]. \tag{3.37}$$

which exists under Condition 2.5. Let $h = h_n \downarrow 0$ satisfy

$$h \to 0, \qquad \frac{\beta_n^{\mathrm{loc}}(\mathbf{x})}{h} \to 0, \qquad \sqrt{n_{\mathrm{eff}}(\mathbf{x})}\,h^2 \to 0 \quad (\text{e.g. } h \asymp n_{\mathrm{eff}}(\mathbf{x})^{-1/3} \text{ when } \beta_n^{\mathrm{loc}}(\mathbf{x}) = o(n_{\mathrm{eff}}(\mathbf{x})^{-1/2})). \tag{3.38}$$

The first two conditions control FD bias $O(h^2)$ and LOO perturbation $O(\beta_n^{\mathrm{loc}}(\mathbf{x})/h)$; the last aligns with the finite-difference LSE block ((iii), Equation (2.43), Equation (2.51)). As for the case of the "meat", we begin by showing consistency of an α-oracle finite difference, i.e. when $a = \boldsymbol{\alpha}_0(\mathbf{x})$ (the nuisance remains the plug-in $\widehat{\boldsymbol{\eta}}$ as in Equation (3.28)).

Proposition 3.11 (Partial-Oracle Finite-Difference for the Bread)**.** (→ Proof, p. 50)

Assume Conditions 2.4 and 2.5 and (iii) and Equation (3.38). Then

$$\left\| \widehat{\mathbf{\Gamma}}_{\mathbf{x}}\big(\boldsymbol{\alpha}_0(\mathbf{x})\big) - \mathbf{\Gamma}_{\mathbf{x}} \right\|_{\mathrm{op}} = O_{\mathbb{P}}\Big(n_{\mathrm{eff}}\,(\mathbf{x})^{-1/2} \Big) + O_{\mathbb{P}}\big(h^2 \big) + O_{\mathbb{P}}\big(\beta_n^{\mathrm{loc}}\,(\mathbf{x})\,/h \big), \tag{3.39}$$

and in particular $\widehat{\mathbf{\Gamma}}_{\mathbf{x}}\big(\boldsymbol{\alpha}_0(\mathbf{x})\big) \xrightarrow{\mathbb{P}} \mathbf{\Gamma}_{\mathbf{x}}$.

The α-oracle version is once again infeasible; we now show that centering at the plug-in $\widehat{\boldsymbol{\alpha}}(\mathbf{x})$ is asymptotically equivalent and inherits the same rate.

Theorem 3.12 (Feasible Finite-Difference via Plug-In)**.** (→ Proof, p. 50)

Assume Conditions 2.4, 2.5 and 2.7 and (iii) and Equation (3.38). Define the feasible estimator by $\widehat{\mathbf{\Gamma}}_{\mathbf{x}} \coloneqq \widehat{\mathbf{\Gamma}}_{\mathbf{x}}(\widehat{\boldsymbol{\alpha}}(\mathbf{x}))$ *with columns from Equation (3.35) at* $a = \widehat{\boldsymbol{\alpha}}(\mathbf{x})$*. Then*

$$\left\| \widehat{\mathbf{\Gamma}}_{\mathbf{x}}\big(\widehat{\boldsymbol{\alpha}}(\mathbf{x})\big) - \widehat{\mathbf{\Gamma}}_{\mathbf{x}}\big(\boldsymbol{\alpha}_0(\mathbf{x})\big) \right\|_{\mathrm{op}} = o_{\mathbb{P}}(\,1\,), \tag{3.40}$$

and consequently

$$\left\| \widehat{\mathbf{\Gamma}}_{\mathbf{x}} - \mathbf{\Gamma}_{\mathbf{x}} \right\|_{\mathrm{op}} = O_{\mathbb{P}}\Big(n_{\mathrm{eff}}\,(\mathbf{x})^{-1/2} \Big) + O_{\mathbb{P}}\big(h^2 \big) + O_{\mathbb{P}}\big(\beta_n^{\mathrm{loc}}\,(\mathbf{x})\,/h \big) + o_{\mathbb{P}}(\,1\,), \tag{3.41}$$

so in particular $\widehat{\mathbf{\Gamma}}_{\mathbf{x}} \xrightarrow{\mathbb{P}} \mathbf{\Gamma}_{\mathbf{x}}$.

Computing $\widehat{\mathbf{\Gamma}}_{\mathbf{x}}(a)$ requires $2p_\alpha$ evaluations of $\widehat{\boldsymbol{m}}(\cdot;\mathbf{x})$ at $(a \pm he_k)_{k\leqslant p_\alpha}$, each in $O(n)$ time using the fixed DNN weights $\{w_i(\mathbf{x})\}_{i\leqslant n}$. No recomputation across deletes is needed, so the cost is comparable to a handful of localized score evaluations, showing that the second part of the variance sandwich is computationally simple as well. Importantly, we can also verify an assumption that we made in Condition 2.6 (iii) as it is fulfilled by means of more primitive assumptions and the structure of the DNN weights themselves. Once more, we decided to preserve the assumption for clarity when considering alternative weighting schemes in the future.

Lemma 3.13 (Finite-Difference Bread is LOO Stable)**.** (→ Proof, p. 57)
Fix $j \in [p_\alpha]$ *and let* $G \coloneqq \{\, X \in \mathcal{B}_{\mathbf{x}} \,\}$ *be as in Condition 2.6. Let* $\widehat{\Gamma}_{h,j}(\mathbf{x})$ *be the central finite-difference estimator in (iii), computed from the localized moment map with weights* $w(D;\mathbf{x})$ *and nuisance* $\widehat{\eta}_j$*, and let* $\widehat{\Gamma}_{h,j}(D^{(l)})$ *denote the same object on the replace-one sample* $D^{(l)}$ *with weights* $w(D^{(l)};\mathbf{x})$ *and nuisance* $\widehat{\eta}_j^{(-l)}$*. Assume Conditions 2.4 and 2.5 and Condition 2.6 (i)(ii). Then there exists a finite constant* $L_{j,\mathbf{x}}$ *(depending only on the local score envelope in Condition 2.4 and G) such that*

$$\sup_{D} \max_{l\in[n]} \Big| \big[\widehat{\Gamma}_{h,j}(D) - \widehat{\Gamma}_{h,j}(D^{(l)})\big]\ \mathbb{1}\{\,G\,\} \Big| \leqslant L_{j,\mathbf{x}}\, \frac{\beta_n^{\mathrm{loc}}\,(\mathbf{x})}{h}\ , \tag{3.42}$$

where $\beta_n^{\mathrm{loc}}\,(\mathbf{x})$ *is the localized replace-one modulus in Equation (2.38). Consequently, if in addition* $\sqrt{n_{\mathrm{eff}}\,(\mathbf{x})}\,\beta_n^{\mathrm{loc}}\,(\mathbf{x})\,/h \to 0$ *and* $\sqrt{n_{\mathrm{eff}}\,(\mathbf{x})}\,h^2 \to 0$ *as in (iii), then*

$$\max_{l\in[n]}\ \sqrt{n_{\mathrm{eff}}\,(\mathbf{x})}\,\Big|\widehat{\Gamma}_{h,j}(D) - \widehat{\Gamma}_{h,j}(D^{(l)})\Big| \xrightarrow{\mathbb{P}} 0. \tag{3.43}$$

Remark 3.1 (Derivative-based Bread when the Score is Smooth)**.**
Assume that for u in $\mathcal{B}_{\mathbf{x}}$ *and* $\tilde{\eta}$ *in a neighborhood of* $\eta_{0,j}$*, the map* $a \mapsto \psi_j\,(W, a, \tilde{\eta})$ *is (a.s.) differentiable in a neighborhood of* $a = \alpha_{0,j}\,(\mathbf{x})$ *and differentiation may be interchanged with the conditional expectation defining* m_j *in Equation (2.25), then Condition 2.5 (v) implies*

$$\Gamma_j\,(\mathbf{x}) \coloneqq \partial_a m_j\,(\mathbf{x}; a, \eta_{0,j})\,\big|_{a=\alpha_{0,j}(\mathbf{x})} \coloneqq \mathbb{E}[\partial_a \psi_j\,(W, \alpha_{0,j}\,(\mathbf{x})\,, \eta_{0,j}) \mid X = \mathbf{x}]. \tag{3.44}$$

In this case, a natural alternative to the finite-difference bread is the weighted derivative estimator

$$\widehat{\Gamma}_j^{\mathrm{der}}(\mathbf{x}) \coloneqq \sum_{i=1}^{n} w_{s,i}(\mathbf{x})\ \partial_a \psi_j(W_i, \widehat{\alpha}_j(\mathbf{x}), \widehat{\eta}_j), \qquad j \in [p_\alpha], \tag{3.45}$$

stacked into $\widehat{\Gamma}_{\mathbf{x}}^{\mathrm{der}}$*. This choice removes the* $1/h$ *amplification and the* $O(h^2)$ *bias in Equations (3.38) and (3.41). Unlike the finite-difference case, this derivative-based construction does not require the second-order Lipschitz smoothness in Condition 2.5 (iv); those additional assumptions were imposed solely to deliver the* $O(h^2)$ *finite-difference truncation error and are immaterial here. In terms of strength, the derivative-based route trades the higher-order population smoothness of* $a \mapsto m_j(u; a, \tilde{\eta})$ *in for a more primitive assumption: almost sure differentiability of* $a \mapsto \psi_j(W, a, \tilde{\eta})$ *with an* L_2 *envelope allowing differentiation under the conditional expectation. It is weaker at the level of the population map* m_j *but stronger at the score (primitive) level.*

Consistency and LOO Stability.
Under Conditions 2.4, 2.5 and 2.6 and Lemma 2.4, apply the localized SE bound to the derivative kernel $f(w) = \partial_a \psi_j\{w, \cdot, \cdot\}\ \mathbb{1}\{G\}$*:*

$$\left\| \widehat{\Gamma}_{\mathbf{x}}^{\mathrm{der}} - \Gamma_{\mathbf{x}} \right\|_{\mathrm{op}} = O_{\mathbb{P}}\Big(n_{\mathrm{eff}}(\mathbf{x})^{-1/2} \Big) + o_{\mathbb{P}}(1), \tag{3.46}$$

$$\max_{\ell \in [n]} \left\| \widehat{\Gamma}_{\mathbf{x}}^{\mathrm{der}} - \widehat{\Gamma}_{\mathbf{x}}^{\mathrm{der},(\ell)} \right\|_{\mathrm{op}} = o_{\mathbb{P}}\Big(n_{\mathrm{eff}}(\mathbf{x})^{-1/2} \Big). \tag{3.47}$$

The first display follows from weight negligibility and the L_2*-envelope for* $\partial_a \psi_j$ *(Condition 2.4) together with the nuisance plug-in control from Condition 2.6 (ii). The LOO bound decomposes into a direct term of order* $\max_{i \in [n]} \left| w_{s,i}(\mathbf{x}) \right| = o_{\mathbb{P}}\Big(n_{\mathrm{eff}}^{-1/2} \Big)$ *by Condition 2.6 (i), and an indirect term proportional to* $\left\| \widehat{\eta}_j - \widehat{\eta}_j^{(\ell)} \right\|_{\eta}$*, which is* $o_{\mathbb{P}}\Big(n_{\mathrm{eff}}^{-1/2} \Big)$ *by the localized replace-one stability for the Jacobian block (Condition 2.6 (ii)). No finite-difference tuning is needed, so the rate constraints in Equation (3.38) are dispensed with.*

Practical Note.
When analytic or automatic derivatives of ψ_j *are available (e.g., GLM-type or autodiff scores), the derivative bread is preferable: it eliminates bandwidth* h*, reduces bias, and inherits the same variance scale and LOO stability as the finite-difference bread used in Theorem 3.12 and Lemma 3.13. If only directional/symmetric differentiability is available, the finite-difference bread remains a robust default covered by our main results.*

3.3.3. Building the "Sandwich"
Having shown the consistency of the ingredients, the asymptotic covariance of the normalized statistic $\sqrt{n_{\mathrm{eff}}(\mathbf{x})}\left(\widehat{\boldsymbol{\alpha}}(\mathbf{x}) - \boldsymbol{\alpha}_0(\mathbf{x})\right)$ is estimated by the localized sandwich

$$\widehat{\Omega}(\mathbf{x}) \ \coloneqq\ \widehat{\mathbf{\Gamma}}_{\mathbf{x}}^{-1}\, \widehat{\Sigma}_{\mathbf{x}}\, \widehat{\mathbf{\Gamma}}_{\mathbf{x}}^{-T}, \qquad \mathrm{se}(\widehat{\alpha}_j(\mathbf{x})) \ \coloneqq\ \sqrt{[\widehat{\Omega}(\mathbf{x})]_{jj}/n_{\mathrm{eff}}(\mathbf{x})}\,. \tag{3.48}$$

By Condition 2.5 (v), $\mathbf{\Gamma}_{\mathbf{x}}$ is invertible; thus $\Omega(\mathbf{x}) > 0$ and the inverse and square root in Equation (3.49) are well defined. For numerical stability, it can be advisable to report the symmetrized version $\widehat{\Omega} \leftarrow (\widehat{\Omega} + \widehat{\Omega}^{\top})/2$ in practice as this is asymptotically immaterial but can improve finite-sample behavior.

Theorem 3.14 (Ratio-Consistency for the Sandwich)**.** (→ Proof, p. 51)

If $\widehat{\Sigma}_{\mathbf{x}}$ *is ratio-consistent for* $\Sigma(\mathbf{x})$ (Theorem 3.10) *and* $\widehat{\mathbf{\Gamma}}_{\mathbf{x}} \xrightarrow{\mathbb{P}} \mathbf{\Gamma}_{\mathbf{x}}$ (Theorem 3.12)*, then*

$$\left\| \ \Omega(\mathbf{x})^{-1/2}\ \widehat{\Omega}(\mathbf{x})\ \Omega(\mathbf{x})^{-1/2} - I_{p_\alpha}\ \right\|_{\mathrm{op}} \xrightarrow{\mathbb{P}}\ 0, \tag{3.49}$$

where $\widehat{\Omega}(\mathbf{x}) \coloneqq \widehat{\mathbf{\Gamma}}_{\mathbf{x}}^{-1}\, \widehat{\Sigma}_{\mathbf{x}}\, \widehat{\mathbf{\Gamma}}_{\mathbf{x}}^{-T}$ *and* $\Omega(\mathbf{x}) \coloneqq \mathbf{\Gamma}_{\mathbf{x}}^{-1}\, \Sigma(\mathbf{x})\ \mathbf{\Gamma}_{\mathbf{x}}^{-T}$*. For any fixed* $J \subset [p_\alpha]$*, the same ratio-consistency holds for the principal submatrix* $\widehat{\Omega}(\mathbf{x})_{JJ}$*.*

3.4. Main Theorems - Inference

We derive Wald-type inference from the local Bahadur expansion (Theorem 3.6), the Gaussian limit (Theorem 3.8), and the ratio-consistent sandwich estimator $\widehat{\Omega}(\mathbf{x})$ defined in Equation (3.48). Here $\widehat{\Omega}(\mathbf{x})$ estimates the covariance of the normalized statistic $\sqrt{n_{\text{eff}}(\mathbf{x})}\big(\widehat{\boldsymbol{\alpha}}(\mathbf{x}) - \boldsymbol{\alpha}_0(\mathbf{x})\big)$, so estimator-scale variances are obtained by dividing by $n_{\text{eff}}(\mathbf{x})$. For $j \in [p_\alpha]$, write

$$\operatorname{se}_j(\mathbf{x}) \coloneqq \sqrt{[\widehat{\Omega}(\mathbf{x})]_{jj}/n_{\text{eff}}(\mathbf{x})}, \tag{3.50}$$

and let $z_{1-\delta/2}$ denote the standard-normal quantile.

Theorem 3.15 (Wald inference for pointwise and linear hypotheses)**.** (→ Proof, p. 52)
Assume Conditions 2.2, 2.3, 2.4, 2.5, 2.6 and 2.7 and the ratio-consistency conclusions of Theorems 3.10, 3.12 and 3.14. Then:

(i) ***Pointwise CIs:*** *For any fixed $j \in [p_\alpha]$,*

$$\mathbb{P}\big(\alpha_{0,j}(\mathbf{x}) \in \big[\widehat{\alpha}_j(\mathbf{x}) \pm z_{1-\delta/2}\operatorname{se}_j(\mathbf{x})\big]\big) \longrightarrow 1-\delta. \tag{3.51}$$

(ii) ***Linear Contrasts:*** *For any fixed $\gamma \in \mathbb{R}^{p_\alpha}$, set*

$$\widehat{\Omega}_\gamma(\mathbf{x}) \coloneqq \gamma^\top \widehat{\Omega}(\mathbf{x})\,\gamma, \qquad \widehat{\operatorname{Var}}_\gamma(\mathbf{x}) \coloneqq \widehat{\Omega}_\gamma(\mathbf{x})/n_{\text{eff}}(\mathbf{x}).$$

Then

$$\frac{\sqrt{n_{\text{eff}}(\mathbf{x})}\,\gamma^\top\big(\widehat{\boldsymbol{\alpha}}(\mathbf{x}) - \boldsymbol{\alpha}_0(\mathbf{x})\big)}{\sqrt{\widehat{\Omega}_\gamma(\mathbf{x})}} \xrightarrow{d} \mathcal{N}(0,1), \tag{3.52}$$

and the Wald interval $\gamma^\top\widehat{\boldsymbol{\alpha}}(\mathbf{x}) \pm z_{1-\delta/2}\sqrt{\widehat{\operatorname{Var}}_\gamma(\mathbf{x})}$ is asymptotically exact.

(iii) ***Linear Restrictions:*** *For fixed full-row-rank $A \in \mathbb{R}^{q\times p_\alpha}$ and $b \in \mathbb{R}^q$,*

$$W_n \coloneqq n_{\text{eff}}(\mathbf{x})\ \big(A\widehat{\boldsymbol{\alpha}}(\mathbf{x}) - b\big)^\top\Big(A\widehat{\Omega}(\mathbf{x})A^\top\Big)^{-1}\big(A\widehat{\boldsymbol{\alpha}}(\mathbf{x}) - b\big) \xrightarrow{d} \chi^2_q. \tag{3.53}$$

Corollary 3.16 (Elliptical $(1-\delta)$ Joint CI for a Fixed Index Set)**.**
Under the assumptions of Theorem 3.15, for any fixed finite $J \subset [p_\alpha]$ let A be the selector matrix that maps $\boldsymbol{\alpha} \mapsto (\alpha_j)_{j\in J}$. Then the region

$$\underset{J,1-\delta}{\operatorname{CI}}(\mathbf{x}) \coloneqq \left\{ u \in \mathbb{R}^{|J|} \,:\, n_{\text{eff}}(\mathbf{x})\ \big(A\widehat{\boldsymbol{\alpha}}(\mathbf{x}) - u\big)^\top\Big(A\widehat{\Omega}(\mathbf{x})A^\top\Big)^{-1}\big(A\widehat{\boldsymbol{\alpha}}(\mathbf{x}) - u\big) \leqslant \chi^2_{|J|,1-\delta} \right\} \tag{3.54}$$

is an asymptotically exact $(1-\delta)$ confidence region for $(\alpha_{0,j}(\mathbf{x}))_{j\in J}$.

As mentioned before, using a symmetrized $\widehat{\Omega}(\mathbf{x}) \coloneqq \{\widehat{\Omega}(\mathbf{x}) + \widehat{\Omega}(\mathbf{x})^\top\}/2$ is asymptotically immaterial and can improve numerical stability in finite samples. One-sided intervals and tests follow by replacing $z_{1-\delta/2}$ with the appropriate one-sided quantile. All results are for fixed, finite dimension; high-dimensional simultaneous inference is outside our scope here.

4. Tuning

4.1. Choosing the Kernel-Order

This section discusses the choice of kernel order s under Conditions 2.1 and 2.7, which are the standing conditions throughout the main text. The bias–variance tradeoff motivates the undersmoothing requirement in Condition 2.7; a detailed derivation of the localization bias and the pseudo-target analysis are given in Section Appendix B.1.
The kernel-order s is the single tuning parameter that governs localization under DNN weighting. Intuitively, s acts like an inverse bandwidth: larger s concentrates weight more tightly around $\mathbf{x}$, while smaller s averages over a wider neighborhood. This can be seen from the L-statistic form of the weights in Equation (2.17) and the rank-ratio identity in Equation (2.18), which shows that mass decays more steeply across nearest-neighbor ranks as s grows, concentrating on the very closest points. In short, s plays the role of an inverse bandwidth; larger s yields tighter localization.

Geometry delivered by the DNN localizer.. Two simple geometric facts drive the choice of s.
(i) *Effective sample size.* Let $n_{\text{eff}}(\mathbf{x}) := \|w\|_2^{-2}$ as in Equation (2.9). Proposition 3.1 implies that, with probability $\to 1$ under $s \to \infty$ and $s/n \to 0$,

$$c\ \frac{s}{n} \leqslant \sum_{i=1}^{n} w_{s,i}^2 \leqslant C\ \frac{s}{n} \qquad \Longrightarrow \qquad c_1\ \frac{n}{s} \leqslant n_{\text{eff}}(\mathbf{x}) \leqslant C_1\ \frac{n}{s}.$$

Hence the first–order variance of a localized (centered) average is of order $1/n_{\text{eff}}(\mathbf{x}) \asymp s/n$.
(ii) *Neighborhood radius.* For the $(1-\delta)$ mass radius

$$R_{s,\delta}(\mathbf{x}) := \inf\Big\{ r > 0 : \sum_{i=1}^{n} w_{s,i}\ \mathbb{1}\{\,\|X_i - \mathbf{x}\|_2 \leqslant r\,\} \geqslant 1-\delta \Big\},$$

Proposition 3.2 (in particular Equation (3.3)) yields, for any fixed $\delta \in (0,1)$,

$$R_{s,\delta}(\mathbf{x}) = \Theta_{\mathbb{P}}\big((\log(1/\delta)/s)^{1/k_x}\big) = \Theta_{\mathbb{P}}\big(s^{-1/k_x}\big).$$

Thus the local neighborhood contracts at the rate s^{-1/k_x} as s grows.

Bias–variance tradeoff.. Under the local Lipschitz condition in u from Condition 2.5 (iii), the localization bias at the point target $\alpha_{0,j}(\mathbf{x})$ scales as

$$\text{Bias}(s) \asymp s^{-1/k_x}, \qquad \text{Variance}(s) \asymp \frac{1}{n_{\text{eff}}(\mathbf{x})} \asymp \frac{s}{n}.$$

A back–of–the–envelope balance gives

$$\text{MSE}(s) \approx C_1\ s^{-2/k_x} + C_2\ \frac{s}{n}, \qquad s^\star \asymp n^{\,k_x/(2+k_x)}.$$

Hence $\text{MSE}(s^\star) \asymp n^{-2/(2+k_x)}$.

Undersmoothing and CLT centering.. At the MSE-optimal order $s^\star \asymp n^{k_x/(k_x+2)}$, the scaled localization bias $\sqrt{n_{\text{eff}}(\mathbf{x})}\,|\alpha_{s,j}(\mathbf{x}) - \alpha_{0,j}(\mathbf{x})|$ is $O(1)$, so MSE-optimal tuning is typically insufficient for a CLT centered at the point target $\alpha_{0,j}(\mathbf{x})$. Valid point-target inference therefore requires *undersmoothing*: choosing s strictly larger than the MSE-optimal order. Condition 2.7 makes this precise:

$$s_n \gg n^{k_x/(k_x+2)}$$

ensures $\sqrt{n_{\text{eff}}(\mathbf{x})}\,|\alpha_{s,j}(\mathbf{x}) - \alpha_{0,j}(\mathbf{x})| \to 0$ so that the localization bias is negligible at the CLT scale. In power-law notation $s = n^\kappa$, this amounts to $\kappa > k_x/(k_x+2)$, strictly above the MSE-optimal exponent. The full derivation of the bias order and the analysis under base conditions alone are given in Section Appendix B.1.

Higher–order (multi–scale) variants.. In settings with k_x moderate or when additional smoothness is available, multi–scale (moment–cancelling) radial combinations of DNN kernels further reduce bias—from $O(R_{s,\delta}^2)$ for single–scale in higher dimensions to $O(R_{s,\delta}^K)$ with $K > d/2$—while preserving the same first–order variance normalization and efficiency statements. We focus on single–scale kernels for exposition here; the efficiency theory extends verbatim under the multi–scale localizer.

Practical takeaway.. For pointwise inference at $\mathbf{x}$ centered at $\boldsymbol{\alpha}_0(\mathbf{x})$ (as assumed throughout), choose s in the undersmoothing regime $s \gg n^{k_x/(k_x+2)}$ from Condition 2.7, or use a multiscale debiasing construction. The MSE-optimal order $s^\star \asymp n^{k_x/(2+k_x)}$ is on the boundary of the undersmoothing regime for single-scale DNN and should be regarded as a lower bound on s rather than a target.

5. Application: Conditional Average Treatment Effects

We illustrate the general framework in the canonical setting of heterogeneous treatment effects, which has been central to applied econometrics for several decades. Conditional average treatment effects (CATE) combine the potential outcomes framework with modern orthogonalization ideas, and serve as a natural test case for our theory.

Condition 5.1 (Heterogeneous Treatment Effect DGP)**.**
We observe i.i.d. data

$$D_{[n]} = \{ W_i = (X_i, Z_i, T_i, Y_i) \}_{i=1}^{n}, \tag{5.1}$$

where $X_i \in \mathcal{X} \subset \mathbb{R}^{k_x}$ are the covariates of primary interest, $Z_i \in \mathcal{Z} \subset \mathbb{R}^{d_z}$ are additional controls that may affect treatment assignment and outcomes, $T_i \in \{ 0, 1 \}$ is the treatment indicator, and $Y_i \in \mathcal{Y} \subset \mathbb{R}$ is the observed outcome. The outcome model takes the form

$$Y = \mu_0 (X, Z, T) + \varepsilon, \qquad T \sim \text{Bern} (\pi_0 (X, Z)), \tag{5.2}$$

with $\mathbb{E}[\varepsilon \mid X, Z, T] = 0$ and $\text{Var}(\varepsilon \mid X, Z) = \sigma_\varepsilon^2 (X, Z)$. The nuisance functions are $\boldsymbol{\eta}_0 = (\mu_0, \pi_0)^\top$, where $\mu_0 \colon \mathcal{X} \times \mathcal{Z} \times \{ 0, 1 \} \to \mathbb{R}$ denotes the potential outcome regressions and $\pi_0 \colon \mathcal{X} \times \mathcal{Z} \to (0, 1)$ the propensity score.

As is common in the literature on heterogeneous treatment effects, the data generating process in Condition 5.1 can be interpreted in the potential outcomes framework. The specification emphasizes that we focus on a low-dimensional set of covariates X at which treatment effect heterogeneity is of interest, while allowing for potentially high-dimensional additional controls Z that may confound treatment assignment.

Condition 5.2 (Technical Conditions)**.**
We impose standard regularity:

(i) The space $\mathcal{X}$ is compact with density f_X bounded away from zero and infinity.

(ii) For each $t \in \{ 0, 1 \}$, the maps $x \mapsto \mu_0 (x, z, t)$ and $x \mapsto \pi_0 (x, z)$ are locally continuous at all interior $x \in \mathcal{X}$, uniformly over z in the support of $Z \mid X = x$.

(iii) The conditional variance $\sigma_\varepsilon^2 (x, z)$ is continuous and uniformly bounded on $\mathcal{X} \times \mathcal{Z}$.

Condition 5.2 imposes only mild smoothness and regularity on the regression and propensity functions. In applied econometrics, these assumptions are the local analogue of the standard conditions used in semiparametric estimation of average treatment effects. Compactness of the support of X ensures that neighborhoods around any interior point contain sufficient mass to allow localization. Continuity of μ_0 and π_0 with respect to X is the minimal structure required to control the bias that arises when approximating conditional expectations locally. By not imposing compactness or smoothness on Z, we highlight that our framework can accommodate modern high-dimensional or flexible first-stage methods for nuisance estimation, in line with the recent debiased machine learning literature.

Condition 5.3 (Overlap)**.**
There exists $\mathfrak{p} \in (0, 1/2)$ such that

$$\forall (x, z) \in \mathcal{X} \times \mathcal{Z} : \qquad \mathfrak{p} \leqslant \pi_0 (x, z) \leqslant 1 - \mathfrak{p}. \tag{5.3}$$

This ensures non-trivial treatment assignment locally around every x of interest.

Condition 5.3 is the familiar overlap or common support assumption that underpins identification of treatment effects under unconfoundedness. Applied work often stresses the importance of checking overlap empirically, since in finite samples lack of support leads to instability of weighting estimators. In our localized setting, this assumption ensures that within every neighborhood of an x of interest, there are both treated and untreated units with non-vanishing probability.

Remark 5.1 (Orthogonal Moment for CATE)**.**
For any interior point $x \in \mathcal{X}$, define the pointwise target

$$\tau_0(x) \coloneqq \mathbb{E}[\mu_0(x, Z, 1) - \mu_0(x, Z, 0) \mid X = x]. \tag{5.4}$$

Let the augmented inverse probability weighting (AIPW) pseudo-outcome be

$$\phi(W; \eta) \coloneqq \frac{T}{\pi(X, Z)}(Y - \mu(X, Z, 1)) - \frac{1 - T}{1 - \pi(X, Z)}(Y - \mu(X, Z, 0)) + \mu(X, Z, 1) - \mu(X, Z, 0). \tag{5.5}$$

A Neyman-orthogonal score for the scalar point target at x is

$$\psi_{\mathrm{CATE}}(W; a, \eta) \coloneqq \phi(W; \eta) - a, \qquad a \in \mathbb{R}. \tag{5.6}$$

This moment condition satisfies

$$\mathbb{E}[\psi_{\mathrm{CATE}}(W; \tau_0(x), \boldsymbol{\eta}_0) \mid X = x] = 0, \tag{5.7}$$

so that $\tau_0(x)$ is the unique solution. The score is orthogonal in the sense that small perturbations of the nuisance functions (μ, π) have no first-order effect on the conditional expectation of ψ_{CATE}, making it robust to imperfect first-stage estimation. In our framework, the global conditional expectation is replaced by a DNN-weighted average local to x, which delivers pointwise inference for $\tau_0(x)$ without requiring sample splitting or strong assumptions on the dimension of Z.

Remark 5.1 introduces the orthogonal score that identifies the pointwise conditional treatment effect $\tau_0(x)$. This moment condition is the localized analogue of the augmented inverse probability weighting (AIPW) score that has become the workhorse for treatment effect estimation in applied econometrics. Its key advantage is Neyman orthogonality: small errors in estimating the nuisance functions μ_0 or π_0 do not translate into first-order bias in the estimation of $\tau_0(x)$. This robustness property underlies the use of flexible machine learning methods for first-stage estimation, as emphasized in Chernozhukov et al. (2018), and allows us to target treatment effect heterogeneity in X without restrictive modeling of high-dimensional confounders Z.

5.1. CATE: Estimator and Inference at a Point

As in the theory sections, we use non-bold x for generic pointwise identities and reserve bold $\mathbf{x}$ for the fixed evaluation point used in estimation and inference. Fix an interior point $\mathbf{x} \in \mathcal{X}$ and define the local parameter of interest $\tau_0(\mathbf{x})$. Let $\{ w_{s,i}(\mathbf{x}) \}_{i=1}^n$ be the DNN weights at $\mathbf{x}$. Let $\widehat{\eta} = (\widehat{\mu}, \widehat{\pi})$ denote same-sample first stages. The localized Z-estimator $\widehat{\tau}(\mathbf{x})$ is any solution to

$$\sum_{i=1}^{n} w_{s,i}(\mathbf{x})\ \psi_{\mathrm{CATE}}(W_i;\, \tau,\, \widehat{\eta}) = 0. \tag{5.8}$$

Lemma 5.1 (Closed-form localized CATE Z-solution)**.** (→ Proof, p. **??**)
Under Conditions 5.1, 5.2 and 5.3, the map in Equation (5.8) is strictly decreasing in τ with slope $-\sum_{i=1}^n w_{s,i}(\mathbf{x}) = -1$, hence admits a unique root given by

$$\begin{aligned} \widehat{\tau}(\mathbf{x}) &= \sum_{i=1}^{n} w_{s,i}(\mathbf{x})\ \widehat{\phi}_i, \\ \widehat{\phi}_i &\coloneqq \phi(W_i; \widehat{\eta}). \end{aligned} \tag{5.9}$$

By the local Bahadur expansion, we have

$$\sqrt{n_{\mathrm{eff}}(\mathbf{x})}\left\{\widehat{\tau}(\mathbf{x}) - \tau_0(\mathbf{x})\right\} = \Gamma(\mathbf{x})^{-1}\, \mathbb{G}_{n,w}\left[\psi_{\mathrm{CATE}}(\,\cdot\,; \tau_0(\mathbf{x}), \eta_0)\right] + o_{\mathbb{P}}(1), \tag{5.10}$$

where $\Gamma(\mathbf{x}) = -1$ for the score in Equation (5.6), and $n_{\mathrm{eff}}(\mathbf{x})$ is the effective local sample size. Let $\widehat{\Omega}(\mathbf{x})$ be the sandwich estimator from Equation (3.48), which estimates the covariance of $\sqrt{n_{\mathrm{eff}}(\mathbf{x})}\,\{\widehat{\tau}(\mathbf{x}) - \tau_0(\mathbf{x})\}$, and write

$$\widehat{\mathrm{se}}(\mathbf{x}) \coloneqq \left\{\widehat{\Omega}(\mathbf{x})\right\}_{11}^{1/2} / \sqrt{n_{\mathrm{eff}}(\mathbf{x})}. \tag{5.11}$$

Then Theorem 3.15 implies the standard Wald inference:

$$\mathbb{P}\big(\tau_0(\mathbf{x}) \in \big[\widehat{\tau}(\mathbf{x}) \pm z_{1-\delta/2}\, \widehat{\mathrm{se}}(\mathbf{x})\big]\big) \;\to\; 1-\delta, \tag{5.12}$$

and, more generally, linear contrasts admit the usual normalization with $\gamma^\top \widehat{\Omega}(\mathbf{x})\gamma$. The corresponding estimator-scale variance proxy is $\gamma^\top \widehat{\Omega}(\mathbf{x})\gamma / n_{\mathrm{eff}}(\mathbf{x})$. Entrywise ratio-consistency yields $\widehat{\mathrm{se}}(\mathbf{x})^2 / \big([\Omega(\mathbf{x})]_{11}/n_{\mathrm{eff}}(\mathbf{x})\big) \xrightarrow{\mathbb{P}} 1$, making Equation (5.12) directly feasible. In sum, the CATE application inherits the generic inference guarantees verbatim: the DNN-localized, orthogonal score Equation (5.6) feeds the Bahadur/CLT machinery and the sandwich $\widehat{\Omega}(\mathbf{x})$ to deliver asymptotically exact pointwise Wald CIs at rate $n_{\mathrm{eff}}(\mathbf{x})^{-1/2}$. For practitioners, implementation at a chosen point $\mathbf{x}$ is simple: fit flexible first stages $\widehat{\mu}, \widehat{\pi}$; compute the AIPW pseudo-outcomes $\widehat{\phi}_i$; and take the DNN-weighted average $\widehat{\tau}(\mathbf{x}) = \sum_i w_{s,i}(\mathbf{x})\, \widehat{\phi}_i$ (with weights normalized at $\mathbf{x}$), then form a Wald interval using the sandwich standard error. We recommend reporting the local effective sample size $n_{\mathrm{eff}}(\mathbf{x})$ together with overlap diagnostics $\sum_i w_{s,i}(\mathbf{x})\, T_i$ and $\sum_i w_{s,i}(\mathbf{x})\,(1-T_i)$, and enforcing a minimal $1/\sum_i w_{s,i}(\mathbf{x})^2$ to avoid extreme weight concentration. Orthogonality and our LOO-stability conditions justify same-sample first stages and can deliver tighter pointwise intervals than sample splitting in some scenarios.

5.2. Simulation Study

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6. Application: Partially Linear Quantile Regression

We next consider partially linear quantile regression (PLQR), another canonical setting in applied econometrics where orthogonalization plays a central role. The model combines a parametric component of interest with a nonparametric nuisance and has been extensively studied in distributional analyses. PLQR serves as a natural companion to the CATE example and illustrates how our conditional Z-estimation framework localizes in X while flexibly adjusting for additional covariates Z.

Condition 6.1 (PLQR DGP)**.**
We observe i.i.d. data

$$D_{[n]} = \{ W_i = (Y_i, D_i, X_i, Z_i) \}_{i=1}^{n}, \tag{6.1}$$

where $Y_i \in \mathbb{R}$ is the outcome, $D_i \in \mathbb{R}^p$ is a regressor of primary interest, $X_i \in \mathcal{X} \subset \mathbb{R}^{k_x}$ are the covariates at which we localize, and $Z_i \in \mathcal{Z} \subset \mathbb{R}^{d_z}$ are additional controls.
For a fixed quantile index $\tau \in (0, 1)$, the conditional quantile function satisfies

$$Q_\tau (Y \mid D, X, Z) = D'\theta_0 (X) \; + \; g_0 (X, Z) , \tag{6.2}$$

where $\theta_0 \colon \mathcal{X} \to \mathbb{R}^p$ is the target coefficient map and $g_0 \colon \mathcal{X} \times \mathcal{Z} \to \mathbb{R}$ is a nuisance function. For any evaluation point $x \in \mathcal{X}$, the pointwise parameter of interest is the coefficient vector $\theta_0 (x)$.

Condition 6.1 is the standard partially linear quantile specification augmented with auxiliary controls Z. Conceptually, X plays the same role as in the CATE example: it is the low-dimensional conditioning argument that drives localization and interpretation, while Z collects additional covariates used for flexible adjustment.

Condition 6.2 (Technical Conditions)**.**
We impose the following local regularity:

(i) The space $\mathcal{X}$ is compact with marginal density f_X bounded away from zero and infinity.

(ii) The function $g_0 (x, z)$ is continuous in x at interior points of $\mathcal{X}$, uniformly over z in the support of $Z \mid X = x$.

(iii) The conditional density $f_{Y \mid D,X,Z} (y \mid d, x, z)$ exists, is continuous in y, and is bounded away from zero in a neighborhood of $y = d'\theta_0 (x) + g_0 (x, z)$.

(iv) The regression of D on (X, Z), $m_0 (x, z) \coloneqq \mathbb{E}[D \mid X = x, Z = z]$, is continuous in x at interior points of $\mathcal{X}$, uniformly over z.

(v) **Relevance:** For each interior $x \in \mathcal{X}$ of interest,

$$\mathbb{E}\Big[(D - m_0 (X, Z)) (D - m_0 (X, Z))^\top \;\Big|\; X = x \Big] \tag{6.3}$$

is positive definite.

Condition 6.2 mirrors the usual PLQR assumptions but is stated locally in x to match our conditional inference target. Compactness and continuity ensure well-behaved localization; the density condition guarantees that the τ-quantile is well defined and estimable; the relevance condition rules out degeneracy after partialling out (X, Z). Notably, we do not require compactness or smoothness in Z beyond integrability, which underscores that adjustment in Z can be high-dimensional and learned flexibly.

Remark 6.1 (Orthogonal Moment for PLQR with Z)**.**
Define the score-stage nuisance

$$\eta_{\text{sc}} \coloneqq (g, m), \qquad \eta_{0,\text{sc}} \coloneqq (g_0, m_0),$$

where $g (x, z)$ approximates $g_0 (x, z)$ and $m (x, z) \coloneqq \mathbb{E}[D \mid X = x, Z = z]$. For any interior point $x \in \mathcal{X}$, an orthogonal score for the pointwise target $\theta_0(x)$ at quantile τ is

$$\psi_{\tau,\text{PLQR}} (W; \theta, \eta_{\text{sc}}) = (\tau - \mathbb{1}\{ Y \leqslant D'\theta + g (X, Z) \}) (D - m (X, Z)) . \tag{6.4}$$

This moment condition satisfies

$$\mathbb{E}[\psi_{\tau,\mathrm{PLQR}}\left(W;\theta_0\left(x\right),\eta_{0,\mathrm{sc}}\right)\mid X=x]=0, \tag{6.5}$$

so that $\theta_0\left(x\right)$ is identified as the unique solution. The score is Neyman-orthogonal: small perturbations in either g or m have no first-order effect on the conditional expectation of $\psi_{\tau,\mathrm{PLQR}}$, which makes plug-in estimation with flexible, possibly high-dimensional first stages in (X,Z) well behaved. The conditional density q_0 is a separate (derived) density nuisance that enters the Jacobian/bread and sandwich formulas below, but not the identifying score moment itself. In our framework, the conditional expectation is estimated via DNN localization in X only, with weights that do not depend on Z, thereby separating interpretability (local in X) from adjustment (flexible in Z).

Remark 6.1 is the quantile analogue of the residual-on-residual orthogonal moment in partially linear mean models and directly parallels the CATE construction. If desired, an efficiency-enhanced variant divides the score by the density nuisance $q_0\left(D,X,Z\right)\coloneqq f_{Y\mid D,X,Z}\left(D'\theta_0(X)+g_0\left(X,Z\right)\mid D,X,Z\right)$; we keep the basic orthogonal score for clarity and because it suffices for our conditional inference results.

6.1. PLQR: Estimator and Inference at a Point

As in the theory sections, we use non-bold x for generic pointwise identities and reserve bold $\mathbf{x}$ for the fixed evaluation point used in estimation and inference. Fix an interior point $\mathbf{x}\in\mathcal{X}$ and let $\{w_{s,i}\left(\mathbf{x}\right)\}_{i=1}^{n}$ denote the DNN weights at $\mathbf{x}$. The local parameter of interest is $\theta_0\left(\mathbf{x}\right)\in\mathbb{R}^p$. We distinguish the score-stage nuisance $\eta_{0,\mathrm{sc}}=(g_0,m_0)$ from the density nuisance q_0: the former enters the orthogonal score, while the latter enters the Jacobian/bread and sandwich formulas. Write $\eta_{\mathrm{sc}}=(g,m)$, with $g(x,z)\approx g_0(x,z)$ and $m(x,z)\coloneqq\mathbb{E}[D\mid X=x,Z=z]$. For a given quantile index $\tau\in(0,1)$, consider the orthogonal score from Remark 6.1,

$$\psi_{\tau,\mathrm{PLQR}}\left(W;\theta,\eta_{\mathrm{sc}}\right)\coloneqq\left(\tau-\mathbb{1}\{\ Y\leqslant D'\theta+g(X,Z)\ \}\right)\left(D-m(X,Z)\right), \tag{6.6}$$

which satisfies $\mathbb{E}[\psi_{\tau,\mathrm{PLQR}}\left(W;\theta_0(\mathbf{x}),\eta_{0,\mathrm{sc}}\right)\mid X=\mathbf{x}]=0$. Let $\widehat{\eta}_{\mathrm{sc}}=(\widehat{g},\widehat{m})$ be same-sample first stages for the score nuisance. The localized Z-estimator $\widehat{\theta}(\mathbf{x})$ solves

$$\sum_{i=1}^{n}w_{s,i}\left(\mathbf{x}\right)\ \psi_{\tau,\mathrm{PLQR}}\left(W_i;\ \theta,\ \widehat{\eta}_{\mathrm{sc}}\right)\ =0. \tag{6.7}$$

The density nuisance q_0 enters next through the local Jacobian (bread) and its plug-in estimator. Let

$$\Gamma(\mathbf{x})\coloneqq\mathbb{E}\Big[q_0(D,X,Z)\left(D-m_0(X,Z)\right)\left(D-m_0(X,Z)\right)^{\top}\ \Big|\ X=\mathbf{x}\Big], \tag{6.8}$$

where $q_0(d,x,z)\coloneqq f_{Y\mid D,X,Z}\left(d'\theta_0(x)+g_0(x,z)\mid d,x,z\right)$. Under Conditions 6.1 and 6.2, $\Gamma(\mathbf{x})$ is positive definite by relevance and the density condition. Then the generic local Bahadur expansion yields

$$\sqrt{n_{\mathrm{eff}}\left(\mathbf{x}\right)}\left\{\widehat{\theta}(\mathbf{x})-\theta_0(\mathbf{x})\right\}=\Gamma(\mathbf{x})^{-1}\,\mathbb{G}_{n,w}\left[\psi_{\tau,\mathrm{PLQR}}\left(\,\cdot\,;\theta_0(\mathbf{x}),\eta_{0,\mathrm{sc}}\right)\right]+o_{\mathbb{P}}(1). \tag{6.9}$$

Let $\tilde{D}_i\coloneqq D_i-\widehat{m}(X_i,Z_i)$, $\widehat{u}_i(\theta)\coloneqq Y_i-D_i'\theta-\widehat{g}(X_i,Z_i)$, and choose any consistent estimator $\widehat{q}_i$ of the density nuisance $q_0(D_i,X_i,Z_i)$ (e.g., kernel density of $\widehat{u}_i(\widehat{\theta}(\mathbf{x}))$ at zero, locally in (D,X,Z)). Define

$$\widehat{J}(\mathbf{x})\coloneqq\sum_{i=1}^{n}w_{s,i}\left(\mathbf{x}\right)\ \widehat{q}_i\,\tilde{D}_i\tilde{D}_i^{\top},\qquad\widehat{S}\left(\mathbf{x}\right)\coloneqq\sum_{i=1}^{n}w_{s,i}\left(\mathbf{x}\right)\ \tau(1-\tau)\,\tilde{D}_i\tilde{D}_i^{\top}. \tag{6.10}$$

Let $\widehat{\Omega}(\mathbf{x})$ denote the sandwich from Equation (3.48) specialized as $\widehat{\Omega}(\mathbf{x})=\widehat{J}(\mathbf{x})^{-1}\widehat{S}\left(\mathbf{x}\right)\widehat{J}(\mathbf{x})^{-1}$; this estimates the covariance of $\sqrt{n_{\mathrm{eff}}\left(\mathbf{x}\right)}\left\{\widehat{\theta}(\mathbf{x})-\theta_0(\mathbf{x})\right\}$. For any fixed contrast $\gamma\in\mathbb{R}^p$,

$$\mathbb{P}\left(\gamma^{\top}\theta_0(\mathbf{x})\in\left[\gamma^{\top}\widehat{\theta}(\mathbf{x})\ \pm\ z_{1-\delta/2}\ \sqrt{\gamma^{\top}\widehat{\Omega}(\mathbf{x})\gamma/n_{\mathrm{eff}}\left(\mathbf{x}\right)}\right]\right)\to1-\delta, \tag{6.11}$$

and, in particular, (j,j)-entries give feasible standard errors $\sqrt{[\widehat{\Omega}(\mathbf{x})]_{jj}/n_{\mathrm{eff}}\left(\mathbf{x}\right)}$ for components of $\theta_0(\mathbf{x})$.

Practically, inference at a chosen $\mathbf{x}$ is performed as follows: fit flexible first stages $\widehat{g}, \widehat{m}$; form residuals $\tilde{D}_i$ and the DNN weights $w_{s,i}(\mathbf{x})$; solve the weighted moment equation Equation (6.7) (e.g., via a smoothed score or a standard IVQR/GMM routine with instruments $\tilde{D}_i$); and report Wald intervals using the sandwich in Equations (6.10)(6.11). Report $n_{\text{eff}}(\mathbf{x})$ alongside basic diagnostics below.

- **Relevance diagnostics:** Check local invertibility by reporting the smallest eigenvalue of $\sum_i w_{s,i}(\mathbf{x})\ \tilde{D}_i\tilde{D}_i^\top$ at $\mathbf{x}$ (and its condition number).
- **Solver guidance:** A smoothed indicator (e.g., logistic or Gaussian cdf with small bandwidth) yields stable Newton steps for Equation (6.7) and leaves the limit unchanged.

7. Conclusion

We develop a localized, orthogonal Z-estimation framework for componentwise targets $\{\alpha_{0,j}(\mathbf{x})\}_{j\in[p_\alpha]}$ based on distributional nearest-neighbor (DNN) weights. Under the *Target Parameter*, *Distribution*, *Score Function*, and *Conditional Moment* conditions, the DNN weights act as an approximate identity around $\mathbf{x}$ and deliver Hájek dominance, which yields an asymptotically linear representation of the estimators. Neyman-orthogonality isolates the effect of nuisance estimation, so that regularized or machine-learned nuisances contribute only higher-order terms. For each j, we establish pointwise consistency and a central limit theorem at $\mathbf{x}$. The corresponding variance admits a simple, fully automatic estimator based on the jackknife, which adds negligible computational cost. Taken together, these results provide a practical recipe for local inference with minimal tuning and clear robustness to high-dimensional nuisance learning. The next section outlines several natural extensions, uniform statements and diagnostics, a localized GMM formulation, and tree-based proximity weights, that we leave for future work.

7.1. Outlook

The results developed here suggest several natural extensions that retain the same localization mechanism and orthogonal score structure. Our working hypothesis is that the core ingredients (approximate-identity behavior of the weights, local invertibility, and the envelope/entropy controls on the score class) continue to govern bias, variance, and asymptotic linearity beyond the baseline setup. We highlight three directions at a conceptual level: uniform-in-x statements and simple diagnostics of localization quality; a localized GMM formulation that stacks score components and accommodates cross-component correlation; and tree-based proximity weights that induce data-adaptive neighborhoods around $\mathbf{x}$. Detailed algorithms, tuning choices, and proofs are left to future work.

Uniform Statements and Diagnostics: Beyond pointwise inference at a fixed $\mathbf{x}$, several applications call for statements that are uniform over a compact region in x and for simple diagnostics of localization quality. We anticipate extending our arguments to deliver simultaneous coverage over x together with basic stability checks (e.g., reporting how estimates and standard errors respond to small changes in localization strength). Detailed procedures are outside the scope of this paper.

A localized GMM Variant: A natural next step is to replace the p_α separate, componentwise problems with a single localized GMM estimator for the full vector $\boldsymbol{\alpha}_0(\mathbf{x})$. Conceptually, nothing about the DNN localization changes: we still weight observations by proximity to $\mathbf{x}$. What changes is the objective, we stack the score components and choose one $\mathbf{a}$ that best balances them jointly, allowing the weighting matrix to learn the cross-component correlation structure. This opens the door to efficiency via optimal local weighting (a two-step update using the weighted outer product of moments), and overidentification in a clean way (adding transformed or auxiliary moments and running a localized Sargan-Hansen check). On the theory side, the upgrade is mostly linear-algebraic: we swap division by scalar sensitivities for a joint sensitivity matrix and carry the same bias/variance and CLT arguments. The rates and localization story remain the same; the gain is a unified estimator, a clearer route to efficiency when scores are correlated, and a platform for richer moment sets. We leave a full development of this GMM variant, and its empirical illustrations, for future work.

Random Forest Weighted Variant: Another natural direction is to replace the DNN weights with random-forest-induced weights. Concretely, one could fit an honest forest and define $w_i(\mathbf{x})$ as the fraction of trees in which X_i

and the query point $\mathbf{x}$ fall in the same terminal node, yielding a data-adaptive, anisotropic neighborhood around $\mathbf{x}$ without hand-tuning a metric or radius. A key attraction is local feature selection in the x-coordinates: because splits concentrate on predictive coordinates near $\mathbf{x}$, the forest implicitly shrinks the neighborhood along relevant directions and spreads it along irrelevant ones. We hypothesize that the honesty and regularity of the forest could deliver the same U-statistic structure and approximate-identity behavior needed for our linearization and CLT, while the adaptive geometry could improve stability when k_x is large and many coordinates are irrelevant. This line also opens the door to localized variable-importance maps (or split-frequency summaries) around $\mathbf{x}$, useful as a diagnostic or to report an "active set" of covariates driving the estimate locally. A full development, including rates, efficiency, and empirical illustrations of local feature selection, remains for future work.

Appendix A. Additional Notation for Proofs

Throughout the proofs we present, we will make use of additional notation compared to the main text of this paper to improve the readability of the arguments. First, taking inspiration from classical empirical process theory, we introduce notions of point-of-interest specific weighted empirical processes with respect to the DNN weights $w_{s,i}(\mathbf{x})$. Let $f : \mathcal{W} \to \mathbb{R}$ be a measurable function, then we define the following processes.

$$\mathbb{T}_{n,w}[f] \coloneqq \sqrt{n_{\text{eff}}(\mathbf{x})} \sum_{i=1}^{n} w_{s,i}(\mathbf{x}) \; f(W_i) \tag{A.1}$$

$$\mathbb{G}_{n,w}[f] \coloneqq \sqrt{n_{\text{eff}}(\mathbf{x})} \sum_{i=1}^{n} w_{s,i}(\mathbf{x}) \; (f(W_i) \; - \; \mathbb{E}[f(W) \mid X = \mathbf{x}]) \tag{A.2}$$

$$\mathbb{H}_{n,w}[f] \coloneqq \sqrt{n_{\text{eff}}(\mathbf{x})} \sum_{i=1}^{n} w_{s,i}(\mathbf{x}) \; (f(W_i) - \mathbb{E}[f(W) \mid X = X_i]) \tag{A.3}$$

$$\mathbb{D}_{n,w}[f] \coloneqq \sqrt{n_{\text{eff}}(\mathbf{x})} \sum_{i=1}^{n} w_{s,i}(\mathbf{x}) \; (\mathbb{E}[f(W) \mid X = X_i] - \mathbb{E}[f(W) \mid X = \mathbf{x}]) \tag{A.4}$$

We assume (X, W) admit a regular conditional distribution of W given X and for each integrable f we fix a Borel version of the conditional expectation. The objects defined above are then well-defined for any $f \in L^1(P)$, which allows us to make use of the following useful identities.

$$\begin{aligned} \mathbb{T}_{n,w}[f] &= \mathbb{G}_{n,w}[f] + \sqrt{n_{\text{eff}}(\mathbf{x})}\, \mathbb{E}[f(W) \mid X = \mathbf{x}] \\ &= \mathbb{H}_{n,w}[f] + \mathbb{D}_{n,w}[f] + \sqrt{n_{\text{eff}}(\mathbf{x})}\, \mathbb{E}[f(W) \mid X = \mathbf{x}] \end{aligned} \tag{A.5}$$

For $a \in \mathcal{A}_j(\mathbf{x})$ and $\tilde{\eta} \in \mathcal{H}_j$, define the score difference relative to the oracle score by

$$\Delta\psi_j(W;\; a, \tilde{\eta}) \coloneqq \psi_j(W; a, \tilde{\eta}) - \psi_j(W; \alpha_{0,j}(\mathbf{x}), \eta_{0,j}). \tag{A.6}$$

We will continue to use m_j for the conditional moment equation as defined in Equation (2.25), i.e.

$$m_j(u; a, \tilde{\eta}) = \mathbb{E}\big[\psi_j\left(W, a, \tilde{\eta}\left(Z^j\right)\right) \;\big|\; X = u\big]. \tag{A.7}$$

Furthermore, we define the conditional moment difference relative to its oracle version by

$$\Delta m_j(u;\; a, \tilde{\eta}) \coloneqq m_j(u; a, \tilde{\eta}) - m_j(u; \alpha_{0,j}(\mathbf{x}), \eta_{0,j}). \tag{A.8}$$

Another important tool that we will use throughout the Appendix is to split sums into "good" and "bad" events according to whether the subsample under consideration contains an observation that falls into the neighborhood $\mathcal{B}_{\mathbf{x}}$. To this end, we define $A(D_\ell) \coloneqq \{\exists i \in \ell :\; X_i \in \mathcal{B}_{\mathbf{x}}\}$ for any subset ℓ of $[n]$. Similarly, for any individual observation W_i we define the event $G_i \coloneqq \{X_i \in \mathcal{B}_{\mathbf{x}}\}$ or $G \coloneqq \{X \in \mathcal{B}_{\mathbf{x}}\}$. As we will often condition on these events, we will furthermore introduce the following variations of our localized empirical process notation. For any measurable $f : \mathcal{W} \to \mathbb{R}$, write $f^{(g)}(w) \coloneqq f(w)\; \mathbb{1}\{G\}$ and, by convention,

$$\mathbb{T}^G_{n,w}[f] \equiv \mathbb{T}_{n,w}\big[f^{(g)}\big], \quad \mathbb{G}^G_{n,w}[f] \equiv \mathbb{G}_{n,w}\big[f^{(g)}\big], \quad \mathbb{H}^G_{n,w}[f] \equiv \mathbb{H}_{n,w}\big[f^{(g)}\big], \quad \mathbb{D}^G_{n,w}[f] \equiv \mathbb{D}_{n,w}\big[f^{(g)}\big]. \tag{A.9}$$

By Lemma Appendix E.5 (weight measurability and tower property), for any integrable h,

$$\mathbb{E}[w_{s,i}(\mathbf{x})\; h(W_i)\; \mathbb{1}\{G_i\}] = \mathbb{E}[w_{s,i}(\mathbf{x})\;\; \mathbb{1}\{G_i\}\, \mathbb{E}[h(W) \mid X = X_i]]. \tag{A.10}$$

Because $\mathbb{1}\{G\} = 1$ when conditioning on $X = \mathbf{x}$ by the definition of G,

$$\mathbb{E}[h(W)\; \mathbb{1}\{G\} \mid X = \mathbf{x}] = \mathbb{E}[h(W) \mid X = \mathbf{x}]. \tag{A.11}$$

Finally, the weighted-operator identity holds in its gated form:

$$\mathbb{G}^G_{n,w}[f] = \mathbb{H}^G_{n,w}[f] + \mathbb{D}^G_{n,w}[f]. \tag{A.12}$$

Appendix B. Pseudo-Target Analysis and Bias-Centering

Appendix B.1. Pseudo-Target Analysis and Bias-Centering

This section collects the analysis for the localized pseudo-target $\boldsymbol{\alpha}_s(\mathbf{x})$, which is the natural population target of the DNN Z-estimator under Condition 2.1 alone (without undersmoothing). We derive the localization bias, show how it enters the centering decomposition, and state pseudo-target-centered versions of the main results. Corollary Appendix B.3 at the end of this section confirms that Condition 2.7 is the minimal additional condition needed to center all main-text results at $\boldsymbol{\alpha}_0(\mathbf{x})$.

Appendix B.1.1. The Localized Pseudo-Target

Definition Appendix B.1 (Localized Pseudo-Target)**.**
For each $j \in [p_\alpha]$, define the *localized population moment*

$$m_{j,s}^{\mathrm{loc}}(a;\mathbf{x}) \;\coloneqq\; \mathbb{E}\big[\tilde{\tau}_{s,\mathbf{x}}(X)\,\psi_j\big(W;\,a,\,\eta_{0,j}\big)\big] \;=\; \mathbb{E}\big[\tilde{\tau}_{s,\mathbf{x}}(X)\,m_j\big(X;\,a,\,\eta_{0,j}\big)\big], \tag{B.1}$$

and let $\alpha_{s,j}(\mathbf{x})$ be the unique solution to $m_{j,s}^{\mathrm{loc}}(\alpha_{s,j}(\mathbf{x});\mathbf{x}) = 0$ in $\mathcal{A}_j(\mathbf{x})$. Write $\boldsymbol{\alpha}_s(\mathbf{x}) \coloneqq (\alpha_{s,1}(\mathbf{x}),\ldots,\alpha_{s,p_\alpha}(\mathbf{x}))^\top$. We call $\boldsymbol{\alpha}_s(\mathbf{x})$ the *localized pseudo-target* at $\mathbf{x}$. It is the natural population target of the DNN Z-estimator: the estimator $\widehat{\boldsymbol{\alpha}}(\mathbf{x})$ approximates $\boldsymbol{\alpha}_s(\mathbf{x})$ at the $n_{\mathrm{eff}}(\mathbf{x})^{-1/2}$ rate, while $\boldsymbol{\alpha}_s(\mathbf{x})$ approximates $\boldsymbol{\alpha}_0(\mathbf{x})$ at the slower localization-bias rate $O(s^{-\beta/k_x})$ (see below).

The estimator is built from weights that place most mass on a shrinking neighborhood of $\mathbf{x}$ (radius $\asymp s^{-1/k_x}$ by Proposition 3.2), not on the event $\{X = \mathbf{x}\}$ itself. So the first-order stochastic term is naturally centered around a localized population moment. As in kernel and k-NN smoothing, the difference between the localized target and the point target is a *bias term* that must be controlled relative to the CLT scale.

Appendix B.1.2. Localization Bias

Step 1: Localization bias in the moment.. At the true point target $a = \alpha_{0,j}(\mathbf{x})$, we have $m_j(\mathbf{x};\alpha_{0,j}(\mathbf{x}),\eta_{0,j}) = 0$ by Condition 2.5 (vi), hence

$$m_{j,s}^{\mathrm{loc}}\big(\alpha_{0,j}(\mathbf{x});\mathbf{x}\big) = \mathbb{E}\Big[\tilde{\tau}_{s,\mathbf{x}}(X)\Big(m_j\big(X;\alpha_{0,j}(\mathbf{x}),\eta_{0,j}\big) - m_j\big(\mathbf{x};\alpha_{0,j}(\mathbf{x}),\eta_{0,j}\big)\Big)\Big]. \tag{B.2}$$

Under Lipschitz continuity of m_j in u (Condition 2.5 (iii)),

$$\big|\,m_j\big(u;\alpha_{0,j}(\mathbf{x}),\eta_{0,j}\big) - m_j\big(\mathbf{x};\alpha_{0,j}(\mathbf{x}),\eta_{0,j}\big)\,\big| \leqslant L_{u,j}\|\,u-\mathbf{x}\,\|_2, \tag{B.3}$$

so the localized moment drift is of order the local radius:

$$\big|\,m_{j,s}^{\mathrm{loc}}\big(\alpha_{0,j}(\mathbf{x});\mathbf{x}\big)\,\big| = O\big(s^{-1/k_x}\big). \tag{B.4}$$

Step 2: Translate moment bias into parameter bias.. By local invertibility (Condition 2.5 (v)) and second-order smoothness in a (Condition 2.5 (iv)), a first-order expansion around $\alpha_{0,j}(\mathbf{x})$ gives

$$0 = m_{j,s}^{\mathrm{loc}}\big(\alpha_{s,j}(\mathbf{x});\mathbf{x}\big) = m_{j,s}^{\mathrm{loc}}\big(\alpha_{0,j}(\mathbf{x});\mathbf{x}\big) + \Gamma_j(\mathbf{x})\big(\alpha_{s,j}(\mathbf{x}) - \alpha_{0,j}(\mathbf{x})\big) + O\big(|\alpha_{s,j}(\mathbf{x}) - \alpha_{0,j}(\mathbf{x})|^2\big), \tag{B.5}$$

with $\Gamma_j(\mathbf{x})$ bounded away from zero. Therefore the *parameter bias* inherits the same order as the localized moment drift:

$$|\alpha_{s,j}(\mathbf{x}) - \alpha_{0,j}(\mathbf{x})| = O\big(s^{-1/k_x}\big). \tag{B.6}$$

Step 3: Centering decomposition.. The full decomposition at the CLT scale is

$$\sqrt{n_{\mathrm{eff}}(\mathbf{x})}\big(\widehat{\alpha}_j(\mathbf{x}) - \alpha_{0,j}(\mathbf{x})\big) = \underbrace{\sqrt{n_{\mathrm{eff}}(\mathbf{x})}\big(\widehat{\alpha}_j(\mathbf{x}) - \alpha_{s,j}(\mathbf{x})\big)}_{\xrightarrow{d}\,\mathcal{N}(0,\,\omega_j^2(\mathbf{x}))} + \underbrace{\sqrt{n_{\mathrm{eff}}(\mathbf{x})}\big(\alpha_{s,j}(\mathbf{x}) - \alpha_{0,j}(\mathbf{x})\big)}_{\text{localization bias}}. \tag{B.7}$$

Using $n_{\mathrm{eff}}(\mathbf{x}) \asymp n/s$ (Proposition 3.1) and Equation (B.6),

$$\sqrt{n_{\mathrm{eff}}(\mathbf{x})}\,\big|\,\alpha_{s,j}(\mathbf{x}) - \alpha_{0,j}(\mathbf{x})\,\big| = O\bigg(\sqrt{\frac{n}{s}}\,s^{-1/k_x}\bigg). \tag{B.8}$$

Thus a CLT centered at $\alpha_{0,j}(\mathbf{x})$ requires this to tend to zero, which is exactly Condition 2.7.

Boundary case: MSE-optimal tuning.. Balancing squared bias $O(s^{-2/k_x})$ against variance $O(s/n)$ gives $s^\star \asymp n^{k_x/(k_x+2)}$. At this order, Equation (B.8) is $O(1)$, so MSE-optimal tuning is typically insufficient for a CLT centered at $\alpha_{0,j}(\mathbf{x})$. Undersmoothing (choosing s strictly above $s^\star$) or multi-scale debiasing restores valid centering.

Interpretation in power-law form.. If $s = n^\kappa$ (up to slowly varying factors), Condition 2.1 requires $0 < \kappa < 1$, while the undersmoothing condition Condition 2.7 further requires $\kappa > k_x/(k_x+2)$.

Appendix B.1.3. Pseudo-Target Centered Results

The following results are the pseudo-target-centered counterparts of those in the main text. They hold under Condition 2.1 alone, without Condition 2.7. The existence of an approximate root near $\alpha_{s,j}(\mathbf{x})$ follows from the same sign-change bracketing argument as Proposition 3.5, applied to the localized moment map $m^{\mathrm{loc}}_{j,s}(\cdot;\mathbf{x})$ at the pseudo-target in place of the point-target moment; we take that as established.

Theorem Appendix B.1 (Local Bahadur Representation (Pseudo-Target Centering))**.**
Assume Conditions 2.4, 2.5 and 2.6 hold (without Condition 2.7). Fix $j \in [p_\alpha]$, write $G := \{ X \in \mathcal{B}_{\mathbf{x}} \}$, and let $\widehat{\alpha}_j(\mathbf{x})$ be any solution to the localized estimating equation with $|\widehat{\alpha}_j(\mathbf{x}) - \alpha_{s,j}(\mathbf{x})| \leqslant \delta_n \downarrow 0$ with probability tending to one. Then

$$\sqrt{n_{\mathrm{eff}}(\mathbf{x})}\,\big(\widehat{\alpha}_j(\mathbf{x}) - \alpha_{s,j}(\mathbf{x})\big) = \Gamma_j(\mathbf{x})^{-1}\,\mathbb{G}_{n,w}\Big[\psi_j\big(\cdot;\alpha_{s,j}(\mathbf{x}),\eta_{0,j}\big)\,\mathbb{1}\{X \in G\}\Big] + o_{\mathbb{P}}(1). \tag{B.9}$$

The score $\psi_j(\cdot;\alpha_{s,j}(\mathbf{x}),\eta_{0,j})$ can be replaced by $\psi_j(\cdot;\alpha_{0,j}(\mathbf{x}),\eta_{0,j})$ up to an additional $o_{\mathbb{P}}(1)$ term whenever $\sqrt{n_{\mathrm{eff}}(\mathbf{x})}\,|\alpha_{s,j}(\mathbf{x}) - \alpha_{0,j}(\mathbf{x})| \to 0$, which holds under Condition 2.7 (Corollary Appendix B.3).

Theorem Appendix B.2 (Asymptotic Normality (Pseudo-Target Centering))**.**
Assume Conditions 2.2, 2.3, 2.4, 2.5 and 2.6 hold (without Condition 2.7). Then

$$\sqrt{n_{\mathrm{eff}}(\mathbf{x})}\,\big(\widehat{\boldsymbol{\alpha}}(\mathbf{x}) - \boldsymbol{\alpha}_s(\mathbf{x})\big) = \Gamma(\mathbf{x})^{-1}\,\mathbb{G}_{n,w}\big[\boldsymbol{\psi}(\,\cdot\,;\boldsymbol{\alpha}_s(\mathbf{x}),\boldsymbol{\eta}_0)\,\mathbb{1}\{X \in G\}\big] + o_{\mathbb{P}}(1), \tag{B.10}$$

and, consequently,

$$\sqrt{n_{\mathrm{eff}}(\mathbf{x})}\,\big(\widehat{\boldsymbol{\alpha}}(\mathbf{x}) - \boldsymbol{\alpha}_s(\mathbf{x})\big) \xrightarrow{d} \mathcal{N}\big(0,\Omega(\mathbf{x})\big). \tag{B.11}$$

The full centering decomposition at the point-target scale is

$$\sqrt{n_{\mathrm{eff}}(\mathbf{x})}\,\big(\widehat{\boldsymbol{\alpha}}(\mathbf{x}) - \boldsymbol{\alpha}_0(\mathbf{x})\big) = \underbrace{\sqrt{n_{\mathrm{eff}}(\mathbf{x})}\,\big(\widehat{\boldsymbol{\alpha}}(\mathbf{x}) - \boldsymbol{\alpha}_s(\mathbf{x})\big)}_{\xrightarrow{d}\,\mathcal{N}(0,\Omega(\mathbf{x}))} + \underbrace{\sqrt{n_{\mathrm{eff}}(\mathbf{x})}\,\big(\boldsymbol{\alpha}_s(\mathbf{x}) - \boldsymbol{\alpha}_0(\mathbf{x})\big)}_{\textit{localization bias}}. \tag{B.12}$$

Under Condition 2.5 (iii), the bias term is $O(\sqrt{n/s}\cdot s^{-1/k_x})$ at the CLT scale.

Appendix B.1.4. Bridging Pseudo-Target and Point-Target

Corollary Appendix B.3 (Point-Target CLT under Undersmoothing)**.**
Adopt the setting of Theorem Appendix B.2 and suppose Condition 2.7 holds, so that $|\alpha_{s,j}(\mathbf{x}) - \alpha_{0,j}(\mathbf{x})| = O(s^{-1/k_x})$ by Equation (B.6) and

$$s \;\gg\; n^{k_x/(k_x+2)}. \tag{B.13}$$

Then the bias in Equation (B.12) satisfies $\sqrt{n_{\mathrm{eff}}(\mathbf{x})}\,(\boldsymbol{\alpha}_s(\mathbf{x}) - \boldsymbol{\alpha}_0(\mathbf{x})) \to 0$, which is exactly the content of Theorem 3.8 in the main text.

What can still be asserted without undersmoothing?. Even when Condition 2.7 fails, Theorem Appendix B.2 delivers a valid Gaussian limit for the estimator centered at the localized pseudo-target $\boldsymbol{\alpha}_s(\mathbf{x})$. What fails is the replacement of $\boldsymbol{\alpha}_s(\mathbf{x})$ by $\boldsymbol{\alpha}_0(\mathbf{x})$: in that case inference based on the point-target centering would be biased at the CLT scale, and the bias is of order $O(1)$ at the MSE-optimal tuning. The main text avoids this complication by maintaining Condition 2.7 as a standing condition.

Appendix C. Proofs of Main Theorems

Appendix C.1. Proof of Theorems related to Asymptotic Normality

Proof of Proposition 3.4

Proof. Let $s \to \infty$ with $s = o(n)$.

Step 1: First projection = Localizer × Score.
By Lemma 3.3,

$$h_{j,s}^{\langle 1\rangle}(W) = \tilde{\tau}_{s,x}^{\circ}(X)\, \psi_j\big(W; \alpha_{0,j}(x), \eta_{0,j}\big) \quad \text{for each } j, \tag{C.1}$$

hence $h_{\gamma,s}^{\langle 1\rangle}(W) = \tilde{\tau}_{s,x}^{\circ}(X)\, \psi_\gamma\big(W; \vec{\alpha}_0(x), \vec{\eta}_0\big)$ with $\psi_\gamma := \sum_j \gamma_j \psi_j$. Condition 2.4 (ii) supplies a measurable envelope $F \in L_2(P)$ that uniformly bounds the scores at the truth; by linearity, $\mathbb{E}[\psi_\gamma(W; \vec{\alpha}_0(x), \vec{\eta}_0)^2] \leqslant \mathbb{E}[F^2] < \infty$ uniformly over $\|\gamma\|_2 = 1$. This gives a γ-uniform L_2 bound for $h_{\gamma,s}^{\langle 1\rangle}$.

Step 2: DNN Geometry: Scale of the Localizer and $n_{\text{eff}}(\mathbf{x})$.
The Steele L-form implies the familiar effective-mass accounting:

$$\sum_{i=1}^{n} w_{s,i}(x)^2 \asymp \frac{s}{n} \qquad \Longleftrightarrow \qquad n_{\text{eff}}(\mathbf{x}) \asymp \frac{n}{s}, \tag{C.2}$$

and DNN weights are negligible and LOO-stable ($\max_i |w_{s,i}(x)| \leqslant s/n$ and $\sqrt{n_{\text{eff}}(\mathbf{x})}\, \beta_n^{\text{loc}}(x) \to 0$). These are standard and recorded in Lemma 2.3 and the surrounding discussion.

Step 3: Juergens (2025): Variance Bookkeeping → Dominance.
Write $V_r(\gamma) := \text{Var}\Big(h_{\gamma,s}^{(r)}(D_{[r]})\Big)$ for the Hoeffding-r kernels of $h_{\gamma,s}$. The generalized H-decomposition gives

$$\zeta_{ss}(\gamma) = \sum_{r=1}^{s} \binom{s}{r} V_r(\gamma) \quad \text{with} \quad V_1(\gamma) = \zeta_{1s}(\gamma). \tag{C.3}$$

Juergens (2025) tracks these components and shows that Hájek dominance is equivalent to the kernel-variance ratio condition

$$\frac{s}{n}\Big(\frac{\zeta_{ss}(\gamma)}{s\, \zeta_{1s}(\gamma)} - 1\Big) \to 0, \tag{C.4}$$

which then yields the variance transfer

$$\frac{n}{s^2}\, \frac{\text{Var}(\overline{m}_{\gamma,s}(x))}{\zeta_{1s}(\gamma)} \to 1 \quad \text{(Hájek variance dominance).} \tag{C.5}$$

We appeal to this equivalence and use it only as a black-box transfer once Equation (C.4) is verified for the DNN kernel.

Step 4: Verifying Equation (C.4) for DNN.
With the L_2 envelope from Step 1 and the DNN weight geometry from Step 2, the interaction-suppression argument from Juergens (2025) applies verbatim: an r-fold Hoeffding interaction requires r distinct points to fall in the $O_{\mathbb{P}}\big(s^{-1/k_x}\big)$ DNN neighborhood and carry nonnegligible weight. Occupancy plus Cauchy-Schwarz therefore give the uniform bound, for $r \geqslant 2$,

$$V_r(\gamma) \lesssim \Big(\frac{s}{n}\Big)^{r-1} V_1(\gamma) \qquad \text{uniformly over } \|\gamma\|_2 = 1, \tag{C.6}$$

with constants depending only on $\mathbb{E}[F^2]$ and the local design, not on γ. Summing in r,

$$\frac{\zeta_{ss}(\gamma)}{s\, \zeta_{1s}(\gamma)} - 1 = \sum_{r=2}^{s} \frac{\binom{s}{r} V_r(\gamma)}{s\, V_1(\gamma)} \lesssim \sum_{r=2}^{s} \frac{\binom{s}{r}}{s} \Big(\frac{s}{n}\Big)^{r-1} \leqslant \sum_{r=2}^{\infty} \frac{1}{r!} \Big(\frac{s}{n}\Big)^{r-1}. \tag{C.7}$$

Multiplying by s/n shows Equation (C.4). This is precisely the kernel-variance argument used in Juergens (2025); here the only DNN-specific inputs are the product form and the L_2 envelope from Step 1 together with the weight scale/negligibility from Step 2.

Step 5: Conclusion and uniformity in γ.
Condition Equation (C.4) is exactly the claim in Equation (3.11), hence Equation (C.5) gives Hájek dominance for $\overline{m}_{\gamma,s}(x)$ and thus asymptotic linearity with influence $h^{\langle 1\rangle}_{\gamma,s}(W)$ at the natural $\sqrt{n_{\text{eff}}(\mathbf{x})}$ scale. Uniformity over $\|\gamma\|_2 = 1$ follows from the uniform L_2 envelope and the linearity of ψ_γ; the vector statement then follows by Cramér-Wold. □

Proof of Proposition 3.5
Proof.

Step 1: Population Sign Change on a $n_{\text{eff}}(\mathbf{x})^{-1/2}$ Bracket.
By Condition 2.5 (iv) and (v), the population map $a \mapsto m_j\big(\mathbf{x}; a, \eta_{0,j}\big)$ admits a second-order expansion at $a = \alpha_{0,j}(\mathbf{x})$:

$$m_j\big(\mathbf{x}; a, \eta_{0,j}\big) = \Gamma_j(\mathbf{x})\,\big(a - \alpha_{0,j}(\mathbf{x})\big) + R_j\big(\mathbf{x}; a\big), \qquad \big|R_j(\mathbf{x}; a)\,\big| \leqslant \tfrac{1}{2}\,\overline{L}\,\big|\,a - \alpha_{0,j}(\mathbf{x})\,\big|^2, \tag{C.8}$$

for some $\overline{L}$ determined by the local bounds in (iv). Let $a_\pm \coloneqq \alpha_{0,j}(\mathbf{x}) \pm M/\sqrt{n_{\text{eff}}(\mathbf{x})}$. Then

$$m_j\big(\mathbf{x}; a_\pm, \eta_{0,j}\big) = \pm\,\Gamma_j(\mathbf{x})\,\frac{M}{\sqrt{n_{\text{eff}}(\mathbf{x})}} + O_{\mathbb{P}}\bigg(\frac{M^2}{n_{\text{eff}}(\mathbf{x})}\bigg), \tag{C.9}$$

so for fixed M and large n,

$$\operatorname{sgn}\big(m_j(\mathbf{x}; a_-, \eta_{0,j})\big) = -\operatorname{sgn}\big(\Gamma_j(\mathbf{x})\big), \qquad \operatorname{sgn}\big(m_j(\mathbf{x}; a_+, \eta_{0,j})\big) = \operatorname{sgn}\big(\Gamma_j(\mathbf{x})\big). \tag{C.10}$$

Step 2: Empirical-Population Coupling on a Shrinking Neighborhood.
Define the oracle-localized process

$$S_{n,j}(a) \coloneqq \frac{1}{\sqrt{n_{\text{eff}}(\mathbf{x})}} \sum_{i=1}^{n} w_{s,i}(\mathbf{x})\;\psi_j\big(W_i; a, \eta_{0,j}\big)\;\mathbb{1}\{\,X_i \in \mathcal{B}_{\mathbf{x}}\,\}. \tag{C.11}$$

By Lemmas 2.2 and 2.4 together with Conditions 2.4 and 2.6, we have the shrinking-set uniform bound

$$\sup_{|a-\alpha_{0,j}(\mathbf{x})|\leqslant M/\sqrt{n_{\text{eff}}(\mathbf{x})}} \Big|\; S_{n,j}(a) - \sqrt{n_{\text{eff}}(\mathbf{x})}\; m_j\big(\mathbf{x}; a, \eta_{0,j}\big)\;\Big| = o_{\mathbb{P}}(1). \tag{C.12}$$

The plug-in effect is negligible at the same (unscaled) level by Neyman orthogonality (iii) and the same localized equicontinuity control:

$$\sup_{|a-\alpha_{0,j}(\mathbf{x})|\leqslant M/\sqrt{n_{\text{eff}}(\mathbf{x})}} \Big|\; F_{n,j}(a) - S_{n,j}(a)\;\Big| = o_{\mathbb{P}}(1). \tag{C.13}$$

Combining Equations (C.12)(C.13) yields, uniformly on the bracket,

$$F_{n,j}(a) = \sqrt{n_{\text{eff}}(\mathbf{x})}\; m_j\big(\mathbf{x}; a, \eta_{0,j}\big) + o_{\mathbb{P}}(1). \tag{C.14}$$

Evaluating Equation (C.14) at $a_\pm$ and using Equation (C.9),

$$F_{n,j}(a_\pm) = \pm\,\Gamma_j(\mathbf{x})\,M \;+\; o_{\mathbb{P}}(1), \tag{C.15}$$

so, with probability tending to one,

$$\operatorname{sgn}\big(F_{n,j}(a_-)\big) = -\operatorname{sgn}\big(\Gamma_j(\mathbf{x})\big), \qquad \operatorname{sgn}\big(F_{n,j}(a_+)\big) = \operatorname{sgn}\big(\Gamma_j(\mathbf{x})\big). \tag{C.16}$$

The same conclusion holds after subtracting any $\Delta_{n,j} = o_{\mathbb{P}}\Big(b_n^{-1}/\sqrt{n_{\text{eff}}(\mathbf{x})}\Big)$, since $o_{\mathbb{P}}\Big(b_n^{-1}/\sqrt{n_{\text{eff}}(\mathbf{x})}\Big) = o_{\mathbb{P}}(1)$.

Step 3: Crossing Selector and Existence.
Define the left-continuous crossing selector on the bracket

$$\tilde{\alpha}_j(\mathbf{x}) \coloneqq \inf\Big\{a \in \big[\alpha_{0,j}(\mathbf{x}) - M/\sqrt{n_{\text{eff}}(\mathbf{x})},\, \alpha_{0,j}(\mathbf{x}) + M/\sqrt{n_{\text{eff}}(\mathbf{x})}\big] :\; F_{n,j}(a) \geqslant \Delta_{n,j}\Big\}. \tag{C.17}$$

By Equation (C.16) the set is nonempty and contained in the bracket, so $\tilde{\alpha}_j(\mathbf{x})$ is well-defined. This gives the near-root inequality

$$F_{n,j}\big(\tilde{\alpha}_j(\mathbf{x}) - \big) \;<\; \Delta_{n,j} \;\leqslant\; F_{n,j}\big(\tilde{\alpha}_j(\mathbf{x})\big), \tag{C.18}$$

hence $\big|\tilde{\alpha}_j(\mathbf{x}) - \alpha_{0,j}(\mathbf{x})\big| \leqslant M/\sqrt{n_{\text{eff}}(\mathbf{x})}$ almost surely and therefore

$$\big|\tilde{\alpha}_j(\mathbf{x}) - \alpha_{0,j}(\mathbf{x})\big| = O_{\mathbb{P}}\Big(n_{\text{eff}}(\mathbf{x})^{-1/2}\Big). \tag{C.19}$$

Step 4: High-Probability Local Monotonicity and Localization Rate.
By Equation (C.14) and Condition 2.5 (iv) and (v), for any $a_1 < a_2$ in the bracket,

$$\begin{aligned} F_{n,j}(a_2) - F_{n,j}(a_1) &= \sqrt{n_{\text{eff}}(\mathbf{x})}\,\Big(m_j(\mathbf{x}; a_2, \eta_{0,j}) - m_j(\mathbf{x}; a_1, \eta_{0,j})\Big) + o_{\mathbb{P}}(1) \\ &= \sqrt{n_{\text{eff}}(\mathbf{x})}\int_{a_1}^{a_2} \partial_\alpha m_j(\mathbf{x}; t, \eta_{0,j})\,\mathrm{d}t + o_{\mathbb{P}}(1) \\ &\geqslant \tfrac{c}{2}\sqrt{n_{\text{eff}}(\mathbf{x})}\,(a_2 - a_1) + o_{\mathbb{P}}(1), \end{aligned} \tag{C.20}$$

since $\partial_\alpha m_j(\mathbf{x}; t, \eta_{0,j})$ stays within $o(1)$ of $\Gamma_j(\mathbf{x})$ on the bracket by Condition 2.5 (iv). Thus $F_{n,j}$ is strictly monotone on the bracket with probability tending to one, and the crossing in Step 3 is unique on that event.
Taking $a_1 = \alpha_{0,j}(\mathbf{x})$ and $a_2 = \tilde{\alpha}_j(\mathbf{x})$ in Equation (C.20), and using $F_{n,j}\big(\alpha_{0,j}(\mathbf{x})\big) = \sqrt{n_{\text{eff}}(\mathbf{x})}\,m_j\big(\mathbf{x}; \alpha_{0,j}(\mathbf{x}), \eta_{0,j}\big) + o_{\mathbb{P}}(1) = o_{\mathbb{P}}(1)$ from Equation (C.14), we obtain

$$\tfrac{c}{2}\sqrt{n_{\text{eff}}(\mathbf{x})}\,\big|\tilde{\alpha}_j(\mathbf{x}) - \alpha_{0,j}(\mathbf{x})\big| \;\leqslant\; \big|F_{n,j}\big(\tilde{\alpha}_j(\mathbf{x})\big) - F_{n,j}\big(\alpha_{0,j}(\mathbf{x})\big)\big| + o_{\mathbb{P}}(1) \;=\; \big|\Delta_{n,j}\big| + o_{\mathbb{P}}(1) \;=\; o_{\mathbb{P}}(1), \tag{C.21}$$

which sharpens the localization to $\big|\tilde{\alpha}_j(\mathbf{x}) - \alpha_{0,j}(\mathbf{x})\big| = O_{\mathbb{P}}\Big(n_{\text{eff}}(\mathbf{x})^{-1/2}\Big)$.

Step 5: Exact Root under Continuity.
If, in addition, $a \mapsto \psi_j(W; a, \widehat{\eta}_j)$ is almost surely continuous (so $F_{n,j}$ is continuous), then Equation (C.16) and the intermediate value theorem yield an exact empirical root for any fixed level, in particular for $\Delta_{n,j} \equiv 0$. □

Proof of Theorem 3.6
Proof. To prove Theorem 3.6, we will proceed along the lines of the proof of Theorem 2 of Belloni et al. (2015) with modifications to adjust for the localized problem we consider and the LOO condition we use to avoid entropy based arguments.

Step 1: Set-up and Target Expansion.
Let $\tilde{\boldsymbol{\alpha}}(\mathbf{x}) = (\tilde{\alpha}_j(\mathbf{x}))_{j=1}^{p_\alpha}$ be any estimator such that with probability $1 - o(1)$,

$$\big\|\tilde{\alpha}(\mathbf{x}) - \boldsymbol{\alpha}_s(\mathbf{x})\big\|_\infty = \max_{j \in [p_\alpha]} \big|\tilde{\alpha}_j(\mathbf{x}) - \alpha_{s,j}(\mathbf{x})\big| \leqslant C\rho_n \tag{C.22}$$

(The pseudo-target $\boldsymbol{\alpha}_s(\mathbf{x})$ satisfies $\left\| \boldsymbol{\alpha}_s(\mathbf{x}) - \boldsymbol{\alpha}_0(\mathbf{x}) \right\| = O(s^{-\beta/k_x})$, so this is a localized consistency assumption; see Definition Appendix B.1.) We wish to show that with probability $1 - o(1)$ and uniformly in $j \in [p_\alpha]$,

$$\begin{aligned}
\frac{1}{\sqrt{n_{\text{eff}}(\mathbf{x})}} \mathbb{T}_{n,w}\left[\psi_j(\cdot;\tilde{\alpha}_j(\mathbf{x}),\widehat{\eta}_j)\right] &= \sum_{i=1}^{n} w_{s,i}(\mathbf{x})\,\psi_j\left(W_i,\tilde{\alpha}_j(\mathbf{x}),\widehat{\eta}_j\left(Z_i^j\right)\right) \\
&= \sum_{i=1}^{n} w_{s,i}(\mathbf{x})\,\psi_j\left(W_i,\alpha_{s,j}(\mathbf{x}),\eta_{0,j}\left(Z_i^j\right)\right) + \Gamma_j\left(\tilde{\alpha}_j(\mathbf{x}) - \alpha_{s,j}(\mathbf{x})\right) + o_{\mathbb{P}}\left(\frac{1}{\sqrt{n_{\text{eff}}(\mathbf{x})}}\right) + o_{\mathbb{P}}\left(\frac{b_n^{-1}}{\sqrt{n_{\text{eff}}(\mathbf{x})}}\right) \\
&= \frac{1}{\sqrt{n_{\text{eff}}(\mathbf{x})}} \mathbb{T}_{n,w}\left[\psi_j(\cdot;\alpha_{s,j}(\mathbf{x}),\eta_{0,j})\right] + \Gamma_j\left(\tilde{\alpha}_j(\mathbf{x}) - \alpha_{s,j}(\mathbf{x})\right) + o_{\mathbb{P}}\left(\frac{1}{\sqrt{n_{\text{eff}}(\mathbf{x})}}\right) + o_{\mathbb{P}}\left(\frac{b_n^{-1}}{\sqrt{n_{\text{eff}}(\mathbf{x})}}\right).
\end{aligned} \tag{C.23}$$

For $a \in \mathcal{A}_j(\mathbf{x})$ and $\tilde{\eta} \in \mathcal{H}_j$, define the score difference by

$$\Delta\psi_j(W;\ a,\tilde{\eta}) \coloneqq \psi_j(W;a,\tilde{\eta}) - \psi_j(W;\alpha_{s,j}(\mathbf{x}),\eta_{0,j}). \tag{C.24}$$

Using add-and-subtract at $(\alpha_{s,j}(\mathbf{x}),\eta_{0,j})$ and Equations (A.1)(A.2), we obtain the exact decomposition

$$\begin{aligned}
&\mathbb{T}_{n,w}\left[\psi_j(\cdot;\tilde{\alpha}_j(\mathbf{x}),\widehat{\eta}_j)\right] \\
&\quad= \underbrace{\mathbb{G}_{n,w}\left[\psi_j(\cdot;\alpha_{s,j}(\mathbf{x}),\eta_{0,j})\right]}_{\mathbf{I}_j} + \underbrace{\mathbb{G}_{n,w}\left[\Delta\psi_j(\cdot;\tilde{\alpha}_j(\mathbf{x}),\widehat{\eta}_j)\right]}_{\mathbf{III}_j} + \underbrace{\sqrt{n_{\text{eff}}(\mathbf{x})}\,\mathbb{E}[\Delta\psi_j(W;\tilde{\alpha}_j(\mathbf{x}),\widehat{\eta}_j) \mid X=\mathbf{x}]}_{\mathbf{II}_j} + \underbrace{\sqrt{n_{\text{eff}}(\mathbf{x})}\,m_j(\mathbf{x};\alpha_{s,j}(\mathbf{x}),}_{\mathbf{B}_j}
\end{aligned} \tag{C.25}$$

Here $\mathbf{I}_j$ is the stochastic fluctuation at the pseudo-target (via Equation (A.2)), $\mathbf{III}_j$ is the localized fluctuation over the neighborhood $\left|a - \alpha_{s,j}(\mathbf{x})\right| \leqslant C\rho_n$, $\mathbf{II}_j$ is the local drift increment from $(\alpha_{s,j},\eta_{0,j})$ to $(\tilde{\alpha}_j,\widehat{\eta}_j)$, and $\mathbf{B}_j$ is the residual pointwise-centering term at $\alpha_{s,j}$. For later uniformity statements we work with the localized score class

$$\mathcal{F} \coloneqq \left\{ z \mapsto \psi_j(z;a,\tilde{\eta}) - \psi_j(z;\alpha_{s,j}(\mathbf{x}),\eta_{0,j}) \ :\ j \in [p_\alpha], \tilde{\eta} \in \mathcal{H}_j, a \in \mathcal{A}_j(\mathbf{x}), \left|a - \alpha_{s,j}(\mathbf{x})\right| \leqslant C\rho_n \right\}, \tag{C.26}$$

whose envelope is the F in Condition 2.4 (ii). Similarly, we define a gated version,

$$\mathcal{F}^G \coloneqq \left\{ z \mapsto (\psi_j(z;a,\tilde{\eta}) - \psi_j(z;\alpha_{s,j}(\mathbf{x}),\eta_{0,j}))\,\mathbb{1}\{G\} \ :\ j \in [p_\alpha], \tilde{\eta} \in \mathcal{H}_j, a \in \mathcal{A}_j(\mathbf{x}), \left|a - \alpha_{s,j}(\mathbf{x})\right| \leqslant C\rho_n \right\}, \tag{C.27}$$

whose envelope is once again F. When analyzing $\mathbf{I}_j$ and $\mathbf{III}_j$ below we will localize functions via the gate $G = \{X \in \mathcal{B}_\mathbf{x}\}$ by working with $\mathbb{G}_{n,w}^G$ on $\mathcal{F}^G$ and an outside remainder $R^{\text{out}}(f)$; see Equations (A.9)(A.12) in the Appendix for the precise identities. We will show in Steps 1 and 3 that $R^{\text{out}}(f)$ is negligible for the specific functions used there, while the conditioning term $\mathbf{II}_j$ is unaffected by gating (cf. Equation (A.11)).

Step 2: Bounding $\mathbf{III}_j$.
By construction of $\tilde{\alpha}_j(\mathbf{x})$ and $\widehat{\eta}_j$, we have with probability $1 - o(1)$

$$f_{j,\text{plug}}(\cdot) \coloneqq \Delta\psi_j(\cdot;\tilde{\alpha}_j(\mathbf{x}),\widehat{\eta}_j) \in \mathcal{F} \quad \text{and} \quad f_{j,\text{plug}}^G(\cdot) \coloneqq \Delta\psi_j(\cdot;\tilde{\alpha}_j(\mathbf{x}),\widehat{\eta}_j)\ \mathbb{1}\{G\} \in \mathcal{F}_j^G, \tag{C.28}$$

where $\mathcal{F}$ and $\mathcal{F}^G$ are the localized (ungated) and gated classes from Equations (C.26)(C.27) with envelope F.

Step 2.A: Gate Switch and Outside Remainder.
Write

$$\mathbf{III}_j = \mathbb{G}_{n,w}\left[f_{j,\text{plug}}\right] = \underbrace{\mathbb{G}_{n,w}^G\left[f_{j,\text{plug}}\right]}_{\text{gated fluctuation}} + \underbrace{\mathbb{G}_{n,w}\left[f_{j,\text{plug}}\ \mathbb{1}\{G^c\}\right]}_{\text{outside remainder } R_j^{\text{out}}}. \tag{C.29}$$

We claim $R_j^{\text{out}} = o_{\mathbb{P}}\left(b_n^{-1}\right)$ uniformly in j. To see this, use the subsample representation of the DNN weights and apply Lemma Appendix E.8) with $h(W) = \Delta\psi_j(W;\tilde{\alpha}_j,\widehat{\eta}_j)$ (or its centered version, as allowed by the lemma). For any $\epsilon > 0$,

$$\mathbb{P}\Big(\ |R_j^{\text{out}}| > \epsilon\Big) \leqslant \frac{C_{1/\mathfrak{q}}}{\epsilon}\,(1-\pi_\mathbf{x})^s \ \longrightarrow\ 0, \tag{C.30}$$

and because $(1-\pi_\mathbf{x})^s$ decays exponentially while b_n grows subexponentially, the same bound implies $R_j^{\text{out}} = o_{\mathbb{P}}\left(b_n^{-1}\right)$ uniformly in j.

Step 2.B: LSE on the Gated Class.
On the event Equation (C.28), $f^G_{j,\text{plug}} \in \mathcal{F}^G_j$. Although Lemma Appendix E.10 is written with anchor $\alpha_{0,j}(\mathbf{x})$, its proof uses only local envelope/Lipschitz control and LOO stability, so the same argument applies with any fixed anchor in the same neighborhood, in particular $\alpha_{s,j}(\mathbf{x})$. Hence, for $\mathcal{F}^G_j$,

$$\sup_{f\in\mathcal{F}^G_j} |\ \mathbb{G}^G_{n,w}[f]\ | = o_{\mathbb{P}}\big(b_n^{-1}\big), \tag{C.31}$$

uniformly over $j \in [p_\alpha]$.

Step 2.C: Conclusion.
Combining Equations (C.29)(C.31),

$$\sup_{j\in[p_\alpha]} |\,\mathbf{III}_j\,| \leqslant \sup_{j\in[p_\alpha]} \sup_{f\in\mathcal{F}^G_j} |\ \mathbb{G}^G_{n,w}[f]\ | \ + \ \sup_{j\in[p_\alpha]} |\,R^{\text{out}}_j\,| = o_{\mathbb{P}}\big(b_n^{-1}\big). \tag{C.32}$$

Intuitively, $\mathbf{III}_j$ is the weighted fluctuation of a score difference over a tiny local class. We first gate to the $\mathbf{x}$-neighborhood; the mass outside G is exponentially small in s, so the outside remainder vanishes even at the b_n^{-1} scale. Inside G, mean-square continuity (MSC) plus leave-one-out stability give localized stochastic equicontinuity, so the gated fluctuation is uniformly $o_{\mathbb{P}}\big(b_n^{-1}\big)$.

Step 3: Expanding $\mathbf{II}_j$.
Recall from Equation (C.25) that

$$\mathbf{II}_j = \sqrt{n_{\text{eff}}(\mathbf{x})}\ \mathbb{E}[\Delta\psi_j(W;\tilde{\alpha}_j(\mathbf{x}),\widehat{\eta}_j) \mid X=\mathbf{x}] = \sqrt{n_{\text{eff}}(\mathbf{x})}\ \Big[m_j(\mathbf{x};\tilde{\alpha}_j(\mathbf{x}),\widehat{\eta}_j) - m_j(\mathbf{x};\alpha_{s,j}(\mathbf{x}),\eta_{0,j})\Big]. \tag{C.33}$$

To linearize around $\alpha_{s,j}$ while preserving orthogonality at $\alpha_{0,j}$, apply Lemma Appendix E.6 twice (at $(\tilde{\alpha}_j,\widehat{\eta}_j)$ and $(\alpha_{s,j},\eta_{0,j})$), then subtract:

$$\begin{aligned} m_j(\mathbf{x};\tilde{\alpha}_j(\mathbf{x}),\widehat{\eta}_j) &= \Gamma_j(\mathbf{x})\,(\tilde{\alpha}_j(\mathbf{x}) - \alpha_{0,j}(\mathbf{x})) + R_j(\mathbf{x},\tilde{\alpha}_j(\mathbf{x}),\widehat{\eta}_j)\,, \\ m_j(\mathbf{x};\alpha_{s,j}(\mathbf{x}),\eta_{0,j}) &= \Gamma_j(\mathbf{x})\,(\alpha_{s,j}(\mathbf{x}) - \alpha_{0,j}(\mathbf{x})) + R_j(\mathbf{x},\alpha_{s,j}(\mathbf{x}),\eta_{0,j})\,. \end{aligned} \tag{C.34}$$

Therefore

$$m_j(\mathbf{x};\tilde{\alpha}_j(\mathbf{x}),\widehat{\eta}_j) - m_j(\mathbf{x};\alpha_{s,j}(\mathbf{x}),\eta_{0,j}) = \Gamma_j(\mathbf{x})\,(\tilde{\alpha}_j(\mathbf{x}) - \alpha_{s,j}(\mathbf{x})) + \overline{R}_j(\mathbf{x},\tilde{\alpha}_j(\mathbf{x}),\widehat{\eta}_j)\,, \tag{C.35}$$

where $\overline{R}_j \coloneqq R_j(\mathbf{x},\tilde{\alpha}_j,\widehat{\eta}_j) - R_j(\mathbf{x},\alpha_{s,j},\eta_{0,j})$. with the uniform bound

$$\big|\overline{R}_j(\mathbf{x},\tilde{\alpha}_j(\mathbf{x}),\widehat{\eta}_j)\big| \leqslant L_v\left(\big|\tilde{\alpha}_j(\mathbf{x}) - \alpha_{0,j}(\mathbf{x})\big|^2 + \|\widehat{\eta}_j - \eta_{0,j}\|_\eta^2 + \big|\alpha_{s,j}(\mathbf{x}) - \alpha_{0,j}(\mathbf{x})\big|^2\right). \tag{C.36}$$

Combining **??**–C.36 gives the drift expansion at the $\mathbb{T}_{n,w}$-scale,

$$\mathbf{II}_j = \sqrt{n_{\text{eff}}(\mathbf{x})}\ \Gamma_j(\mathbf{x})\,(\tilde{\alpha}_j(\mathbf{x}) - \alpha_{s,j}(\mathbf{x})) + \sqrt{n_{\text{eff}}(\mathbf{x})}\ \overline{R}_j(\mathbf{x},\tilde{\alpha}_j(\mathbf{x}),\widehat{\eta}_j)\,, \tag{C.37}$$

and, equivalently, at the $(1/\sqrt{n_{\text{eff}}(\mathbf{x})})$-scale used in the target display,

$$\frac{1}{\sqrt{n_{\text{eff}}(\mathbf{x})}}\,\mathbf{II}_j = \Gamma_j(\mathbf{x})\,(\tilde{\alpha}_j(\mathbf{x}) - \alpha_{s,j}(\mathbf{x})) + \overline{R}_j(\mathbf{x},\tilde{\alpha}_j(\mathbf{x}),\widehat{\eta}_j)\,. \tag{C.38}$$

Since Equation (C.36) is quadratic in the local errors, any rates ρ_n and $r_{n,j}$ with

$$\sqrt{n_{\text{eff}}(\mathbf{x})}\ \left(\rho_n^2 + r_{n,j}^2 + \big|\alpha_{s,j}(\mathbf{x}) - \alpha_{0,j}(\mathbf{x})\big|^2\right) = o\left(b_n^{-1}\right) \tag{C.39}$$

imply

$$\sqrt{n_{\text{eff}}(\mathbf{x})}\ \overline{R}_j(\mathbf{x},\tilde{\alpha}_j(\mathbf{x}),\widehat{\eta}_j) = o_{\mathbb{P}}\big(b_n^{-1}\big), \qquad \overline{R}_j(\mathbf{x},\tilde{\alpha}_j(\mathbf{x}),\widehat{\eta}_j) = o_{\mathbb{P}}\big(b_n^{-1}\big), \tag{C.40}$$

uniformly across $j \in [p_\alpha]$, so that $\mathbf{II}_j$ contributes exactly the linear term $\Gamma_j(\mathbf{x})\,(\tilde{\alpha}_j(\mathbf{x}) - \alpha_{s,j}(\mathbf{x}))$ at the $(1/\sqrt{n_{\text{eff}}(\mathbf{x})})$-scale, with a negligible remainder. Intuitively, the localized moment is first-order linear in the target a around $\alpha_{s,j}(\mathbf{x})$ with slope $\Gamma_j(\mathbf{x})$, and orthogonal to nuisance perturbations, so $\widehat{\eta}_j$ only enters at second order. That is why the drift is precisely the Jacobian times the small parameter shift, plus a quadratic remainder that vanishes at the $\sqrt{n_{\text{eff}}(\mathbf{x})}$ scale under Equation (C.39).

Step 4: Analyzing $\mathbf{I}_j$.
Let $f_j(w) \coloneqq \psi_j\,(w, \alpha_{s,j}(\mathbf{x}), \eta_{0,j})$. By Equation (A.2),

$$\mathbf{I}_j = \mathbb{G}_{n,w}\,[f_j]\,. \tag{C.41}$$

Here $m_j(\mathbf{x}; \alpha_{s,j}(\mathbf{x}), \eta_{0,j})$ need not be zero; the decomposition into centered and drift pieces controls this shift explicitly.

Step 4.A: Gate Switch and Operator Identity.
Let $G = \{X \in \mathcal{B}_{\mathbf{x}}\}$ and $G_i = \{X_i \in \mathcal{B}_{\mathbf{x}}\}$. Define the "outside" remainder

$$R_j^{\text{out}} \coloneqq \sqrt{n_{\text{eff}}\,(\mathbf{x})}\sum_{i=1}^{n} w_{s,i}\,(\mathbf{x})\;\; f_j(W_i)\;\; \mathbb{1}\{\,G_i^c\,\}. \tag{C.42}$$

By the Appendix identities Equations (A.9)(A.12),

$$\mathbf{I}_j = \mathbb{G}_{n,w}^{G}[f_j] + R_j^{\text{out}} = \underbrace{\mathbb{H}_{n,w}^{G}[f_j]}_{\mathbf{I}_j^{\text{center},G}} + \underbrace{\mathbb{D}_{n,w}^{G}[f_j]}_{\mathbf{I}_j^{\text{drift},G}} + R_j^{\text{out}}. \tag{C.43}$$

Step 4.B: Centered Term drives the Limit.
For each i, $m_j(u; \alpha_{s,j}(\mathbf{x}), \eta_{0,j}) \coloneqq \mathbb{E}[f_j(W) \mid X = u]$, hence

$$\mathbb{E}\big[\big\{f_j(W_i) - m_j(X_i; \alpha_{s,j}(\mathbf{x}), \eta_{0,j})\big\}\;\mathbb{1}\{\,G_i\,\}\;\big|\;X_{[n]}\big] = 0, \tag{C.44}$$

so $\mathbb{E}\Big[\mathbf{I}_j^{\text{center},G}\;\Big|\;X_{[n]}\Big] = 0$. We defer its limit law to the CLT step; here we only record the decomposition.

Step 4.C: Local Drift is Negligible.
By the Lipschitz-in-u argument of Lemma Appendix E.9 (which is anchor-invariant for fixed a in the same local neighborhood), gating can only reduce the drift and

$$\frac{1}{\sqrt{n_{\text{eff}}\,(\mathbf{x})}}\,\big|\,\mathbb{D}_{n,w}^{G}\,[f_j]\,\big| \leqslant L_u\sum_{i=1}^{n} w_{s,i}\,(\mathbf{x})\;\;\|\,X_i - \mathbf{x}\,\|_2\;\mathbb{1}\{\,G_i\,\} = o_{\mathbb{P}}(\,1\,), \tag{C.45}$$

and, in the CLT scaling, $|\,\mathbb{D}_{n,w}^{G}[f_j]\,| = o_{\mathbb{P}}(\,1\,)$ uniformly in j by the same mass-radius control used in Equation (E.94).

Step 4.D: Bad-Event Remainder is Negligible.
Using the subsample representation of the DNN weights, apply Lemma Appendix E.8 with $h = f_j$ or its centered version. For any $\epsilon > 0$,

$$\mathbb{P}\Big(|\,R_j^{\text{out}}\,| > \epsilon\Big) \leqslant \frac{C_{1/\mathfrak{q}}}{\epsilon}\;(1 - \pi_{\mathbf{x}})^{s} \longrightarrow 0, \tag{C.46}$$

uniformly over $j \in [p_\alpha]$. Hence $R_j^{\text{out}} = o_{\mathbb{P}}(\,1\,)$; at the CLT scale it remains negligible since $(1 - \pi_{\mathbf{x}})^{s}$ decays exponentially in s while $\sqrt{n_{\text{eff}}\,(\mathbf{x})} \asymp \sqrt{n/s}$ grows subexponentially.

Step 4.E: Conclusion.
Combining Equations (C.43)(C.46),

$$\mathbf{I}_j = \mathbf{I}_j^{\text{center},G} + o_{\mathbb{P}}(\,1\,) \qquad \text{uniformly in } j \in [p_\alpha]. \tag{C.47}$$

Since $\mathbf{I}_j^{\text{center},G} - \mathbf{I}_j^{\text{center}} = \mathbb{H}_{n,w}^{G}[f_j] - \mathbb{H}_{n,w}[f_j] = \sqrt{n_{\text{eff}}\,(\mathbf{x})}\sum_i w_{s,i}\,(\mathbf{x})\,\big\{f_j(W_i) - m_j(X_i; \alpha_{s,j}, \eta_{0,j})\big\}\;\mathbb{1}\big\{\,G_i^c\,\big\} = o_{\mathbb{P}}(\,1\,)$ by the same bad-event bound, we may equivalently write

$$\mathbf{I}_j = \mathbf{I}_j^{\text{center}} + o_{\mathbb{P}}(\,1\,), \tag{C.48}$$

and the limit distribution is governed by the centered mean-zero fluctuation; see Theorem 3.8.

Step 5: Collecting terms.
By Equation (C.25) we have, for each $j \in [p_\alpha]$,

$$\mathbb{T}_{n,w}\Big[\psi_j(\cdot;\tilde{\alpha}_j(\mathbf{x}),\widehat{\eta}_j)\Big] = \mathbf{I}_j + \mathbf{II}_j + \mathbf{III}_j + \mathbf{B}_j. \tag{C.49}$$

On the high-probability event $\big(\tilde{\alpha}_j(\mathbf{x}),\widehat{\eta}_j\big) \in \mathcal{P}_j(\mathbf{x})$ and working with the gate $G = \{X \in \mathcal{B}_\mathbf{x}\}$ (standing conventions; cf. Equations (A.9)(A.12)), the three components satisfy:

(i) **Plug-in Fluctuation:** By Step 2, uniformly in j,

$$|\,\mathbf{III}_j\,| = o_\mathbb{P}\big(\,b_n^{-1}\,\big). \tag{C.50}$$

(ii) **Local Drift:** By Step 3, using Equation (C.37) and Equation (C.40), uniformly in j,

$$\mathbf{II}_j = \sqrt{n_{\text{eff}}\,(\mathbf{x})}\;\Gamma_j(\mathbf{x})\Big(\tilde{\alpha}_j(\mathbf{x}) - \alpha_{s,j}(\mathbf{x})\Big) + o_\mathbb{P}\big(\,b_n^{-1}\,\big). \tag{C.51}$$

(iii) **Pseudo-target-level Fluctuation:** By Step 4, see Equation (C.47),

$$\mathbf{I}_j = \mathbf{I}_j^{\text{center}} + o_\mathbb{P}(\,1\,), \tag{C.52}$$

where $\mathbf{I}_j^{\text{center}} \coloneqq \mathbb{H}_{n,w}[f_j]$ with $f_j(w) = \psi_j\,(w,\alpha_{s,j}(\mathbf{x}),\eta_{0,j})$. By the bad-event bound in Step 4, $\mathbb{H}_{n,w}[f_j]$ and $\mathbb{H}^G_{n,w}[f_j]$ differ by $o_\mathbb{P}(\,1\,)$, so either version can be used.

(iv) **Centering Offset:**

$$\mathbf{B}_j = \sqrt{n_{\text{eff}}\,(\mathbf{x})}\;m_j\,(\mathbf{x};\alpha_{s,j}(\mathbf{x}),\eta_{0,j})\,. \tag{C.53}$$

Combining Equations (C.49)(C.52) yields the collected expansion at the $\mathbb{T}_{n,w}$ scale:

$$\mathbb{T}_{n,w}\Big[\psi_j(\cdot;\tilde{\alpha}_j(\mathbf{x}),\widehat{\eta}_j)\Big] = \mathbb{G}_{n,w}\left[\psi_j\,(\cdot,\alpha_{s,j}(\mathbf{x}),\eta_{0,j})\right] + \sqrt{n_{\text{eff}}\,(\mathbf{x})}\;\Gamma_j(\mathbf{x})\Big(\tilde{\alpha}_j(\mathbf{x}) - \alpha_{s,j}(\mathbf{x})\Big) + \sqrt{n_{\text{eff}}\,(\mathbf{x})}\;m_j\,(\mathbf{x};\alpha_{s,j}(\mathbf{x}),\eta_{0,j}) + o_\mathbb{P}(\,1\,) + o_\mathbb{P}\big(\,b_n^{-1}\,\big) \tag{C.54}$$

uniformly over $j \in [p_\alpha]$. Equivalently, dividing by $\sqrt{n_{\text{eff}}\,(\mathbf{x})}$ gives the $(1/\sqrt{n_{\text{eff}}\,(\mathbf{x})})$-scale form used in the target display:

$$\begin{aligned}\frac{1}{\sqrt{n_{\text{eff}}\,(\mathbf{x})}}\;\mathbb{T}_{n,w}\Big[\psi_j(\cdot;\tilde{\alpha}_j(\mathbf{x}),\widehat{\eta}_j)\Big] = {}& \frac{1}{\sqrt{n_{\text{eff}}\,(\mathbf{x})}}\;\mathbb{G}_{n,w}\left[\psi_j\,(\cdot,\alpha_{s,j}(\mathbf{x}),\eta_{0,j})\right] + \Gamma_j(\mathbf{x})\Big(\tilde{\alpha}_j(\mathbf{x}) - \alpha_{s,j}(\mathbf{x})\Big) \\ & + m_j\,(\mathbf{x};\alpha_{s,j}(\mathbf{x}),\eta_{0,j}) + o_\mathbb{P}\left(\frac{1}{\sqrt{n_{\text{eff}}\,(\mathbf{x})}}\right) + o_\mathbb{P}\left(\frac{b_n^{-1}}{\sqrt{n_{\text{eff}}\,(\mathbf{x})}}\right).\end{aligned} \tag{C.55}$$

In particular, Equations (C.54)(C.55) establish local asymptotic linearity: the plug-in score equals the pseudo-target-level centered fluctuation plus the Jacobian times the local parameter shift, up to negligible remainders, with an explicit centering-offset term at $\alpha_{s,j}$.

Intuition:

(i) The DNN weights concentrate almost all mass in a small neighborhood of $\mathbf{x}$, so the plug-in score behaves like a local average with effective size $n_{\text{eff}}\,(\mathbf{x})$.

(ii) The fluctuation $\mathbf{III}_j$ is a small, stability-controlled wobble that comes only from plugging in $(\tilde{\alpha}_j, \widehat{\eta}_j)$, and it vanishes uniformly by localized stochastic equicontinuity.

(iii) The drift $\mathbf{II}_j$ reduces to a first-order shift $\Gamma_j(\mathbf{x})\big(\tilde{\alpha}_j(\mathbf{x}) - \alpha_{s,j}(\mathbf{x})\big)$, while orthogonality makes all nuisance perturbations second order and therefore negligible at the $\sqrt{n_{\text{eff}}\,(\mathbf{x})}$ scale.

(iv) What remains is $\mathbf{I}_j$, the pseudo-target-level centered fluctuation, plus the explicit offset $\mathbf{B}_j = \sqrt{n_{\text{eff}}\,(\mathbf{x})}\,m_j(\mathbf{x};\alpha_{s,j},\eta_{0,j})$.

(v) The rare subsamples that miss the local ball contribute an exponentially small remainder and do not affect the limit.

Together these pieces yield a local, first-order linear representation with a pseudo-target-level stochastic term, a deterministic Jacobian shift, and an explicit centering-offset term.

Step 6: Explicit Local Bahadur Representation.
By Condition 2.5 (v), the identification slope is uniformly bounded away from zero,

$$\inf_{j\in[p_\alpha]} |\,\Gamma_j(\mathbf{x})\,| \geqslant c > 0. \tag{C.56}$$

By Proposition 3.5, for any tolerance $\Delta_{n,j} = o_{\mathbb{P}}\Big(b_n^{-1}/\sqrt{n_{\text{eff}}\,(\mathbf{x})}\Big)$ there exists (with probability tending to one) a unique $\tilde{\alpha}_j(\mathbf{x})$ in an $O_{\mathbb{P}}\Big(n_{\text{eff}}\,(\mathbf{x})^{-1/2}\Big)$ neighborhood of $\alpha_{s,j}(\mathbf{x})$ such that

$$\frac{1}{\sqrt{n_{\text{eff}}\,(\mathbf{x})}}\ \sum_{i=1}^{n} w_{s,i}\,(\mathbf{x})\ \psi_j\big(W_i;\tilde{\alpha}_j(\mathbf{x}),\widehat{\eta}_j\big)\ \mathbb{1}\{\,X_i\in\mathcal{B}_{\mathbf{x}}\,\}\ =\ \Delta_{n,j}, \tag{C.57}$$

where $\Delta_{n,j}\equiv 0$ if the estimating equation is solved exactly. We henceforth treat $\Delta_{n,j}$ as the (asymptotically negligible) solver tolerance permitted by Proposition 3.5.
Combining Equation (C.57) with the expansion in Equation (C.55) and solving for $\tilde{\alpha}_j(\mathbf{x})-\alpha_{s,j}(\mathbf{x})$ yields, uniformly in $j\in[p_\alpha]$,

$$\sqrt{n_{\text{eff}}\,(\mathbf{x})}\ \big(\tilde{\alpha}_j(\mathbf{x})-\alpha_{s,j}(\mathbf{x})\big)\ =\ -\,\Gamma_j(\mathbf{x})^{-1}\ \mathbb{G}_{n,w}\Big[\psi_j\big(\cdot,\alpha_{s,j}(\mathbf{x}),\eta_{0,j}\big)\Big]-\sqrt{n_{\text{eff}}\,(\mathbf{x})}\ \Gamma_j(\mathbf{x})^{-1}\ m_j\,(\mathbf{x};\alpha_{s,j}(\mathbf{x}),\eta_{0,j})+o_{\mathbb{P}}(\,1\,)+o_{\mathbb{P}}\big(\,b_n^{-1}\,\big). \tag{C.58}$$

Equivalently, in unscaled form,

$$\tilde{\alpha}_j(\mathbf{x})-\alpha_{s,j}(\mathbf{x})\ =\ -\,\Gamma_j(\mathbf{x})^{-1}\ \frac{1}{\sqrt{n_{\text{eff}}\,(\mathbf{x})}}\ \mathbb{G}_{n,w}\Big[\psi_j\big(\cdot,\alpha_{s,j}(\mathbf{x}),\eta_{0,j}\big)\Big]-\Gamma_j(\mathbf{x})^{-1}\ m_j\,(\mathbf{x};\alpha_{s,j}(\mathbf{x}),\eta_{0,j})+o_{\mathbb{P}}\left(\ \frac{1}{\sqrt{n_{\text{eff}}\,(\mathbf{x})}}\ \right)+o_{\mathbb{P}}\left(\ \frac{b_n^{-1}}{\sqrt{n_{\text{eff}}\,(\mathbf{x})}}\ \right) \tag{C.59}$$

If the pointwise-centering offset satisfies

$$\sqrt{n_{\text{eff}}\,(\mathbf{x})}\ m_j\,(\mathbf{x};\alpha_{s,j}(\mathbf{x}),\eta_{0,j})\ =\ o(1), \tag{C.60}$$

then Equations (C.58)(C.59) reduce to the pseudo-target-centered representation stated in Theorem 3.6.
It is often convenient to write the influence-function form. Define

$$\varphi_j(w;\mathbf{x})\ \coloneqq\ -\,\Gamma_j(\mathbf{x})^{-1}\ \Big\{\psi_j\big(w,\alpha_{s,j}(\mathbf{x}),\eta_{0,j}\big)\ -\ m_j\big(\mathbf{x};\alpha_{s,j}(\mathbf{x}),\eta_{0,j}\big)\Big\}. \tag{C.61}$$

Since $\mathbb{G}_{n,w}[f]\ =\ \sqrt{n_{\text{eff}}\,(\mathbf{x})}\sum_{i=1}^{n} w_{s,i}\,(\mathbf{x})\,\big(f(W_i)-\mathbb{E}[f(W)\mid X=\mathbf{x}]\big)$, Equation (C.58) is equivalent to

$$\sqrt{n_{\text{eff}}\,(\mathbf{x})}\ \big(\tilde{\alpha}_j(\mathbf{x})-\alpha_{s,j}(\mathbf{x})\big)\ =\ \sum_{i=1}^{n} w_{s,i}\,(\mathbf{x})\ \ \varphi_j(W_i;\mathbf{x})\ -\ \sqrt{n_{\text{eff}}\,(\mathbf{x})}\ \Gamma_j(\mathbf{x})^{-1}\ m_j\,(\mathbf{x};\alpha_{s,j}(\mathbf{x}),\eta_{0,j})\ +\ o_{\mathbb{P}}(\,1\,)\ +\ o_{\mathbb{P}}\big(\,b_n^{-1}\,\big), \tag{C.62}$$

uniformly over $j\in[p_\alpha]$.

(i) By Step 4, $\mathbb{G}_{n,w}$ may be replaced by its gated version $\mathbb{G}^{G}_{n,w}$ in Equations (C.58)(C.59) at an $o_{\mathbb{P}}(\,1\,)$ cost.

(ii) If $p_\alpha>1$ and the moments are stacked with Jacobian $J(\mathbf{x})\coloneqq\operatorname{diag}\big(\Gamma_1(\mathbf{x}),\ldots,\Gamma_{p_\alpha}(\mathbf{x})\big)$, the vector form is

$$\begin{aligned}\sqrt{n_{\text{eff}}\,(\mathbf{x})}\ \big(\tilde{\boldsymbol{\alpha}}(\mathbf{x})-\boldsymbol{\alpha}_s(\mathbf{x})\big)\ &=\ -\,J(\mathbf{x})^{-1}\ \mathbb{G}_{n,w}\Big[\boldsymbol{\psi}\big(\cdot,\boldsymbol{\alpha}_s(\mathbf{x}),\boldsymbol{\eta}_0\big)\Big]\ -\ \sqrt{n_{\text{eff}}\,(\mathbf{x})}\ J(\mathbf{x})^{-1}\ \boldsymbol{m}\,(\mathbf{x};\boldsymbol{\alpha}_s(\mathbf{x}),\boldsymbol{\eta}_0)+\mathbf{r}_n,\\ \|\,\mathbf{r}_n\,\|_\infty\ &=\ o_{\mathbb{P}}(\,1\,)+o_{\mathbb{P}}\big(\,b_n^{-1}\,\big),\end{aligned} \tag{C.63}$$

with the obvious definitions of the stacked score and nuisances.

□

Proof of Theorem 3.8

Proof. Write $U_{n,s,\omega}(\mathbf{x})$ for the complete generalized U-statistic with kernel $h_{n,s,x}$ from Equation (3.4), and let

$$\zeta_{1,\omega}(\mathbf{x}) \coloneqq \mathrm{Var}(\mathbb{E}[h_{n,s,x}(Z_1,\ldots,Z_s;\omega) \mid Z_1]) \quad \text{and} \quad \zeta_s(\mathbf{x}) \coloneqq \mathrm{Var}(h_{n,s,x}(Z_1,\ldots,Z_s;\omega)). \tag{C.64}$$

By Proposition 3.4, $\frac{s}{n}\left(\frac{\zeta_s(\mathbf{x})}{s\,\zeta_{1,\omega}(\mathbf{x})}\right) \xrightarrow{\mathbb{P}} 1$; by our standing asymptotics $s = o(n)$. Thus the central condition of Peng et al. (2022) Theorem 1 holds, yielding

$$\frac{U_{n,s,\omega}(\mathbf{x}) - \theta(\mathbf{x})}{\sqrt{s^2\,\zeta_{1,\omega}(\mathbf{x})/n}} \xrightarrow{d} \mathcal{N}(0,1). \tag{C.65}$$

By the H-decomposition, this limit transfers to the first projection $h^{\langle 1\rangle}_{n,s,x}(W)$; in our notation this equals the oracle linear statistic $L_{n,s}(\mathbf{x})$ introduced in Definition 3.2. Stacking coordinates and combining with the Bahadur representation in Equation (C.58),

$$\sqrt{n_{\mathrm{eff}}(\mathbf{x})}\big(\widehat{\alpha}(\mathbf{x}) - \alpha_0(\mathbf{x})\big) = \Gamma(\mathbf{x})^{-1} L_{n,s}(\mathbf{x}) + o_{\mathbb{P}}(1), \tag{C.66}$$

Slutsky and Cramér-Wold give the stated Gaussian limit with covariance $\Omega(\mathbf{x}) \coloneqq \Gamma(\mathbf{x})^{-1}\Sigma(\mathbf{x})\Gamma(\mathbf{x})^{-1}$, as recorded in Theorem 3.8. □

Appendix C.2. Proof of Theorems related to Variance Estimation

Proof of Proposition 3.9

Proof. Write the localized empirical moment as in Equation (3.28) and define its mean by

$$\widehat{m}_j(a;\mathbf{x}) \coloneqq \sum_{i=1}^{n} w_{s,i}(\mathbf{x}) \cdot \psi_j\left(W_i; a, \widehat{\eta}_j\left(Z_i^j\right)\right), \qquad \overline{m}_j(a;\mathbf{x}) \coloneqq \mathbb{E}[\widehat{m}_j(a;\mathbf{x})], \tag{C.67}$$

and let $\widehat{\boldsymbol{m}}(a;\mathbf{x})$ and $\overline{\boldsymbol{m}}(a;\mathbf{x})$ stack $j = 1,\dots,p_\alpha$. Recall the target covariance (cf. Equation (3.29))

$$\Sigma(\mathbf{x}) \coloneqq \mathrm{Var}\Big(\big(\tilde{\tau}_{s,\mathbf{x}} - \mathbb{E}[\tilde{\tau}_{s,\mathbf{x}}]\big)\,\boldsymbol{\psi}\left(W;\boldsymbol{\alpha}_0(\mathbf{x}),\boldsymbol{\eta}_0\right)\Big) = \lim_{n\to\infty} n_{\mathrm{eff}}(\mathbf{x})\ \mathrm{Var}(\widehat{\boldsymbol{m}}(\boldsymbol{\alpha}_0(\mathbf{x});\mathbf{x})). \tag{C.68}$$

Form delete-1 localized moments and pseudo-values as in Equations (3.30) and (3.31), and define the scaled jackknife meat estimator via Equation (3.32).

Step 1: Hájek Linearization.
Fix $\gamma \in \mathbb{R}^{p_\alpha}$ with $\|\gamma\|_2 = 1$, and write

$$\widehat{m}_\gamma(\boldsymbol{\alpha}_0;\mathbf{x}) \coloneqq \langle \gamma, \widehat{\boldsymbol{m}}(\boldsymbol{\alpha}_0;\mathbf{x})\rangle. \tag{C.69}$$

By Hájek dominance for the DNN kernel (Proposition 3.4), we have the uniform linearization

$$\widehat{m}_\gamma(\boldsymbol{\alpha}_0;\mathbf{x}) = \frac{1}{n}\sum_{i=1}^{n}\xi_{\gamma,i} + r_{\gamma,n}, \qquad \xi_{\gamma,i} \coloneqq s\cdot h_{\gamma,s}^{\langle 1\rangle}(W_i), \qquad \sup_{\|\gamma\|_2=1} |r_{\gamma,n}| = o_{\mathbb{P}}\Big(n_{\mathrm{eff}}(\mathbf{x})^{-1/2}\Big), \tag{C.70}$$

where $h_{\cdot,s}^{\langle 1\rangle}(W)$ is the first Hájek projection of the DNN L-form; see the display after Equation (3.29) and the sentence "Under Hájek dominance, $\mathrm{Var}(\widehat{\boldsymbol{m}}(\boldsymbol{\alpha}_0;\mathbf{x})) \asymp n_{\mathrm{eff}}(\mathbf{x})^{-1}$."

Step 2: Jackknife Reproduces the First Projection.
Form the delete-1 pseudo-values $P_{\gamma,i}(a) \coloneqq \langle \gamma, \boldsymbol{P}_i(a)\rangle$ using Equation (3.31), and set $a = \boldsymbol{\alpha}_0(\mathbf{x})$ (partial-oracle; nuisance fixed at $\widehat{\boldsymbol{\eta}}$). By Juergens (2025), under Hájek dominance with $s\to\infty$ and $s = o(n)$, the delete-1 jackknife variance reproduces the variance of the first projection uniformly over linear combinations. Concretely,

$$\max_{1\leqslant i\leqslant n}\sup_{\|\gamma\|_2=1}|P_{\gamma,i}^{\alpha\text{-orc}} - \xi_{\gamma,i}| = o_{\mathbb{P}}(1), \qquad \frac{1}{n}\sum_{i=1}^{n}\Big(P_{\gamma,i}^{\alpha\text{-orc}} - \overline{P}_{\gamma}^{\alpha\text{-orc}}\Big)^2 = \frac{1}{n}\sum_{i=1}^{n}\Big(\xi_{\gamma,i} - \overline{\xi}_{\gamma}\Big)^2 + o_{\mathbb{P}}\Big(n_{\mathrm{eff}}(\mathbf{x})^{-1}\Big), \tag{C.71}$$

and hence

$$\widehat{\Omega}_{\mathbf{x},\gamma}^{\mathrm{JK}} \coloneqq \langle \gamma, \widehat{\Omega}_{\mathbf{x}}^{\mathrm{JK}}\gamma\rangle = \frac{1}{n}\sum_{i=1}^{n}\Big(\xi_{\gamma,i} - \overline{\xi}_{\gamma}\Big)^2 + o_{\mathbb{P}}\Big(n_{\mathrm{eff}}(\mathbf{x})^{-1}\Big) \quad \text{uniformly in } \|\gamma\|_2 = 1. \tag{C.72}$$

The Steele L-form underlying the DNN weights guarantees that delete-1 updates are rank-wise and hence L-statistic-like.

Step 3: Keeping $\widehat{\eta}$ Fixed Across Deletes is Harmless.
By Condition 2.6 (ii),

$$\max_{1\leqslant i\leqslant n}\ \|\ \widehat{m}_j(\boldsymbol{\alpha}_{0,j}(\mathbf{x});\mathbf{x}) - \widehat{m}_j^{(-i)}(\boldsymbol{\alpha}_{0,j}(\mathbf{x});\mathbf{x})\ \|_{L^2(G)} = o_{\mathbb{P}}\Big(n_{\mathrm{eff}}(\mathbf{x})^{-1/2}\Big), \tag{C.73}$$

and, together with the Efron-Stein envelope for localized weighted averages (Appendix C), this implies that replacing fully LOO nuisances by the fixed full-sample $\widehat{\eta}_j$ perturbs pseudo-values by at most $o_{\mathbb{P}}\Big(n_{\mathrm{eff}}(\mathbf{x})^{-1/2}\Big)$, uniformly in γ. Thus Equations (C.71)(C.72) hold for the implementable pseudo-values with $\widehat{\eta}$ kept fixed.

Step 4: Quadratic-Form Convergence and Ratio-Consistency.
From Equation (C.70),

$$\operatorname{Var}(\widehat{m}_\gamma(\boldsymbol{\alpha}_0;\mathbf{x})) = \frac{1}{n}\operatorname{Var}(\xi_{\gamma,1}) + o_{\mathbb{P}}\Big(n_{\text{eff}}(\mathbf{x})^{-1}\Big). \tag{C.74}$$

Hence by Equation (C.68),

$$\gamma^\top \Sigma(\mathbf{x})\,\gamma = \lim_{n\to\infty} n_{\text{eff}}(\mathbf{x}) \cdot \frac{1}{n}\operatorname{Var}(\xi_{\gamma,1}). \tag{C.75}$$

Combining Equation (C.72) with a law of large numbers and Equation (C.75),

$$\gamma^\top \widehat{\Sigma}_{\mathbf{x}}\gamma = n_{\text{eff}}(\mathbf{x})\ \widehat{\Omega}^{\text{JK}}_{\mathbf{x},\gamma} = n_{\text{eff}}(\mathbf{x}) \cdot \frac{1}{n}\sum_{i=1}^{n}\Big(\xi_{\gamma,i} - \bar{\xi}_\gamma\Big)^2 + o_{\mathbb{P}}(1) \xrightarrow{\mathbb{P}} \gamma^\top \Sigma(\mathbf{x})\,\gamma, \tag{C.76}$$

uniformly over $\|\gamma\|_2 = 1$. Therefore,

$$\sup_{\|\gamma\|_2=1,\ \gamma^\top\Sigma(\mathbf{x})\gamma>0}\left|\frac{\gamma^\top\widehat{\Sigma}_{\mathbf{x}}\gamma}{\gamma^\top\Sigma(\mathbf{x})\,\gamma} - 1\right| \xrightarrow{\mathbb{P}} 0, \tag{C.77}$$

i.e. matrix ratio-consistency as in Definition 3.3. If $\Sigma(\mathbf{x}) \succ 0$, this is equivalent to

$$\Big\|\Sigma(\mathbf{x})^{-1/2}\,\widehat{\Sigma}_{\mathbf{x}}\,\Sigma(\mathbf{x})^{-1/2} - I\Big\|_{\text{op}} \xrightarrow{\mathbb{P}} 0. \tag{C.78}$$

Remarks on Inputs Used.

(i) The Hájek-dominance linearization Equation (C.70) matches the variance scale and the identification of $\Sigma(\mathbf{x})$ in Equation (C.68).

(ii) The L-statistic structure (Steele L-form) underpins Equations (C.71)(C.72); see Juergens (2025).

(iii) Local LOO stability, Condition 2.6, is used only to justify keeping $\widehat{\eta}_j$ fixed in Equation (3.30), via Equation (C.73) and the Efron-Stein envelope.

This proves the claim. □

Proof of Theorem 3.10
Proof. Let $\tilde{\boldsymbol{P}}_i(a) \coloneqq \boldsymbol{P}_i(a) - \overline{\boldsymbol{P}}(a)$ denote the centered jackknife pseudo-values defined in Equation (3.31), and recall

$$\widehat{\Omega}^{\text{JK}}_{\mathbf{x}}(a) = \frac{n-1}{n}\cdot\frac{1}{n}\sum_{i=1}^{n}\tilde{\boldsymbol{P}}_i(a)\ \tilde{\boldsymbol{P}}_i(a)^\top, \qquad \widehat{\Sigma}_{\mathbf{x}}(a) = n_{\text{eff}}(\mathbf{x})\ \widehat{\Omega}^{\text{JK}}_{\mathbf{x}}(a) \tag{C.79}$$

from Equation (3.32). We prove the two claims in Equation (3.34) in two steps.

Step 1: Plug-in Equivalence at the Level of Pseudo-Values.
Write $\Delta\boldsymbol{\alpha} \coloneqq \widehat{\boldsymbol{\alpha}}(\mathbf{x}) - \boldsymbol{\alpha}_0(\mathbf{x})$ and define

$$\Delta_i \ \coloneqq\ \tilde{\boldsymbol{P}}_i\big(\widehat{\boldsymbol{\alpha}}(\mathbf{x})\big)\ -\ \tilde{\boldsymbol{P}}_i\big(\boldsymbol{\alpha}_0(\mathbf{x})\big) \qquad (i = 1,\dots,n). \tag{C.80}$$

Expanding Equation (3.31),

$$\Delta_i = n\Big(\widehat{\boldsymbol{m}}(\widehat{\boldsymbol{\alpha}};\mathbf{x}) - \widehat{\boldsymbol{m}}(\boldsymbol{\alpha}_0;\mathbf{x})\Big) - (n-1)\Big(\widehat{\boldsymbol{m}}^{(-i)}(\widehat{\boldsymbol{\alpha}};\mathbf{x}) - \widehat{\boldsymbol{m}}^{(-i)}(\boldsymbol{\alpha}_0;\mathbf{x})\Big) - \Big(\overline{\boldsymbol{P}}(\widehat{\boldsymbol{\alpha}}) - \overline{\boldsymbol{P}}(\boldsymbol{\alpha}_0)\Big). \tag{C.81}$$

The last term centers across i and will cancel when we average. Subtract and add the full-sample increment in the second bracket to obtain

$$\Delta_i = (n-1)\left\{\big(\widehat{\boldsymbol{m}}(\widehat{\boldsymbol{\alpha}}) - \widehat{\boldsymbol{m}}(\boldsymbol{\alpha}_0)\big) - \big(\widehat{\boldsymbol{m}}^{(-i)}(\widehat{\boldsymbol{\alpha}}) - \widehat{\boldsymbol{m}}^{(-i)}(\boldsymbol{\alpha}_0)\big)\right\}\ -\ \Big(\overline{\boldsymbol{P}}(\widehat{\boldsymbol{\alpha}}) - \overline{\boldsymbol{P}}(\boldsymbol{\alpha}_0)\Big). \tag{C.82}$$

By Condition 2.5 and Lemma 2.4, the increments

$$\widehat{\boldsymbol{m}}(\widehat{\boldsymbol{\alpha}}) - \widehat{\boldsymbol{m}}(\boldsymbol{\alpha}_0) \quad \text{and} \quad \widehat{\boldsymbol{m}}^{(-i)}(\widehat{\boldsymbol{\alpha}}) - \widehat{\boldsymbol{m}}^{(-i)}(\boldsymbol{\alpha}_0) \tag{C.83}$$

are (uniformly in i) Lipschitz in $\Delta\boldsymbol{\alpha}$ with $L^2(G)$-envelopes bounded in probability. The difference between the full and delete-i increments in Equation (C.82) is therefore governed by the difference of weights $w - w^{(-i)}$. Applying the centered leave-one-out stability from Condition 2.6 (ii) to the array

$$Z_r \;\coloneqq\; \frac{\boldsymbol{\psi}\big(W_r; \boldsymbol{\alpha}_0 + t_r\Delta\boldsymbol{\alpha}, \widehat{\boldsymbol{\eta}}\big) - \boldsymbol{\psi}\big(W_r; \boldsymbol{\alpha}_0, \widehat{\boldsymbol{\eta}}\big)}{\|\,\Delta\boldsymbol{\alpha}\,\|}\; \mathbb{1}\{\,X_r \in G\,\}, \qquad t_r \in [0,1], \tag{C.84}$$

and using the local $L^2(G)$ Lipschitz bound from Condition 2.5, we obtain the uniform estimate

$$\max_{1\leqslant i\leqslant n}\; \sqrt{n_{\text{eff}}\,(\mathbf{x})}\; \Big\| \big(\widehat{\boldsymbol{m}}(\widehat{\boldsymbol{\alpha}}) - \widehat{\boldsymbol{m}}(\boldsymbol{\alpha}_0)\big) - \big(\widehat{\boldsymbol{m}}^{(-i)}(\widehat{\boldsymbol{\alpha}}) - \widehat{\boldsymbol{m}}^{(-i)}(\boldsymbol{\alpha}_0)\big) \Big\| = O_{\mathbb{P}}\big(\beta_n^{\text{loc}}(\mathbf{x})\,\|\,\Delta\boldsymbol{\alpha}\,\|\big). \tag{C.85}$$

Here $\beta_n^{\text{loc}}(\mathbf{x}) \to 0$ is the modulus from Condition 2.6. Because $\overline{\boldsymbol{P}}(\cdot)$ is the average over i, the centering term in Equation (C.82) is of the same order as the left-hand side of Equation (C.85) and thus obeys the same bound. Therefore,

$$\max_{1\leqslant i\leqslant n} \|\,\Delta_i\,\| = O_{\mathbb{P}}\left(\frac{\beta_n^{\text{loc}}(\mathbf{x})}{\sqrt{n_{\text{eff}}\,(\mathbf{x})}}\,\|\,\Delta\boldsymbol{\alpha}\,\|\right), \qquad \frac{1}{n}\sum_{i=1}^{n}\|\,\Delta_i\,\|^2 = O_{\mathbb{P}}\left(\frac{\big(\beta_n^{\text{loc}}(\mathbf{x})\big)^2}{n_{\text{eff}}\,(\mathbf{x})}\,\|\,\Delta\boldsymbol{\alpha}\,\|^2\right). \tag{C.86}$$

By Theorem 3.6, $\sqrt{n_{\text{eff}}\,(\mathbf{x})}\,\|\,\Delta\boldsymbol{\alpha}\,\| = O_{\mathbb{P}}(\,1\,)$. Hence

$$n_{\text{eff}}\,(\mathbf{x}) \cdot \frac{1}{n}\sum_{i=1}^{n}\|\,\Delta_i\,\|^2 = O_{\mathbb{P}}\Big(\,\big(\beta_n^{\text{loc}}(\mathbf{x})\big)^2\,\|\,\Delta\boldsymbol{\alpha}\,\|^2\,\Big) = o_{\mathbb{P}}(\,1\,), \tag{C.87}$$

since $\beta_n^{\text{loc}}(\mathbf{x}) \to 0$.

Step 2: From Pseudo-Values to the Covariance and Ratio-Consistency.
By the algebra of quadratic forms,

$$\widehat{\Omega}_{\mathbf{x}}^{\text{JK}}\big(\widehat{\boldsymbol{\alpha}}\big) - \widehat{\Omega}_{\mathbf{x}}^{\text{JK}}\big(\boldsymbol{\alpha}_0\big) = \frac{n-1}{n}\cdot\frac{1}{n}\sum_{i=1}^{n}\Big\{\tilde{\boldsymbol{P}}_i(\widehat{\boldsymbol{\alpha}})\;\tilde{\boldsymbol{P}}_i(\widehat{\boldsymbol{\alpha}})^{\top} - \tilde{\boldsymbol{P}}_i(\boldsymbol{\alpha}_0)\;\tilde{\boldsymbol{P}}_i(\boldsymbol{\alpha}_0)^{\top}\Big\}. \tag{C.88}$$

Using $\big\|\,uu^{\top} - vv^{\top}\,\big\|_{\text{op}} \leqslant \|u-v\|\,\big(\|u\| + \|v\|\big)$ and Cauchy-Schwarz,

$$\begin{aligned}\Big\|\,\widehat{\Omega}_{\mathbf{x}}^{\text{JK}}\big(\widehat{\boldsymbol{\alpha}}\big) - \widehat{\Omega}_{\mathbf{x}}^{\text{JK}}\big(\boldsymbol{\alpha}_0\big)\,\Big\|_{\text{op}} \;&\leqslant\; \frac{1}{n}\sum_{i=1}^{n}\|\,\Delta_i\,\|\,\big(\|\,\tilde{\boldsymbol{P}}_i(\widehat{\boldsymbol{\alpha}})\,\| + \|\,\tilde{\boldsymbol{P}}_i(\boldsymbol{\alpha}_0)\,\|\big)\\ &\leqslant\; \Big(\frac{1}{n}\sum_{i=1}^{n}\|\,\Delta_i\,\|^2\Big)^{1/2}\Big(\frac{1}{n}\sum_{i=1}^{n}\big(\|\,\tilde{\boldsymbol{P}}_i(\widehat{\boldsymbol{\alpha}})\,\| + \|\,\tilde{\boldsymbol{P}}_i(\boldsymbol{\alpha}_0)\,\|\big)^2\Big)^{1/2}.\end{aligned} \tag{C.89}$$

Under Conditions 2.3, 2.4 and 2.5 and Hájek dominance (cf. Step 2 in the proof of Proposition 3.9), the second factor is $O_{\mathbb{P}}(\,1\,)$. Combining with Equation (C.87) and multiplying by $n_{\text{eff}}\,(\mathbf{x})$,

$$\Big\|\,\widehat{\Sigma}_{\mathbf{x}}\big(\widehat{\boldsymbol{\alpha}}\big) - \widehat{\Sigma}_{\mathbf{x}}\big(\boldsymbol{\alpha}_0\big)\,\Big\|_{\text{op}} = n_{\text{eff}}\,(\mathbf{x})\;\Big\|\,\widehat{\Omega}_{\mathbf{x}}^{\text{JK}}\big(\widehat{\boldsymbol{\alpha}}\big) - \widehat{\Omega}_{\mathbf{x}}^{\text{JK}}\big(\boldsymbol{\alpha}_0\big)\,\Big\|_{\text{op}} = o_{\mathbb{P}}(\,1\,), \tag{C.90}$$

which proves the first claim in Equation (3.34). For the ratio-consistency, triangle inequality with Proposition 3.9 yields

$$\Big\|\,\widehat{\Sigma}_{\mathbf{x}}\big(\widehat{\boldsymbol{\alpha}}\big)\;\Sigma(\mathbf{x})^{-1} - I\,\Big\|_{\text{op}} \;\leqslant\; \Big\|\,\widehat{\Sigma}_{\mathbf{x}}\big(\widehat{\boldsymbol{\alpha}}\big) - \widehat{\Sigma}_{\mathbf{x}}\big(\boldsymbol{\alpha}_0\big)\,\Big\|_{\text{op}}\,\Big\|\,\Sigma(\mathbf{x})^{-1}\,\Big\|_{\text{op}} + \Big\|\,\widehat{\Sigma}_{\mathbf{x}}\big(\boldsymbol{\alpha}_0\big)\;\Sigma(\mathbf{x})^{-1} - I\,\Big\|_{\text{op}} \;\xrightarrow{\;\mathbb{P}\;}\; 0. \tag{C.91}$$

This establishes both conclusions in Equation (3.34). □

Proof of Proposition 3.11

Proof. Fix $k \in [p_\alpha]$ and define the population central finite difference

$$\mathbf{\Gamma}_{\mathbf{x}}^{(h)} \coloneqq \left[\frac{\overline{\boldsymbol{m}}\big(\boldsymbol{\alpha}_0(\mathbf{x}) + h\, e_k; \mathbf{x}\big) - \overline{\boldsymbol{m}}\big(\boldsymbol{\alpha}_0(\mathbf{x}) - h\, e_k; \mathbf{x}\big)}{2h} \right]_{k=1}^{p_\alpha}, \tag{C.92}$$

where $\overline{\boldsymbol{m}}(a; \mathbf{x}) \coloneqq \mathbb{E}[\widehat{\boldsymbol{m}}(a; \mathbf{x})]$ as in Equation (3.37). For the kth column, decompose

$$\widehat{\Gamma}_{\cdot k}\big(\boldsymbol{\alpha}_0; \mathbf{x}\big) - \Gamma_{\cdot k}(\mathbf{x}) \;=\; \underbrace{\Big(\widehat{\Gamma}_{\cdot k}\big(\boldsymbol{\alpha}_0; \mathbf{x}\big) - \mathbb{E}\Big[\widehat{\Gamma}_{\cdot k}\big(\boldsymbol{\alpha}_0; \mathbf{x}\big)\Big]\Big)}_{\text{stochastic}} \;+\; \underbrace{\Big(\mathbb{E}\Big[\widehat{\Gamma}_{\cdot k}\big(\boldsymbol{\alpha}_0; \mathbf{x}\big)\Big] - \Gamma_{\cdot k}^{(h)}(\mathbf{x})\Big)}_{=\,0} \;+\; \underbrace{\Big(\Gamma_{\cdot k}^{(h)}(\mathbf{x}) - \Gamma_{\cdot k}(\mathbf{x})\Big)}_{\text{bias}}. \tag{C.93}$$

The middle term is zero by Equation (C.92). We bound the bias and stochastic parts.

Bias: Deterministic FD Truncation.

Let $f_j(a) \coloneqq m_j(\mathbf{x}; a, \eta_{0,j})$. By Condition 2.5 and the symmetric-difference expansion (Equation (2.43)),

$$\Big\| \Gamma_{\cdot k}^{(h)}(\mathbf{x}) - \Gamma_{\cdot k}(\mathbf{x}) \Big\|_2 \;=\; O\left(h^2\right) \qquad \text{uniformly over } k \in [p_\alpha]. \tag{C.94}$$

Stochastic Term: FDLSE at the Oracle Center.

By the finite-difference localized stochastic equicontinuity block (Condition 2.6 (iii)), evaluated at $\boldsymbol{\alpha}_0(\mathbf{x}) \pm h\, e_k$ with the nuisance fixed at $\widehat{\boldsymbol{\eta}}$,

$$\sqrt{n_{\text{eff}}\,(\mathbf{x})}\, \Big\| \widehat{\Gamma}_{\cdot k}\big(\boldsymbol{\alpha}_0; \mathbf{x}\big) - \mathbb{E}\Big[\widehat{\Gamma}_{\cdot k}\big(\boldsymbol{\alpha}_0; \mathbf{x}\big)\Big] \Big\| \;=\; O_{\mathbb{P}}(1) \;+\; O_{\mathbb{P}}\Big(\sqrt{n_{\text{eff}}\,(\mathbf{x})}\, \beta_n^{\text{loc}}\,(\mathbf{x})\,/h \Big). \tag{C.95}$$

Hence

$$\Big\| \widehat{\Gamma}_{\cdot k}\big(\boldsymbol{\alpha}_0; \mathbf{x}\big) - \mathbb{E}\Big[\widehat{\Gamma}_{\cdot k}\big(\boldsymbol{\alpha}_0; \mathbf{x}\big)\Big] \Big\| \;=\; O_{\mathbb{P}}\Big(n_{\text{eff}}\,(\mathbf{x})^{-1/2} \Big) \;+\; O_{\mathbb{P}}\big(\beta_n^{\text{loc}}\,(\mathbf{x})\,/h\big). \tag{C.96}$$

Collect and Stack.

Combining Equation (C.94) and Equation (C.96) in Equation (C.93),

$$\Big\| \widehat{\Gamma}_{\cdot k}\big(\boldsymbol{\alpha}_0; \mathbf{x}\big) - \Gamma_{\cdot k}(\mathbf{x}) \Big\|_2 \;=\; O_{\mathbb{P}}\Big(n_{\text{eff}}\,(\mathbf{x})^{-1/2} \Big) \;+\; O_{\mathbb{P}}\big(h^2\big) \;+\; O_{\mathbb{P}}\big(\beta_n^{\text{loc}}\,(\mathbf{x})\,/h\big), \tag{C.97}$$

uniformly in $k \in [p_\alpha]$. Since $\|\cdot\|_{\text{op}} \leqslant \max_k \|\cdot_{\cdot k}\|_2$, stacking gives

$$\Big\| \widehat{\mathbf{\Gamma}}_{\mathbf{x}}\big(\boldsymbol{\alpha}_0(\mathbf{x})\big) - \mathbf{\Gamma}_{\mathbf{x}} \Big\|_{\text{op}} \;=\; O_{\mathbb{P}}\Big(n_{\text{eff}}\,(\mathbf{x})^{-1/2} \Big) \;+\; O_{\mathbb{P}}\big(h^2\big) \;+\; O_{\mathbb{P}}\big(\beta_n^{\text{loc}}\,(\mathbf{x})\,/h\big), \tag{C.98}$$

which matches Equation (3.39). Under Equation (3.38) these terms vanish, so $\widehat{\mathbf{\Gamma}}_{\mathbf{x}}\big(\boldsymbol{\alpha}_0(\mathbf{x})\big) \xrightarrow{\mathbb{P}} \mathbf{\Gamma}_{\mathbf{x}}$. □

Proof of Theorem 3.12

Proof. Decompose the plug-in estimator into a center-shift term plus the α-oracle term:

$$\widehat{\mathbf{\Gamma}}_{\mathbf{x}}\big(\widehat{\boldsymbol{\alpha}}(\mathbf{x})\big) - \mathbf{\Gamma}_{\mathbf{x}} \;=\; \underbrace{\Big(\widehat{\mathbf{\Gamma}}_{\mathbf{x}}\big(\widehat{\boldsymbol{\alpha}}(\mathbf{x})\big) - \widehat{\mathbf{\Gamma}}_{\mathbf{x}}\big(\boldsymbol{\alpha}_0(\mathbf{x})\big)\Big)}_{\text{plug-in center shift}} \;+\; \underbrace{\Big(\widehat{\mathbf{\Gamma}}_{\mathbf{x}}\big(\boldsymbol{\alpha}_0(\mathbf{x})\big) - \mathbf{\Gamma}_{\mathbf{x}}\Big)}_{\alpha\text{-oracle}}, \tag{C.99}$$

and control the two brackets separately.

Step 1: Plug-in Equivalence.
Write $\Delta\boldsymbol{\alpha} \coloneqq \widehat{\boldsymbol{\alpha}}(\mathbf{x}) - \boldsymbol{\alpha}_0(\mathbf{x})$. For each column $k \in [p_\alpha]$, expand via Equation (3.35):

$$\begin{aligned}\widehat{\Gamma}_{\cdot k}\big(\widehat{\boldsymbol{\alpha}};\mathbf{x}\big) - \widehat{\Gamma}_{\cdot k}\big(\boldsymbol{\alpha}_0;\mathbf{x}\big) = \frac{1}{2h}\Big\{&\big[\widehat{\boldsymbol{m}}(\widehat{\boldsymbol{\alpha}} + he_k;\mathbf{x}) - \widehat{\boldsymbol{m}}(\boldsymbol{\alpha}_0 + he_k;\mathbf{x})\big] \\ &- \big[\widehat{\boldsymbol{m}}(\widehat{\boldsymbol{\alpha}} - he_k;\mathbf{x}) - \widehat{\boldsymbol{m}}(\boldsymbol{\alpha}_0 - he_k;\mathbf{x})\big]\Big\}.\end{aligned} \tag{C.100}$$

Apply the mean value theorem along the segment $\boldsymbol{\alpha}_0 \to \widehat{\boldsymbol{\alpha}}$ to each bracket: there exist $t_\pm \in [0,1]$ such that

$$\frac{\widehat{\boldsymbol{m}}(\widehat{\boldsymbol{\alpha}} \pm he_k;\mathbf{x}) - \widehat{\boldsymbol{m}}(\boldsymbol{\alpha}_0 \pm he_k;\mathbf{x})}{2h} = \frac{1}{2}\,\nabla_a\widehat{\boldsymbol{m}}\big(\boldsymbol{\alpha}_0 \pm he_k + t_\pm\Delta\boldsymbol{\alpha};\mathbf{x}\big)\,\Delta\boldsymbol{\alpha}, \tag{C.101}$$

where $\nabla_a\widehat{\boldsymbol{m}}$ is the sample Jacobian in a. Subtracting the $+$ and $-$ versions in Equation (C.101) and using the local smoothness of the moment map from Condition 2.5 together with the finite-difference LSE block (Equation (2.51)) yields the uniform bound

$$\Big\|\,\widehat{\Gamma}_{\cdot k}\big(\widehat{\boldsymbol{\alpha}};\mathbf{x}\big) - \widehat{\Gamma}_{\cdot k}\big(\boldsymbol{\alpha}_0;\mathbf{x}\big)\,\Big\| \;\leqslant\; C_{\mathbf{x}}\,\|\,\Delta\boldsymbol{\alpha}\,\| \;+\; O_{\mathbb{P}}\!\left(\frac{\beta_n^{\mathrm{loc}}(\mathbf{x})}{h}\,\|\,\Delta\boldsymbol{\alpha}\,\|\right), \tag{C.102}$$

for some constant $C_{\mathbf{x}}$ not depending on n or k. By the local Bahadur rate from Theorem 3.6, $\sqrt{n_{\mathrm{eff}}(\mathbf{x})}\,\|\,\Delta\boldsymbol{\alpha}\,\| = O_{\mathbb{P}}(1)$. Combining with Equation (3.38) (so that $\beta_n^{\mathrm{loc}}(\mathbf{x})/h \to 0$) and stacking columns,

$$\Big\|\,\widehat{\boldsymbol{\Gamma}}_{\mathbf{x}}\big(\widehat{\boldsymbol{\alpha}}(\mathbf{x})\big) - \widehat{\boldsymbol{\Gamma}}_{\mathbf{x}}\big(\boldsymbol{\alpha}_0(\mathbf{x})\big)\,\Big\|_{\mathrm{op}} \;=\; o_{\mathbb{P}}(1), \tag{C.103}$$

which is Equation (3.40).

Step 2: Add the α-oracle Rate and Conclude..
By Proposition 3.11,

$$\Big\|\,\widehat{\boldsymbol{\Gamma}}_{\mathbf{x}}\big(\boldsymbol{\alpha}_0(\mathbf{x})\big) - \boldsymbol{\Gamma}_{\mathbf{x}}\,\Big\|_{\mathrm{op}} \;=\; O_{\mathbb{P}}\Big(n_{\mathrm{eff}}(\mathbf{x})^{-1/2}\Big) + O_{\mathbb{P}}\big(h^2\big) + O_{\mathbb{P}}\big(\beta_n^{\mathrm{loc}}(\mathbf{x})/h\big). \tag{C.104}$$

Applying the triangle inequality to Equation (C.99) with Equation (C.103) and Equation (C.104) yields

$$\Big\|\,\widehat{\boldsymbol{\Gamma}}_{\mathbf{x}} - \boldsymbol{\Gamma}_{\mathbf{x}}\,\Big\|_{\mathrm{op}} \;=\; O_{\mathbb{P}}\Big(n_{\mathrm{eff}}(\mathbf{x})^{-1/2}\Big) + O_{\mathbb{P}}\big(h^2\big) + O_{\mathbb{P}}\big(\beta_n^{\mathrm{loc}}(\mathbf{x})/h\big) + o_{\mathbb{P}}(1), \tag{C.105}$$

which is exactly Equation (3.41). In particular, Equation (3.38) implies $\widehat{\boldsymbol{\Gamma}}_{\mathbf{x}} \xrightarrow{\mathbb{P}} \boldsymbol{\Gamma}_{\mathbf{x}}$. □

Proof of Theorem 3.14
Proof. Write $A \coloneqq \boldsymbol{\Gamma}_{\mathbf{x}}$, $\widehat{A} \coloneqq \widehat{\boldsymbol{\Gamma}}_{\mathbf{x}}$, $S \coloneqq \Sigma(\mathbf{x})$, and $\widehat{S} \coloneqq \widehat{\Sigma}_{\mathbf{x}}$, so $\Omega = A^{-1}SA^{-T}$ and $\widehat{\Omega} = \widehat{A}^{-1}\widehat{S}\widehat{A}^{-T}$. By Theorem 3.12, $\big\|\widehat{A} - A\big\|_{\mathrm{op}} \xrightarrow{\mathbb{P}} 0$ and, since A is invertible by Condition 2.5 (v), $\widehat{A}$ is invertible with probability tending to one and $\widehat{A}^{-1} \xrightarrow{\mathbb{P}} A^{-1}$ by continuity of inversion on GL_{p_α}. By Theorem 3.10 and Definition 3.3,

$$\big\|\;S^{-1/2}\widehat{S}S^{-1/2} - I_{p_\alpha}\;\big\|_{\mathrm{op}} \xrightarrow{\mathbb{P}} 0 \quad\Longrightarrow\quad \big\|\,\widehat{S} - S\,\big\|_{\mathrm{op}} \leqslant \|\,S\,\|_{\mathrm{op}}\,\big\|\;S^{-1/2}\widehat{S}S^{-1/2} - I_{p_\alpha}\;\big\|_{\mathrm{op}} \xrightarrow{\mathbb{P}} 0. \tag{C.106}$$

Hence

$$\begin{aligned}\big\|\,\widehat{\Omega} - \Omega\,\big\|_{\mathrm{op}} \leqslant{}& \big\|\,\widehat{A}^{-1} - A^{-1}\,\big\|_{\mathrm{op}}\,\big\|\,\widehat{S}\,\big\|_{\mathrm{op}}\,\big\|\,\widehat{A}^{-T}\,\big\|_{\mathrm{op}} + \big\|\,A^{-1}\,\big\|_{\mathrm{op}}\,\big\|\,\widehat{S} - S\,\big\|_{\mathrm{op}}\,\big\|\,\widehat{A}^{-T}\,\big\|_{\mathrm{op}} \\ &+ \big\|\,A^{-1}\,\big\|_{\mathrm{op}}\,\|\,S\,\|_{\mathrm{op}}\,\big\|\,\widehat{A}^{-T} - A^{-T}\,\big\|_{\mathrm{op}} \;\xrightarrow{\mathbb{P}}\; 0,\end{aligned} \tag{C.107}$$

where boundedness in probability of the operator norms follows from $\widehat{A}^{-1} \xrightarrow{\mathbb{P}} A^{-1}$ and Equation (C.106). Since Ω is positive definite (A invertible by Condition 2.5 (v) and S positive definite by construction), we have

$$\Big\|\,\Omega^{-1/2}\widehat{\Omega}\,\Omega^{-1/2} - I_{p_\alpha}\,\Big\|_{\mathrm{op}} \;=\; \Big\|\,\Omega^{-1/2}\big(\widehat{\Omega} - \Omega\big)\Omega^{-1/2}\,\Big\|_{\mathrm{op}} \;\leqslant\; \big\|\,\Omega^{-1/2}\,\big\|_{\mathrm{op}}^2\,\Big\|\,\widehat{\Omega} - \Omega\,\Big\|_{\mathrm{op}} \xrightarrow{\mathbb{P}} 0, \tag{C.108}$$

which is Equation (3.49). Finally, for any fixed $J \subset [p_\alpha]$, the operator norm of a principal submatrix is bounded by that of the full matrix, so Equation (3.49) implies the same ratio-consistency for $\widehat{\Omega}(\mathbf{x})_{JJ}$. □

Appendix C.3. Proof of Theorems related to Inference

Proof of Theorem 3.15

Proof. By Theorem 3.8, the localized estimator satisfies the vector CLT

$$\sqrt{n_{\text{eff}}(\mathbf{x})}\,\big(\widehat{\boldsymbol{\alpha}}(\mathbf{x}) - \boldsymbol{\alpha}_0(\mathbf{x})\big) \xrightarrow{d} \mathcal{N}\big(0, \Omega(\mathbf{x})\big), \quad \Omega(\mathbf{x}) \coloneqq \Gamma(\mathbf{x})^{-1}\Sigma(\mathbf{x})\Gamma(\mathbf{x})^{-1}. \tag{C.109}$$

Let $\widehat{\Omega}(\mathbf{x})$ be the localized sandwich from Equation (3.48). By Theorem 3.14 we have matrix ratio-consistency,

$$\Big\| \Omega(\mathbf{x})^{-1/2}\,\widehat{\Omega}(\mathbf{x})\,\Omega(\mathbf{x})^{-1/2} - I_{p_\alpha} \Big\|_{\text{op}} \xrightarrow{\mathbb{P}} 0. \tag{C.110}$$

Step 1: Linear Contrasts.
For any fixed $\gamma \in \mathbb{R}^{p_\alpha}$, Equation (C.109) yields

$$\sqrt{n_{\text{eff}}(\mathbf{x})}\,\gamma^\top\big(\widehat{\boldsymbol{\alpha}}(\mathbf{x}) - \boldsymbol{\alpha}_0(\mathbf{x})\big) \xrightarrow{d} \mathcal{N}\big(0, \text{Var}_\gamma(\mathbf{x})\big), \quad \text{Var}_\gamma(\mathbf{x}) \coloneqq \gamma^\top\Omega(\mathbf{x})\,\gamma. \tag{C.111}$$

By Equation (C.110), $\widehat{\Omega}_\gamma(\mathbf{x}) \coloneqq \gamma^\top\widehat{\Omega}(\mathbf{x})\,\gamma$ satisfies $\widehat{\Omega}_\gamma(\mathbf{x})/\,\text{Var}_\gamma(\mathbf{x}) \xrightarrow{\mathbb{P}} 1$. Define the estimator-scale variance proxy $\widehat{\text{Var}}_\gamma(\mathbf{x}) \coloneqq \widehat{\Omega}_\gamma(\mathbf{x})/n_{\text{eff}}(\mathbf{x})$. Then Slutskys theorem gives

$$\frac{\sqrt{n_{\text{eff}}(\mathbf{x})}\,\gamma^\top\big(\widehat{\boldsymbol{\alpha}}(\mathbf{x}) - \boldsymbol{\alpha}_0(\mathbf{x})\big)}{\sqrt{\widehat{\Omega}_\gamma(\mathbf{x})}} \xrightarrow{d} \mathcal{N}(0,1), \tag{C.112}$$

and the stated Wald interval is asymptotically exact.

Step 2: Pointwise CIs.
Taking $\gamma = e_j$ to apply the result from step 1 and using $\text{se}_j(\mathbf{x}) \coloneqq \sqrt{[\widehat{\Omega}(\mathbf{x})]_{jj}/n_{\text{eff}}(\mathbf{x})}$ from Equation (3.48) yields

$$\mathbb{P}\big(\alpha_{0,j}(\mathbf{x}) \in \big[\widehat{\alpha}_j(\mathbf{x}) \pm z_{1-\delta/2}\,\text{se}_j(\mathbf{x})\big]\big) \to 1 - \delta. \tag{C.113}$$

Step 3: Linear Restrictions.
From Equation (C.109) and the continuous mapping theorem,

$$\sqrt{n_{\text{eff}}(\mathbf{x})}\,A\big(\widehat{\boldsymbol{\alpha}}(\mathbf{x}) - \boldsymbol{\alpha}_0(\mathbf{x})\big) \xrightarrow{d} \mathcal{N}\big(0,\, A\Omega(\mathbf{x})A^\top\big). \tag{C.114}$$

By Equation (C.110) and congruence with fixed full-row-rank A,

$$\Big\| (A\Omega(\mathbf{x})A^\top)^{-1/2}\,(A\widehat{\Omega}(\mathbf{x})A^\top)\,(A\Omega(\mathbf{x})A^\top)^{-1/2} - I_q \Big\|_{\text{op}} \xrightarrow{\mathbb{P}} 0. \tag{C.115}$$

Slutsky then implies

$$W_n \coloneqq n_{\text{eff}}(\mathbf{x})\;\big(A\widehat{\boldsymbol{\alpha}}(\mathbf{x}) - b\big)^\top\Big(A\widehat{\Omega}(\mathbf{x})A^\top\Big)^{-1}\big(A\widehat{\boldsymbol{\alpha}}(\mathbf{x}) - b\big) \xrightarrow{d} \chi^2_q. \tag{C.116}$$

This completes the proof. □

Appendix D. Proofs of Lemmas in the Main Text

Appendix D.1. Proofs of Lemmas in Section 2

Proof of Lemma 2.1

Proof.

Step 1: Existence and Finiteness. Fix an arbitrary $j \in [p_\alpha]$, $a \in \mathcal{A}_j(\mathbf{x})$ and $\tilde{\eta} \in \mathcal{H}_j \cup \{\eta_{0,j}\}$. First, observe that by Condition 2.4 (i), each map $w \mapsto \psi_j(w, a, \tilde{\eta})$ is measurable. By Condition 2.4 (ii) and Hölder's inequality, we can make the following observation.

$$\begin{aligned} \left| m_j(u;\ a, \tilde{\eta}) \right| &= \left| \mathbb{E}\left[\psi_j\left(W, a, \tilde{\eta}\left(Z^j\right)\right) \mid X = u \right] \right| \leqslant \mathbb{E}\left[\left| \psi_j\left(W, a, \tilde{\eta}\left(Z^j\right)\right) \right| \mid X = u \right] \\ &\leqslant \mathbb{E}\left[\left| F(W) \right| \mid X = u \right] \leqslant \mathbb{E}\left[\left| F(W) \right|^{\mathfrak{q}} \,\middle|\, X = u \right]^{1/\mathfrak{q}} \\ &\leqslant \left(\sup_{u \in \mathcal{X}} \mathbb{E}\left[\left| F(W) \right|^{\mathfrak{q}} \,\middle|\, X = u \right] \right)^{1/\mathfrak{q}} \leqslant C_1^{1/\mathfrak{q}} \end{aligned} \tag{D.1}$$

Thus, the conditional moment exists and is finite for P_X-almost every $u \in \mathcal{X}$.

Step 2: Measurability in u. Recall that $\mathcal{X}$ is a Borel subset of $\mathbb{R}^{k_x}$ and that by Condition 2.3, X has a density in $\mathcal{B}_{\mathbf{x}}$. Thus, for any $u \in \mathcal{B}_{\mathbf{x}}$ there exists a regular conditional distribution $P_{W \mid X=u}$. This allows us to choose the version

$$u \mapsto m_j(u;\ a, \tilde{\eta}) = \int \psi_j(w, a, \tilde{\eta})\, P_{W \mid X=u}(\,\mathrm{d}w) \tag{D.2}$$

to be Borel-measurable in u.

Step 3: Measurability in $(a, \tilde{\eta})$ for P_X-a.e. u. By Condition 2.4 (i), $\mathcal{F}_j$ is pointwise measurable. Thus, standard measurable envelope arguments imply that for P_X-almost every $u \in \mathcal{B}_{\mathbf{x}}$, the map

$$(a, \tilde{\eta}) \mapsto \int \psi_j(w, a, \tilde{\eta})\, P_{W \mid X=u}(\,\mathrm{d}w) \tag{D.3}$$

is Borel measurable. The essential step in this argument is to use the envelope that exists by Condition 2.4 (ii) to justify the use of the dominated convergence theorem. □

Proof of Lemma 2.2

Proof. Fix $j \in [p_\alpha]$ and any u in the stated neighborhoods. Throughout we use that all Lipschitz/smoothness statements below hold uniformly over the corresponding local balls from Condition 2.5.

Proof of Lemma 2.2 (i). By Condition 2.5 (ii), for all $\tilde{\eta}, \tilde{\eta}' \in \mathcal{B}_{\eta,j}$ and $u \in \mathcal{B}_{\mathbf{x},j}^{(4)}$,

$$\left| m_j(u;\ \alpha_{0,j}(\mathbf{x}), \tilde{\eta}) - m_j(u;\ \alpha_{0,j}(\mathbf{x}), \tilde{\eta}') \right| \leqslant L_{\eta,j} \left\| \tilde{\eta} - \tilde{\eta}' \right\|_\eta \quad \text{(uniformly in } u\text{)}. \tag{D.4}$$

Squaring and taking the (conditional) L_2 norm yields Equation (2.33) with constant $L_{\eta,j}^2$.

Proof of Lemma 2.2 (ii). Condition 2.5 (iv) gives the operator-Lipschitz bound

$$\left\| Dm_j(u;\ a, \tilde{\eta}) - Dm_j(u;\ a', \tilde{\eta}') \right\|_{\mathrm{op}} \leqslant L_{v,j}\left(\left| a - a' \right| + \left\| \tilde{\eta} - \tilde{\eta}' \right\|_\eta \right) \quad \text{for } u \in \mathcal{B}_{\mathbf{x},j}^{(6)}, \tag{D.5}$$

with $Dm_j(u;\ a, \tilde{\eta})[\Delta_\alpha, \Delta_\eta] = \partial_\alpha m_j(u;\ a, \tilde{\eta})\ \Delta_\alpha + \partial_\eta m_j(u;\ a, \tilde{\eta})[\Delta_\eta]$ by (3.11). Evaluating the operator difference at $(\Delta_\alpha, \Delta_\eta) = (1, 0)$ (which has product norm 1) gives

$$\left| \partial_\alpha m_j(u;\ a, \tilde{\eta}) - \partial_\alpha m_j(u;\ a', \tilde{\eta}') \right| \leqslant L_{v,j}\left(\left| a - a' \right| + \left\| \tilde{\eta} - \tilde{\eta}' \right\|_\eta \right). \tag{D.6}$$

Setting $a = a' = \alpha_{0,j}(\mathbf{x})$ yields the pointwise bound in $\tilde{\eta}$, and squaring gives Equation (2.34).

Proof of Lemma 2.2 (iii). By Condition 2.5 (i), for all $a, a' \in \mathcal{B}_{\alpha,j}$ and $u \in \mathcal{B}^{(3)}_{\mathbf{x},j}$,

$$\left| m_j\left(u;\ a, \eta_{0,j}\right) - m_j\left(u;\ a', \eta_{0,j}\right) \right| \leqslant L_{\alpha,j} \left| a - a' \right|. \tag{D.7}$$

Squaring yields Equation (2.35).

Proof of Lemma 2.2 (iv). For $(a, \tilde{\eta}), (a', \tilde{\eta}') \in \mathcal{O}_j$ and $u \in \mathcal{B}^{(3)}_{\mathbf{x},j} \cap \mathcal{B}^{(4)}_{\mathbf{x},j}$, apply the triangle inequality together with Condition 2.5 (i) and (ii):

$$\begin{aligned} \left| m_j\left(u;\ a, \tilde{\eta}\right) - m_j\left(u;\ a', \tilde{\eta}'\right) \right| &\leqslant \left| m_j\left(u;\ a, \tilde{\eta}\right) - m_j\left(u;\ a', \tilde{\eta}\right) \right| + \left| m_j\left(u;\ a', \tilde{\eta}\right) - m_j\left(u;\ a', \tilde{\eta}'\right) \right| \\ &\leqslant L_{\alpha,j} \left| a - a' \right| + L_{\eta,j} \left\| \tilde{\eta} - \tilde{\eta}' \right\|_{\eta}. \end{aligned} \tag{D.8}$$

Squaring gives Equation (2.36).

Proof of Lemma 2.2 (v). By Condition 2.5 (iii), for $u, u' \in \mathcal{B}^{(5)}_{\mathbf{x},j}$,

$$\left| m_j\left(u;\ \alpha_{0,j}(\mathbf{x}), \eta_{0,j}\right) - m_j\left(u'; \alpha_{0,j}(\mathbf{x}), \eta_{0,j}\right) \right| \leqslant L_{u,j} \left\| u - u' \right\|_2, \tag{D.9}$$

and squaring yields Equation (2.37).

Restriction to Measurable Subsets U. For any measurable $U \subset \mathcal{B}^{(k)}_{\mathbf{x},j}$, multiplying the integrands by $\mathbb{1}\{ X \in U \}$ and integrating over u preserves all displays because

$$\int_U \mathbb{E}\left[\{\cdot\}^2 \,\middle|\, X = u \right] \, \mathrm{d}P_X(u) \leqslant C \int_U \mathrm{d}P_X(u) = C\, \mathbb{P}(X \in U), \tag{D.10}$$

so the right-hand sides pick up the factor $\mathbb{P}(X \in U)$ as stated in Lemma 2.2. □

Proof of Lemma 2.3
Proof.

Step 1: L-form, Monotonicity, and the Maximal Weight.
Order the sample by increasing distance to $\mathbf{x}$ as in Equation (2.16). In the L-form representation Equation (2.17), the weight attached to the i-th nearest neighbor is

$$w_{s,(i)}\left(\mathbf{x}\right) = \frac{s}{n} \prod_{t=0}^{i-2} \frac{n - s - t}{n - 1 - t}, \qquad i = 1, 2, \ldots, n - s + 1, \tag{D.11}$$

with $w_{s,(i)}\left(\mathbf{x}\right) = 0$ for $i > n - s + 1$. Hence $i \mapsto w_{s,(i)}(\mathbf{x})$ is nonincreasing and $w_{s,(1)}(\mathbf{x}) = s/n$. Since $w_i(D; \mathbf{x})$ is obtained by assigning $w_{s,(r)}(\mathbf{x})$ according to the rank $r = \mathrm{rk}(\mathbf{x}; X_i, D)$, we have for every D,

$$\max_{i \in [n]} | \, w_i(D; \mathbf{x}) \, | = \max_r w_{s,(r)}(\mathbf{x}) = \frac{s}{n}. \tag{D.12}$$

By Proposition 3.1, $n_{\mathrm{eff}}\left(\mathbf{x}\right) = \left(\sum_{i=1}^n w_i(D; \mathbf{x})^2 \right)^{-1} \asymp n/s$ with probability tending to one; thus $s/n = n_{\mathrm{eff}}\left(\mathbf{x}\right)^{-1} = o_{\mathbb{P}}\left(n_{\mathrm{eff}}\left(\mathbf{x}\right)^{-1/2} \right)$ under $s/n \to 0$, proving (i).

Step 2: Replace-one L^1 modulus for L-Statistic Weights.
Fix D and $l \in [n]$, and let $r_0 = \mathrm{rk}(\mathbf{x}; X_l, D)$ and $r_0' = \mathrm{rk}(\mathbf{x}; X_l', D^{(l)})$ denote the ranks of the replaced point before/after replacement. The L-form shows that replacing one observation only permutes the fixed nonincreasing sequence $\{w_{s,(r)}(\mathbf{x})\}_r$ across labels, with the following structure: all indices i with ranks strictly between r_0 and r_0' shift by exactly one place, while all other indices keep their rank. Therefore,

$$\sum_{i=1}^{n} | \, w_i(D;\mathbf{x}) - w_i(D^{(l)};\mathbf{x}) \, | = 2 \sum_{r=\min(r_0,r_0')}^{\max(r_0,r_0')-1} \big(w_{s,(r)}(\mathbf{x}) - w_{s,(r+1)}(\mathbf{x}) \big) \leqslant 2 \, w_{s,(1)}(\mathbf{x}) = \frac{2s}{n}, \tag{D.13}$$

since the sum telescopes and $r \mapsto w_{s,(r)}(\mathbf{x})$ is nonincreasing with limit 0. Multiplying by the indicator $\mathbb{1}\{ X_i \in \mathcal{B}_{\mathbf{x}} \}$ can only decrease the sum, so the same bound holds for the localized modulus. Taking the supremum over D and l preserves this deterministic bound, proving $\beta_n^{\mathrm{loc}}(\mathbf{x}) \leqslant 2s/n$.

Step 3: Scaling by $\sqrt{n_{\mathrm{eff}}(\mathbf{x})}$.
By Proposition 3.1, $n_{\mathrm{eff}}(\mathbf{x}) \asymp n/s$ with probability $\to 1$. Hence

$$\sqrt{n_{\mathrm{eff}}(\mathbf{x})} \, \beta_n^{\mathrm{loc}}(\mathbf{x}) \leqslant C \, \sqrt{n/s} \cdot (2s/n) = 2C \, \sqrt{s/n} \longrightarrow 0 \tag{D.14}$$

under $s/n \to 0$. This establishes (ii). □

Proof of Lemma 2.4
Proof.

Step 1: Setup and MSC Size Control. For $\theta = (a, \tilde{\eta})$ with $\| \theta - \theta_0 \| \leqslant \delta_n$, define

$$f_\theta(w) \coloneqq \big(\psi_j(w; a, \tilde{\eta}) - \psi_j(w; a, \eta_{0,j}) \big) \, \mathbb{1}\{ X \in G \}. \tag{D.15}$$

By Lemma 2.2 (iv), uniformly on G,

$$\| f_\theta \|_{L_2(G)} \leqslant L_{\alpha,j} \, \big| a - \alpha_{0,j}(\mathbf{x}) \big| + L_{\eta,j} \, \big\| \tilde{\eta} - \eta_{0,j} \big\|_\eta \lesssim \delta_n \xrightarrow[n\to\infty]{} 0. \tag{D.16}$$

For replace-one datasets $D^{(\ell)}$ from Condition 2.6, let $w^{(\ell)}$ be the associated weights and let $\beta_n^{\mathrm{loc}}(\mathbf{x})$ be the localized replace-one modulus defined in Equation (2.38).

Step 2: Efron-Stein Variance Envelope. By Corollary Appendix E.3, for any square-integrable f supported on G,

$$\mathrm{Var}(\mathbb{G}_{n,w}[f]) \lesssim \Big\{ n_{\mathrm{eff}}(\mathbf{x}) \; \mathbb{E}\Big[\max_{i\in[n]} w_i^2(\mathbf{x}) \Big] \; + \; n_{\mathrm{eff}}(\mathbf{x}) \; \beta_n^{\mathrm{loc}}(\mathbf{x})^2 \; + \; o(1) \Big\} \, \| f \|_{L_2(G)}^2, \tag{D.17}$$

and, under Condition 2.6, the bracket is $o(1)$. We apply Equation (D.17) with $f = f_\theta$ and with the finite-difference kernels below.

Step 3: Moment Block Equation (2.48). Fix $\varepsilon > 0$. For any θ with $\| \theta - \theta_0 \| \leqslant \delta_n$, Chebyshev and Equation (D.17) give

$$\mathbb{P}\Big(\big| \, \mathbb{G}_{n,w}[f_\theta] \, \big| > \varepsilon \Big) \leqslant \varepsilon^{-2} \, o(1) \, \| f_\theta \|_{L_2(G)}^2. \tag{D.18}$$

Taking the supremum over $\| \theta - \theta_0 \| \leqslant \delta_n$ and using Equation (D.16) yields

$$\sup_{\| \theta - \theta_0 \| \leqslant \delta_n} \big| \, \mathbb{G}_{n,w}[f_\theta] \, \big| = o_{\mathbb{P}}(1). \tag{D.19}$$

Step 4: Plug-in Equation (2.49).. Let $A_n \coloneqq \Big\{ \big\| \hat{\eta}_j - \eta_{0,j} \big\|_\eta \leqslant \delta_n \Big\}$. By the nuisance LOO stability in Condition 2.6, $\mathbb{P}(A_n) \to 1$ for any $\delta_n \downarrow 0$. On A_n, Equation (D.16) implies $\big\| f_{(\alpha_{0,j}(\mathbf{x}), \hat{\eta}_j)} \big\|_{L_2(G)} \lesssim \delta_n$, hence Step 3 gives $\mathbb{G}_{n,w}\big[f_{(\alpha_{0,j}(\mathbf{x}), \hat{\eta}_j)} \big] = o_{\mathbb{P}}(1)$, which is exactly Equation (2.49).

Step 5: Jacobian Block (finite differences) Equation (2.51).. Define, for any η,

$$\Delta_{h,j}(w;\eta) \coloneqq \frac{\psi_j\big(w;\alpha_{0,j}(\mathbf{x})+h,\eta\big)-\psi_j\big(w;\alpha_{0,j}(\mathbf{x})-h,\eta\big)}{2h}\,\mathbb{1}\{X\in G\}, \qquad f_D(w)\coloneqq \Delta_{h,j}(w;\widehat{\eta}_j)-\Delta_{h,j}(w;\eta_{0,j}). \tag{D.20}$$

By the Jacobian MSC in $\tilde{\eta}$, Equation (2.34), we have the local Lipschitz control

$$\big\|\,\Delta_{h,j}(\cdot;\tilde{\eta})-\Delta_{h,j}(\cdot;\tilde{\eta}')\,\big\|_{L_2(G)} \leqslant L_{v,j}\,\big\|\,\tilde{\eta}-\tilde{\eta}'\,\big\|_{\eta}\;+\;O(h^2), \tag{D.21}$$

where the $O(h^2)$ term is the uniform central-difference bias. Hence, on A_n, $\|\,f_D\,\|_{L_2(G)} \lesssim \big\|\,\widehat{\eta}_j-\eta_{0,j}\,\big\|_{\eta}\;+\;h^2$.
Apply Equation (D.17) to $f=f_D$. Using the Jacobian LOO stability in Condition 2.6 together with the finite-difference tuning Equation (2.43) (namely $\sqrt{n_{\mathrm{eff}}\,(\mathbf{x})}\,\beta_n^{\mathrm{loc}}\,(\mathbf{x})\,/h\to 0$ and $\sqrt{n_{\mathrm{eff}}\,(\mathbf{x})}\,h^2\to 0$), the bracket in Equation (D.17) is $o(1)$ for f_D, so $\mathbb{G}_{n,w}[f_D]=o_{\mathbb{P}}(\,1\,)$.

Step 6: Decomposition. Finally, decompose

$$\sqrt{n_{\mathrm{eff}}\,(\mathbf{x})}\,\Big|\widehat{\Gamma}_{h,j}(\mathbf{x})-\widehat{\Gamma}^{(0)}_{h,j}(\mathbf{x})\Big| \leqslant \big|\,\mathbb{G}_{n,w}[f_D]\,\big|+\sqrt{n_{\mathrm{eff}}\,(\mathbf{x})}\,\big|\mathbb{E}[f_D(W)\;\mid\;X=\mathbf{x}]\,\big|. \tag{D.22}$$

The centering term is controlled by Equation (D.21) and nuisance LOO stability:

$$\sqrt{n_{\mathrm{eff}}\,(\mathbf{x})}\,\big|\mathbb{E}[f_D(W)\;\mid\;X=\mathbf{x}]\,\big| \lesssim \sqrt{n_{\mathrm{eff}}\,(\mathbf{x})}\,\big\|\,\widehat{\eta}_j-\eta_{0,j}\,\big\|_{\eta}\;+\;\sqrt{n_{\mathrm{eff}}\,(\mathbf{x})}\,h^2=o_{\mathbb{P}}(\,1\,). \tag{D.23}$$

Therefore Equation (2.51) holds, completing the proof. □

Appendix D.2. Proofs of Lemmas in Section 3

Proof of Lemma 3.3

Proof. Fix $j\in[p_\alpha]$ and write $k_{j,s}$ for the symmetric s-variate kernel that generates the localized jth score at the truth $\alpha_{0,j}\,(\mathbf{x})$ (the kernel appearing in Equation (3.5)). By definition of the first Hájek projection,

$$h^{\langle 1\rangle}_{j,s}(W_1)\coloneqq \mathbb{E}[k_{j,s}(W_1,W_2,\ldots,W_s)\mid W_1]-\mathbb{E}[k_{j,s}(W_1,\ldots,W_s)]. \tag{D.24}$$

By construction of the localized kernel (symmetry and linearity of the localizer), we can write

$$k_{j,s}(W_1,\ldots,W_s)=\sum_{r=1}^{s}\tilde{\tau}^{\circ}_{s,\mathbf{x}}(X_r)\;\psi_j\big(W_r;\alpha_{0,j}(\mathbf{x}),\eta_{0,j}\big), \tag{D.25}$$

where $\tilde{\tau}^{\circ}_{s,\mathbf{x}}$ depends only on the covariate X and $\mathbb{E}\big[\tilde{\tau}^{\circ}_{s,\mathbf{x}}(X)\big]=0$ by definition.
Taking conditional expectations of Equation (D.25) given W_1 and using independence across indices together with Condition 2.4, we obtain

$$\mathbb{E}[k_{j,s}(W_1,\ldots,W_s)\mid W_1]=\tilde{\tau}^{\circ}_{s,\mathbf{x}}(X_1)\;\psi_j\big(W_1;\alpha_{0,j}(\mathbf{x}),\eta_{0,j}\big)+\sum_{r=2}^{s}\mathbb{E}\big[\tilde{\tau}^{\circ}_{s,\mathbf{x}}(X_r)\;\psi_j\big(W_r;\alpha_{0,j}(\mathbf{x}),\eta_{0,j}\big)\big]. \tag{D.26}$$

For each $r\geqslant 2$,

$$\mathbb{E}\big[\tilde{\tau}^{\circ}_{s,\mathbf{x}}(X_r)\;\psi_j\big(W_r;\alpha_{0,j}(\mathbf{x}),\eta_{0,j}\big)\big]=\mathbb{E}\big[\tilde{\tau}^{\circ}_{s,\mathbf{x}}(X)\;\mathbb{E}\big[\psi_j\big(W;\alpha_{0,j}(\mathbf{x}),\eta_{0,j}\big)\mid X\big]\big]=0, \tag{D.27}$$

where the last equality follows from Condition 2.4. Hence Equation (D.26) reduces to

$$\mathbb{E}[k_{j,s}(W_1,\ldots,W_s)\mid W_1]=\tilde{\tau}^{\circ}_{s,\mathbf{x}}(X_1)\;\psi_j\big(W_1;\alpha_{0,j}(\mathbf{x}),\eta_{0,j}\big). \tag{D.28}$$

A parallel application of Condition 2.4 and $\mathbb{E}\big[\tilde{\tau}^{\circ}_{s,\mathbf{x}}(X)\big]=0$ shows that the unconditional mean of the kernel vanishes:

$$\mathbb{E}[k_{j,s}(W_1,\ldots,W_s)]=\sum_{r=1}^{s}\mathbb{E}\big[\tilde{\tau}^{\circ}_{s,\mathbf{x}}(X_r)\;\psi_j\big(W_r;\alpha_{0,j}(\mathbf{x}),\eta_{0,j}\big)\big]=0. \tag{D.29}$$

Combining Equation (D.28) and Equation (D.29) with Equation (D.24) yields

$$h_{j,s}^{\langle 1\rangle}(W_1) = \tilde{\tau}_{s,\mathbf{x}}^{\circ}(X_1)\ \psi_j\big(W_1; \alpha_{0,j}(\mathbf{x}), \eta_{0,j}\big), \tag{D.30}$$

which is exactly Equation (3.10). □

Proof of Lemma 3.13
Proof. Write $G \coloneqq \{ X \in \mathcal{B}_{\mathbf{x}} \}$ and correspondingly $G_i \coloneqq \{ X_i \in \mathcal{B}_{\mathbf{x}} \}$. Let

$$\Delta_l \coloneqq \widehat{\Gamma}_{h,j}(D) - \widehat{\Gamma}_{h,j}(D^{(l)}) = \frac{1}{2h}\sum_{i=1}^{n}\Big(w_i(D;\mathbf{x}) - w_i(D^{(l)};\mathbf{x}) \Big)\ \Delta_i(\widehat{\eta}_j) + \frac{1}{2h}\sum_{i=1}^{n} w_i(D^{(l)};\mathbf{x})\ \big[\Delta_i(\widehat{\eta}_j) - \Delta_i(\widehat{\eta}_j^{(-l)})\big], \tag{D.31}$$

where $\Delta_i(\eta) \coloneqq \big[\psi_j(W_i; \alpha_{0,j}(\mathbf{x}) + h, \eta) - \psi_j(W_i; \alpha_{0,j}(\mathbf{x}) - h, \eta)\big]\ \mathbb{1}\{G_i\}$.

Weightperturbation part.
By the mean value theorem in α and Condition 2.4, for each i there exists $\tilde{\alpha}_i \in [\alpha_{0,j}(\mathbf{x}) - h, \alpha_{0,j}(\mathbf{x}) + h]$ with

$$\frac{\Delta_i(\widehat{\eta}_j)}{2h} = \partial_\alpha \psi_j\big(W_i; \tilde{\alpha}_i, \widehat{\eta}_j\big)\ \mathbb{1}\{G_i\}, \qquad \big|\partial_\alpha \psi_j\big(W_i; \tilde{\alpha}_i, \widehat{\eta}_j\big)\big|\ \mathbb{1}\{G_i\} \leqslant L_{j,\mathbf{x}}, \tag{D.32}$$

for some finite envelope $L_{j,\mathbf{x}}$ on G (from the local Lipschitz/boundedness in Condition 2.4). Hence

$$\left|\frac{1}{2h}\sum_{i=1}^{n}\Big(w_i(D;\mathbf{x}) - w_i(D^{(l)};\mathbf{x})\Big)\ \Delta_i(\widehat{\eta}_j)\right| \leqslant L_{j,\mathbf{x}}\sum_{i=1}^{n}\big|\, w_i(D;\mathbf{x}) - w_i(D^{(l)};\mathbf{x})\,\big|\ \mathbb{1}\{G_i\} \leqslant L_{j,\mathbf{x}}\,\beta_n^{\mathrm{loc}}(\mathbf{x}). \tag{D.33}$$

NuisancePerturbation Part.
Define $g_i(a) \coloneqq \psi_j(W_i; a, \widehat{\eta}_j) - \psi_j(W_i; a, \widehat{\eta}_j^{(-l)})$. By the mean value theorem in a,

$$\frac{\Delta_i(\widehat{\eta}_j) - \Delta_i(\widehat{\eta}_j^{(-l)})}{2h} = \partial_\alpha m_j\big(W_i; \overline{\alpha}_i, \widehat{\eta}_j\big) - \partial_\alpha m_j\big(W_i; \overline{\alpha}_i, \widehat{\eta}_j^{(-l)}\big), \qquad \overline{\alpha}_i \in [\alpha_{0,j}(\mathbf{x}) - h, \alpha_{0,j}(\mathbf{x}) + h]. \tag{D.34}$$

Taking $L_2(G)$ norms and using Condition 2.4 (local equicontinuity in α) yields, uniformly over $\overline{\alpha} \in [\alpha_{0,j}(\mathbf{x}) - h, \alpha_{0,j}(\mathbf{x}) + h]$,

$$\Big\|\, \partial_\alpha m_j\big(W; \overline{\alpha}, \widehat{\eta}_j\big) - \partial_\alpha m_j\big(W; \overline{\alpha}, \widehat{\eta}_j^{(-l)}\big)\,\Big\|_{L_2(G)} \leqslant \Big\|\, \partial_\alpha m_j\big(W; \alpha_{0,j}(\mathbf{x}), \widehat{\eta}_j\big) - \partial_\alpha m_j\big(W; \alpha_{0,j}(\mathbf{x}), \widehat{\eta}_j^{(-l)}\big)\,\Big\|_{L_2(G)} + C_{j,\mathbf{x}}\, h, \tag{D.35}$$

for some finite $C_{j,\mathbf{x}}$ depending on the local Lipschitz modulus in Condition 2.4. Since $\sum_{i=1}^{n} w_i(D^{(l)};\mathbf{x})\ \mathbb{1}\{G_i\} \leqslant 1$, Jensens inequality together with Condition 2.6 (ii) implies

$$\left|\frac{1}{2h}\sum_{i=1}^{n} w_i(D^{(l)};\mathbf{x})\ \big[\Delta_i(\widehat{\eta}_j) - \Delta_i(\widehat{\eta}_j^{(-l)})\big]\right| \leqslant \Big\|\, \partial_\alpha m_j\big(W; \overline{\alpha}, \widehat{\eta}_j\big) - \partial_\alpha m_j\big(W; \overline{\alpha}, \widehat{\eta}_j^{(-l)}\big)\,\Big\|_{L_2(G)} = o\Big(n_{\mathrm{eff}}(\mathbf{x})^{-1/2}\Big) + O(h). \tag{D.36}$$

Combine.
From Equations (D.31), (D.33), and (D.36) we obtain

$$|\,\Delta_l\,| \leqslant L_{j,\mathbf{x}}\,\beta_n^{\mathrm{loc}}(\mathbf{x}) + o\Big(n_{\mathrm{eff}}(\mathbf{x})^{-1/2}\Big) + O(h). \tag{D.37}$$

Since $h \leqslant 1$ eventually, $L_{j,\mathbf{x}}\beta_n^{\mathrm{loc}}(\mathbf{x}) \leqslant L_{j,\mathbf{x}}\,\beta_n^{\mathrm{loc}}(\mathbf{x})/h$, which gives the deterministic bound Equation (3.42). Multiplying Equation (D.37) by $\sqrt{n_{\mathrm{eff}}(\mathbf{x})}$ and using the rate conditions $\sqrt{n_{\mathrm{eff}}(\mathbf{x})}\,\beta_n^{\mathrm{loc}}(\mathbf{x})/h \to 0$ and $\sqrt{n_{\mathrm{eff}}(\mathbf{x})}\,h^2 \to 0$ in (iii) (the latter controls the $O(h)$ term uniformly via the same local smoothness), we conclude Equation (3.43). □

Appendix E. Auxiliary Lemmas and Proofs

Appendix E.1. Lemmas: DNN Geometry and Weights

Proof of Proposition 3.1

Proof. To prove Proposition 3.1, we will rely on the L-statistic representation of the DNN weights presented in Equation (2.17). Observe first that $w_{s,(1)} = \binom{n}{s}^{-1}\binom{n-1}{s-1} = s/n$. Recall the following fact presented in Section 2.2

$$\frac{w_{s,(i+1)}}{w_{s,(i)}} = 1 - \frac{s-1}{n-i}, \qquad i = 1, \dots, n-s, \tag{E.1}$$

so, since $s \leqslant n - i \leqslant s, n-1$, the ratio is decreasing in i and

$$1 - \frac{s-1}{s} \leqslant \frac{w_{s,(i+1)}}{w_{s,(i)}} \leqslant 1 - \frac{s-1}{n-1}. \tag{E.2}$$

Thus

$$\frac{s}{n}\left(1 - \frac{s-1}{s}\right)^{i-1} \leqslant w_{s,(i)} \leqslant \frac{s}{n}\left(1 - \frac{s-1}{n-1}\right)^{i-1}. \tag{E.3}$$

We first construct an upper bound to the sum of squared weights.

$$\sum_{i=1}^{n-s+1} w_{s,(i)}^2 \leqslant \left(\frac{s}{n}\right)^2 \sum_{i=0}^{n-s}\left(1 - \frac{s-1}{n-1}\right)^{2i} = \left(\frac{s}{n}\right)^2 \frac{1 - \left(\frac{n-s}{n-1}\right)^{2(n-s+1)}}{1 - \left(\frac{n-s}{n-1}\right)^2}. \tag{E.4}$$

Equivalently,

$$\sum_{i=1}^{n-s+1} w_{s,(i)}^2 = \left(\frac{s}{n}\right)^2 \frac{(n-1)^2 - (n-s)^2\left(\frac{n-s}{n-1}\right)^{2(n-s)}}{(n-1)^2 - (n-s)^2} \lesssim \frac{s}{n}. \tag{E.5}$$

Next, we follow up with a corresponding lower bound. For $i \leqslant \lfloor n/2 \rfloor$, $\frac{w_{s,(i+1)}}{w_{s,(i)}} \geqslant 1 - \frac{2(s-1)}{n}$. Hence for $m \coloneqq \left\lfloor \frac{n}{4s} \right\rfloor$,

$$w_{s,(i)} \geqslant \frac{s}{n}\left(1 - \frac{2(s-1)}{n}\right)^{i-1}, \qquad i = 1, \dots, m, \tag{E.6}$$

and therefore

$$\sum_{i=1}^{n-s+1} w_{s,(i)}^2 \geqslant \sum_{i=1}^{m}\left(\frac{s}{n}\right)^2\left(1 - \frac{2(s-1)}{n}\right)^{2(i-1)} = \left(\frac{s}{n}\right) \cdot \frac{1 - \left(1 - \frac{2(s-1)}{n}\right)^{2m}}{2\frac{s-1}{n} - \left(\frac{s-1}{n}\right)^2} \gtrsim \frac{s}{n}. \tag{E.7}$$

Combining the two displays yields the desired result

$$\sum_{i=1}^{n-s+1} w_{s,(i)}^2 \asymp s/n \tag{E.8}$$

and hence $n_{\text{eff}}(\mathbf{x}) = \left(\sum_i w_{s,(i)}^2\right)^{-1} \asymp n/s$. Under Condition 2.3, nearest-neighbor ranks are almost surely unique, hence $\left\{ w_{s,(i)} \right\}_{i \in [n]}$ and the corresponding sum of squares are purely combinatorial functions of (n, s). The probability qualifier is therefore vacuously true. □

Proof of Proposition 3.2

Proof. Write $\mathcal{B}(\mathbf{x}, r) \coloneqq \{\, u : \| u - \mathbf{x} \|_2 \leqslant r \,\}$ and $N_r \coloneqq \sum_{i=1}^{n} \mathbb{1}\{\, X_i \in \mathcal{B}(\mathbf{x}, r) \,\}$. By the L-statistic form of the DNN weights, the cumulative weight on the first m nearest neighbors is

$$\sum_{i=1}^{m} w_{s,(i)} = \binom{n}{s}^{-1} \sum_{i=1}^{m} \binom{n-i}{s-1} = 1 - \frac{\binom{n-m}{s}}{\binom{n}{s}}. \tag{E.9}$$

Hence, for any $r > 0$,

$$\sum_{i=1}^{n} w_{s,i}\ \mathbb{1}\{\, X_i \in \mathcal{B}(\mathbf{x}, r) \,\} = \sum_{i=1}^{N_r} w_{s,(i)} = 1 - \frac{\binom{n-N_r}{s}}{\binom{n}{s}}. \tag{E.10}$$

Therefore the $(1-\delta)$-mass radius in Equation (3.2) satisfies

$$R_{s,\delta}(\mathbf{x}) = \inf\left\{ r > 0 : \frac{\binom{n-N_r}{s}}{\binom{n}{s}} \leqslant \delta \right\}. \tag{E.11}$$

Step 1: Binomial Tail Sandwich. For any $m \in \{0, \ldots, n-s\}$,

$$\frac{\binom{n-m}{s}}{\binom{n}{s}} = \prod_{j=0}^{s-1} \left(1 - \frac{m}{n-j}\right) \in \left[\left(1 - \frac{m}{n-s+1}\right)^{s}, \left(1 - \frac{m}{n}\right)^{s} \right]. \tag{E.12}$$

Applying Equation (E.12) with $m = N_r$ and using $1 - x \leqslant e^{-x}$ and, for $x \in (0, 1/2)$, $1 - x \geqslant e^{-x/(1-x)} \geqslant e^{-2x}$, we obtain

$$e^{-\frac{2s\,N_r}{n-s+1}} \leqslant \frac{\binom{n-N_r}{s}}{\binom{n}{s}} \leqslant e^{-\frac{s\,N_r}{n}} \qquad \text{whenever} \quad \frac{N_r}{n-s+1} \leqslant \frac{1}{2}. \tag{E.13}$$

Step 2: Local Mass and Binomial Concentration. Let $p(r) \coloneqq \mathbb{P}(X \in \mathcal{B}(\mathbf{x}, r))$. By Condition 2.3 and letting v_{k_x} be the Lebesgue volume of the k_x-dimensional unit ball, there exists $r_0 > 0$ such that for all $r \in (0, r_0]$,

$$f_-\ v_{k_x}\ r^{k_x} \leqslant p(r) \leqslant\ f_+\ v_{k_x}\ r^{k_x}. \tag{E.14}$$

For $\epsilon \in (0,1)$ and any $r \in (0, r_0]$, define the concentration event

$$\mathcal{E}(r, \epsilon) \coloneqq \left\{\, \left| N_r - n\ p(r) \right| \leqslant \epsilon\ n\ p(r) \,\right\}. \tag{E.15}$$

If $N_r \sim \mathrm{Bin}\,(n, p(r))$, the multiplicative Chernoff bound yields

$$\mathbb{P}(\mathcal{E}(r,\epsilon)^c) \leqslant 2\ \exp\left(-\frac{\epsilon^2}{3}\ n\ p(r)\right), \qquad \epsilon \in (0,1). \tag{E.16}$$

Step 3: Upper Bound on $R_{s,\delta}(\mathbf{x})$. Set

$$r_+(s,\delta) \coloneqq \left(\frac{2\log(1/\delta)}{f_-\ v_{k_x}\ s}\right)^{1/k_x}, \qquad \epsilon \coloneqq \frac{1}{2}. \tag{E.17}$$

By Equation (E.14), $p(r_+) \geqslant f_- v_{k_x} r_+^{k_x} = 2\log(1/\delta)/s$, hence

$$n\ p(r_+) \geqslant 2\ \frac{n}{s}\ \log(1/\delta). \tag{E.18}$$

Define the event

$$\mathcal{E}_+ \coloneqq \mathcal{E}\left(r_+, \frac{1}{2}\right) = \left\{\, N_{r_+} \geqslant \frac{1}{2}\ n\ p(r_+) \,\right\}. \tag{E.19}$$

By Equation (E.16),

$$\mathbb{P}\big(\mathcal{E}_+^c\big) \leqslant 2\exp\left(-\frac{1}{12}\, n\, p(r_+)\right) \leqslant 2\exp\left(-\frac{1}{6}\,\frac{n}{s}\,\log(1/\delta)\right) = o(1) \tag{E.20}$$

under $s \to \infty$ and $s/n \to 0$. On $\mathcal{E}_+$ we have $N_{r_+} \geqslant \frac{1}{2} np(r_+) \geqslant \frac{n}{s}\log(1/\delta)$, so by the *right* inequality in Equation (E.13),

$$\frac{\binom{n-N_{r_+}}{s}}{\binom{n}{s}} \leqslant e^{-\,sN_{r_+}/n} \leqslant e^{-\log(1/\delta)} = \delta, \tag{E.21}$$

and by Equation (E.11) this gives $R_{s,\delta}(\mathbf{x}) \leqslant r_+(s,\delta)$ on $\mathcal{E}_+$.

Step 4: Lower Bound on $R_{s,\delta}(\mathbf{x})$. Set

$$r_-(s,\delta) \coloneqq \left(\frac{\log(1/\delta)}{4\, f_+\, v_{k_x}\, s}\right)^{1/k_x}, \qquad \epsilon \coloneqq \frac{1}{2}. \tag{E.22}$$

By Equation (E.14), both

$$p(r_-) \leqslant f_+ v_{k_x} r_-^{k_x} = \frac{\log(1/\delta)}{4s}, \qquad p(r_-) \geqslant f_- v_{k_x} r_-^{k_x} = \frac{f_-}{f_+}\cdot\frac{\log(1/\delta)}{4s}. \tag{E.23}$$

Define the event

$$\mathcal{E}_- \coloneqq \mathcal{E}\left(r_-, \frac{1}{2}\right) = \left\{ N_{r_-} \leqslant \frac{3}{2}\, n\, p(r_-) \right\}. \tag{E.24}$$

By Equation (E.16) and the lower bound on $p(r_-)$,

$$\mathbb{P}\big(\mathcal{E}_-^c\big) \leqslant 2\exp\left(-\frac{1}{12}\, n\, p(r_-)\right) \leqslant 2\exp\left(-\,\frac{f_-}{48\, f_+}\,\frac{n}{s}\,\log(1/\delta)\right) = o(1). \tag{E.25}$$

On $\mathcal{E}_-$ we have

$$N_{r_-} \leqslant \frac{3}{2}\, n\, p(r_-) \leqslant \frac{3}{2}\, n\, \frac{\log(1/\delta)}{4s} = \frac{3n}{8}\cdot\frac{\log(1/\delta)}{s}. \tag{E.26}$$

For fixed $\delta \in (0,1)$ and large s, this implies

$$\frac{N_{r_-}}{n-s+1} \leqslant \frac{3}{8}\cdot\frac{\log(1/\delta)}{s}\cdot\frac{n}{n-s+1} \leqslant \frac{1}{2}, \tag{E.27}$$

so the *left* inequality in Equation (E.13) applies and yields

$$\frac{\binom{n-N_{r_-}}{s}}{\binom{n}{s}} \geqslant e^{-\,\frac{2sN_{r_-}}{n-s+1}} \geqslant e^{-\frac{3}{4}\cdot\frac{n}{n-s+1}\log(1/\delta)} = \delta^{\frac{3}{4}\cdot\frac{n}{n-s+1}} > \delta \tag{E.28}$$

for all large n (since $s/n \to 0$). By Equation (E.11), the mass inside $\mathcal{B}(\mathbf{x}, r_-)$ is $< 1-\delta$, hence $R_{s,\delta}(\mathbf{x}) \geqslant r_-(s,\delta)$ on $\mathcal{E}_-$.

Conclusion. Since $\mathbb{P}\big(\mathcal{E}_+^c \cup \mathcal{E}_-^c\big) \leqslant \mathbb{P}\big(\mathcal{E}_+^c\big) + \mathbb{P}\big(\mathcal{E}_-^c\big) = o(1)$ by Equations (E.20) and (E.25), both bounds hold with probability $\to 1$. Absorbing numeric constants into c_δ, C_δ (depending only on (k_x, δ, f_-, f_+)) gives

$$c_\delta\left(\frac{\log(1/\delta)}{s}\right)^{1/k_x} \leqslant R_{s,\delta}(\mathbf{x}) \leqslant C_\delta\left(\frac{\log(1/\delta)}{s}\right)^{1/k_x}, \tag{E.29}$$

which is Equation (3.3). In particular, for each fixed $\delta \in (0,1)$, $R_{s,\delta}(\mathbf{x}) = \Theta\big(s^{-1/k_x}\big)$. □

Lemma Appendix E.1 (Probability that a Random s-Subset misses $\mathcal{B}_{\mathbf{x}}$)**.** (→ Proof, p. 61)
Write

$$N_0(D_{[n]}) \coloneqq \sum_{i=1}^{n} \mathbb{1}\{\, X_i \in \mathcal{B}_{\mathbf{x}} \,\}, \qquad \pi_{\mathbf{x}} \coloneqq \mathbb{P}(X \in \mathcal{B}_{\mathbf{x}}). \tag{E.30}$$

Let L be uniformly distributed over all s-subsets of $[n]$ (independent of $D_{[n]}$), and define

$$A(D_\ell) \coloneqq \big\{ \exists\, i \in \ell :\ X_i \in \mathcal{B}_{\mathbf{x}} \big\}. \tag{E.31}$$

Then

$$\mathbb{P}\big(A\,(D_L)^c \,\big|\, D_{[n]}\big) = \binom{n - N_0(D_{[n]})}{s} \binom{n}{s}^{-1}. \tag{E.32}$$

Moreover, taking expectations,

$$\mathbb{P}(A\,(D_L)^c) = \frac{\mathbb{E}\big[\ \big(n - N_0(D_{[n]})\big)_s\ \big]}{(n)_s} = (1 - \pi_{\mathbf{x}})^s, \tag{E.33}$$

where $(m)_s \coloneqq m(m-1)\cdots(m-s+1)$ denotes the falling factorial.

Proof of Lemma Appendix E.1
Proof. Conditional on $D_{[n]}$, the number $N_0(D_{[n]})$ of indices with $X_i \in \mathcal{B}_{\mathbf{x}}$ is fixed. A uniformly drawn s-subset L misses $\mathcal{B}_{\mathbf{x}}$ iff all s chosen indices lie among the $n - N_0(D_{[n]})$ indices with $X_i \notin \mathcal{B}_{\mathbf{x}}$. Hence the hypergeometric expression Equation (E.32).
For the unconditional probability, take expectations. Using falling factorials,

$$\mathbb{P}(A\,(D_L)^c) = \mathbb{E}\left[\binom{n - N_0(D_{[n]})}{s} \binom{n}{s}^{-1} \right] = \frac{\mathbb{E}\big[\ \big(n - N_0(D_{[n]})\big)_s\ \big]}{(n)_s}. \tag{E.34}$$

Since $N_0(D_{[n]}) \sim \mathrm{Bin}\,(n, \pi_{\mathbf{x}})$ (row-wise i.i.d.), we have $n - N_0(D_{[n]}) \sim \mathrm{Bin}\,(n, 1 - \pi_{\mathbf{x}})$ and the binomial factorial-moment identity

$$\mathbb{E}[(X)_s] = (n)_s p^s \qquad \text{for} X \sim \mathrm{Bin}(n, p) \tag{E.35}$$

(see, e.g., Potts (1953)) yields

$$\mathbb{E}\big[\ \big(n - N_0(D_{[n]})\big)_s\ \big] = (n)_s (1 - \pi_{\mathbf{x}})^s, \tag{E.36}$$

which gives Equation (E.33). □

Appendix E.2. Lemmas: Stability and Equicontinuity

Lemma Appendix E.2 (Replace-one Increment Bound for the Localized Process)**.** (→ Proof, p. 62)
Fix $j \in [p_\alpha]$, let $G \coloneqq \{\, X \in \mathcal{B}_{\mathbf{x}} \,\}$ from Condition 2.6, and recall the following definition of the localized empirical process

$$\mathbb{G}_{n,w}[f] \coloneqq \sqrt{n_{\mathrm{eff}}\,(\mathbf{x})} \Big\{ \sum_{i=1}^{n} w_{s,i}\,(\mathbf{x})\ f_D(W_i) - \mathbb{E}[f_D(W) \mid X = \mathbf{x}] \Big\} \tag{E.37}$$

as in Equation (2.47), where $f_D \colon \mathcal{W} \to \mathbb{R}$ is any square-integrable function that may depend on D (e.g., via first-stage fits) and the conditional expectation exists and is finite. For each $l \in [n]$, let $D^{(l)}$ be the data set $D = \{\, W_1, \ldots, W_n \,\}$ with W_l replaced by an i.i.d. copy W_l' and set

$$\Delta_l(f) \coloneqq \mathbb{G}_{n,w}[f]\,(D) - \mathbb{G}_{n,w}[f]\,\big(D^{(l)}\big). \tag{E.38}$$

Then, for any square-integrable f_D supported on G (or already multiplied by $\mathbb{1}\{\, X \in G \,\}$),

$$\big|\, \Delta_l(f) \,\big| \leqslant A_l(f) + B_l(f) + C_l(f), \tag{E.39}$$

where, writing $w_{s,i}(\mathbf{x}) \coloneqq w_{s,i}(\mathbf{x})$ *and* $w_i^{(l)}(\mathbf{x}) \coloneqq w_{s,i}(\mathbf{x})(D^{(l)}; x)$,

$$\textit{(A) Direct change:} \quad A_l(f) \coloneqq \sqrt{n_{\text{eff}}(\mathbf{x})}\Big(|w_l| \left| f_D(W_l) \right| + \left| w_l^{(l)} \right| \left| f_{D^{(l)}}(W_l') \right| \Big), \tag{E.40}$$

$$\textit{(B) Weight perturbation:} \quad B_l(f) \coloneqq \sqrt{n_{\text{eff}}(\mathbf{x})} \sum_{i=1}^{n} \left| w_{s,i}(\mathbf{x}) - w_i^{(l)}(\mathbf{x}) \right| \left| f_D(W_i) \right| \mathbb{1}\{ X_i \in G \}, \tag{E.41}$$

$$\textit{(C) Centering change:} \quad C_l(f) \coloneqq \sqrt{n_{\text{eff}}(\mathbf{x})} \left| \mathbb{E}[f_D(W) \mid X = \mathbf{x}] - \mathbb{E}[f_{D^{(l)}}(W) \mid X = \mathbf{x}] \right|. \tag{E.42}$$

Moreover, with the localized replace-one modulus $\beta_n^{\text{loc}}(\mathbf{x})$ *from Equation (2.38) and the rates in Condition 2.6, we have the following uniform bounds:*

$$\sup_{l \in [n]} \mathbb{E}\Big[A_l(f)^2 \Big] \lesssim n_{\text{eff}}(\mathbf{x}) \ \mathbb{E}\Big[\max_{i \leqslant n} w_i^2(\mathbf{x}) \Big] \cdot \| f \|_{L_2(G)}^2, \tag{E.43}$$

$$\sup_{l \in [n]} \mathbb{E}\Big[B_l(f)^2 \Big] \lesssim n_{\text{eff}}(\mathbf{x}) \ \big(\beta_n^{\text{loc}}(\mathbf{x}) \big)^2 \cdot \| f \|_{L_2(G)}^2, \tag{E.44}$$

$$\sup_{l \in [n]} \mathbb{E}\Big[C_l(f)^2 \Big] = o(1) \cdot \| f \|_{L_2(G)}^2. \tag{E.45}$$

Proof of Lemma Appendix E.2

Proof. The decomposition $\left| \Delta_l(f) \right| \leqslant A_l(f) + B_l(f) + C_l(f)$ is a triangle inequality after adding/subtracting

$$\sqrt{n_{\text{eff}}(\mathbf{x})} \sum_i w_i^{(l)}(\mathbf{x}) f_D(W_i)$$

and noting the centering term changes when D is replaced by $D^{(l)}$.

(A) Direct Change. Only the l-th summand differs directly across D and $D^{(l)}$. Squaring and taking expectation, by symmetry and Cauchy-Schwarz,

$$\mathbb{E}\Big[A_l(f)^2 \Big] \lesssim n_{\text{eff}}(\mathbf{x}) \ \mathbb{E}\Big[w_l^2 \ f_D(W_l)^2 + (w_l^{(l)})^2 \ f_{D^{(l)}}(W_l')^2 \Big] \lesssim n_{\text{eff}}(\mathbf{x}) \ \mathbb{E}\Big[\max_{i \in [n]} w_i^2(\mathbf{x}) \Big] \| f \|_{L_2(G)}^2, \tag{E.46}$$

yielding Equation (E.43). The rate $\max_{i \in [n]} \left| w_{s,i}(\mathbf{x})(\mathbf{x}) \right| = o\big(n_{\text{eff}}(\mathbf{x})^{-1/2} \big)$ from Condition 2.6 (i) will be plugged in later.

(B) Weight Perturbation. By the weighted Cauchy-Schwarz inequality,

$$B_l(f) \leqslant \sqrt{n_{\text{eff}}(\mathbf{x})} \left(\sum_i \left| w_{s,i}(\mathbf{x}) - w_i^{(l)}(\mathbf{x}) \right| \right)^{1/2} \left(\sum_i \left| w_{s,i}(\mathbf{x}) - w_i^{(l)}(\mathbf{x}) \right| f_D(W_i)^2 \ \mathbb{1}\{ X_i \in G \} \right)^{1/2}. \tag{E.47}$$

Taking squares and expectations and using the definition of $\beta_n^{\text{loc}}(\mathbf{x})$ in Equation (2.38) and the indicator restriction to G, we get

$$\mathbb{E}\Big[B_l(f)^2 \Big] \lesssim n_{\text{eff}}(\mathbf{x}) \ \big(\beta_n^{\text{loc}}(\mathbf{x}) \big)^2 \| f \|_{L_2(G)}^2, \tag{E.48}$$

which is Equation (E.44). The localized stability rate $\sqrt{n_{\text{eff}}(\mathbf{x})}\, \beta_n^{\text{loc}}(\mathbf{x}) \to 0$ is in Condition 2.6 (i).

(C) Centering Change. By Condition 2.6 (ii) and (iii), the replace-one change in the conditional mean at x is $o\big(n_{\text{eff}}(\mathbf{x})^{-1/2}\big)$ in $L_2(G)$. Multiplying by $\sqrt{n_{\text{eff}}(\mathbf{x})}$ yields Equation (E.45). □

Corollary Appendix E.3 (Concrete Efron-Stein variance bound)**.**
Let $T(D) \coloneqq \mathbb{G}_{n,w}[f](D)$ and $\Delta_l(f)$ as above. By the Efron-Stein inequality,

$$\operatorname{Var}(\mathbb{G}_{n,w}[f]) \leqslant \frac{1}{2}\sum_{l\in[n]}\mathbb{E}\Big[\big(\Delta_l(f)\big)^2\Big]. \tag{E.49}$$

Combining with Lemma Appendix E.2 and using Condition 2.6 gives

$$\operatorname{Var}(\mathbb{G}_{n,w}[f]) \lesssim \Big\{n_{\text{eff}}(\mathbf{x})\ \mathbb{E}\Big[\max_{i\in[n]} w_i^2(\mathbf{x})\Big] + n_{\text{eff}}(\mathbf{x})\ (\beta_n^{\text{loc}}(\mathbf{x}))^2 + o(1)\Big\}\,\|f\|_{L_2(G)}^2, \tag{E.50}$$

and, in particular, under Condition 2.6 (i) we have $\operatorname{Var}(\mathbb{G}_{n,w}[f]) = o(1)\cdot\|f\|_{L_2(G)}^2$.

Lemma Appendix E.4 (LSE for Localized Score Differences)**.** (→ Proof, p. 63)
Assume Conditions 2.4, 2.5 and 2.6 and hence Lemmas 2.2 and 2.4 hold. Fix $j \in [p_\alpha]$ and $G \coloneqq \{X \in \mathcal{B}_{\mathbf{x}}\}$. For a shrinking radius $\delta_n \downarrow 0$ define the localized score-difference class

$$\mathcal{F}_n \coloneqq \Big\{ f_{a,\tilde\eta}(w) \coloneqq \big(\psi_j(w;a,\tilde\eta) - \psi_j(w;a,\eta_{0,j})\big)\ \mathbb{1}\{X\in G\} : \big|a-\alpha_{0,j}(\mathbf{x})\big| \leqslant \delta_n, \big\|\tilde\eta-\eta_{0,j}\big\|_\eta \leqslant \delta_n\Big\}. \tag{E.51}$$

Then:

(i) ***Weighted-process identity:*** *For all $f \in \mathcal{F}_n$,*

$$\mathbb{G}_{n,w}[f] = \mathbb{T}_{n,w}[f]\ -\ \sqrt{n_{\text{eff}}(\mathbf{x})}\,\mathbb{E}[f(W)\mid X=\mathbf{x}]. \tag{E.52}$$

(ii) ***LSE bound (uniform over $\mathcal{F}_n$):*** *By Lemma 2.4 (applied in the weight form),*

$$\sup_{f\in\mathcal{F}_n}\big|\,\mathbb{G}_{n,w}[f]\,\big| = o_{\mathbb{P}}(1). \tag{E.53}$$

In particular, combining Equations (E.52) and (E.53),

$$\sup_{f\in\mathcal{F}_n}\Big|\,\mathbb{T}_{n,w}[f]\ -\ \sqrt{n_{\text{eff}}(\mathbf{x})}\,\mathbb{E}[f(W)\mid X=\mathbf{x}]\,\Big| = o_{\mathbb{P}}(1). \tag{E.54}$$

Proof of Lemma Appendix E.4
Proof. By construction, each $f \in \mathcal{F}_n$ is a localized score difference gated to $G \coloneqq \{X\in\mathcal{B}_{\mathbf{x}}\}$:

$$f_{a,\tilde\eta}(W) \coloneqq \big(\psi_j(W;a,\tilde\eta) - \psi_j(W;a,\eta_{0,j})\big)\,\mathbb{1}\{X\in G\}, \qquad \big|a-\alpha_{0,j}(\mathbf{x})\big| \leqslant \delta_n,\ \ \big\|\tilde\eta-\eta_{0,j}\big\|_\eta \leqslant \delta_n. \tag{E.55}$$

The weightedprocess identity in the lemma statement holds for every f:

$$\mathbb{G}_{n,w}[f]\ =\ \mathbb{T}_{n,w}[f]\ -\ \sqrt{n_{\text{eff}}(\mathbf{x})}\,\mathbb{E}[f(W)\mid X=\mathbf{x}], \tag{E.56}$$

which is exactly Equation (E.52).

Step 1: Local L_2 modulus on G.
By Lemma 2.2 and Condition 2.5 (local moment control on G), there exists a finite constant $C_G < \infty$ (independent of n) such that, for all $(a, \tilde{\eta})$ satisfying the constraints in Equation (E.55),

$$\mathbb{E}\big[f_{a,\tilde{\eta}}(W)^2 \mid X \in G\big] \;\leqslant\; C_G\Big(\big|\, a - \alpha_{0,j}(\mathbf{x})\,\big| + \big\|\,\tilde{\eta} - \eta_{0,j}\,\big\|_{\eta}\Big)^2 \;\leqslant\; C_G\,\delta_n^2. \tag{E.57}$$

Hence, the class has a vanishing localized envelope:

$$\sup_{f\in\mathcal{F}_n} \|\, f\,\|_{L_2(G)} \;\lesssim\; \delta_n \;\to\; 0. \tag{E.58}$$

Step 2: Pre-Gaussian (increment/variance) control on G.
Let $\rho_n(f,g)^2 \coloneqq \mathrm{Var}(\mathbb{G}_{n,w}[f-g])$ denote the canonical semimetric of the gated, weighted process. The replaceone/EfronStein bound used in Lemma 2.4 yields, uniformly for squareintegrable f supported on G,

$$\mathrm{Var}(\mathbb{G}_{n,w}[f]) \;\lesssim\; \Big(n_{\mathrm{eff}}\,(\mathbf{x})\;\mathbb{E}\Big[\max_{i\in[n]} w_{s,i}\,(\mathbf{x})^2\Big] \;+\; n_{\mathrm{eff}}\,(\mathbf{x})\;\beta_n^{\mathrm{loc}}\,(\mathbf{x})^2 \;+\; o(1)\Big)\,\|\, f\,\|^2_{L_2(G)}. \tag{E.59}$$

By Condition 2.6(i), $\max_{i\in[n]} w_{s,i}\,(\mathbf{x}) = o_{\mathbb{P}}\Big(\, n_{\mathrm{eff}}\,(\mathbf{x})^{-1/2}\,\Big)$ and $\sqrt{n_{\mathrm{eff}}\,(\mathbf{x})}\,\beta_n^{\mathrm{loc}}\,(\mathbf{x}) \xrightarrow{\mathbb{P}} 0$, so the bracket in Equation (E.59) is $o(1)$. Consequently,

$$\rho_n(f,g)^2 \;\lesssim\; c_n\,\|\, f-g\,\|^2_{L_2(G)} \quad \text{with} \quad c_n = o(1). \tag{E.60}$$

Step 3: Local SE on $\mathcal{F}_n$.
The shrinking $L_2(G)$diameter in Equation (E.58) together with the subGaussian increment control Equation (E.60) matches the hypotheses of Lemma 2.4 (weight form, gated to G). Therefore,

$$\sup_{f\in\mathcal{F}_n}\big|\,\mathbb{G}_{n,w}[f]\,\big| \;=\; o_{\mathbb{P}}(\,1\,), \tag{E.61}$$

which is exactly Equation (E.53).

Step 4: Conclusion for $\mathbb{T}_{n,w}$.
Combining Equations (E.56) and (E.61) gives

$$\sup_{f\in\mathcal{F}_n}\Big|\,\mathbb{T}_{n,w}[f] \;-\; \sqrt{n_{\mathrm{eff}}\,(\mathbf{x})}\;\mathbb{E}[f(W)\mid X=\mathbf{x}]\,\Big| \;=\; o_{\mathbb{P}}(\,1\,), \tag{E.62}$$

i.e. Equation (E.54). This completes the proof. □

Appendix E.3. Lemmas: Linearization and Localized Empirical Processes

Lemma Appendix E.5 (Weight Measurability and Conditional Reduction)**.** (→ Proof, p. 65)
Fix $s \in \mathbb{N}$ and a subset $\ell \subset [n]$ with $|\,\ell\,| = s$. Let $D_\ell = \{W_j : j \in \ell\}$ and let $w_i^{(\ell)}\,(\mathbf{x})$ denote the DNN weight assigned to observation i at target $\mathbf{x}$ when the weights are computed from D_ℓ (with any tie-breaking that is measurable w.r.t. $\sigma\,(X_j : j \in \ell)$). Then $w_i^{(\ell)}\,(\mathbf{x})$ is measurable with respect to $\sigma\,(X_j : j \in \ell)$.
Hence, for any integrable $h(W)$,

$$\mathbb{E}\Big[w_i^{(\ell)}\,(\mathbf{x})\;\; h\,(W_i)\Big] = \mathbb{E}\Big[w_i^{(\ell)}\,(\mathbf{x})\;\; \mathbb{E}[h(W)\mid X = X_i]\Big]. \tag{E.63}$$

In particular, with $h\,(W_i) = \psi_j\,(W_i, a, \tilde{\eta})$,

$$\mathbb{E}\Big[w_i^{(\ell)}\,(\mathbf{x})\;\; \psi_j\,(W_i, a, \tilde{\eta})\Big] = \mathbb{E}\Big[w_i^{(\ell)}\,(\mathbf{x})\;\; m_j\,(X_i; a, \tilde{\eta})\Big]. \tag{E.64}$$

Proof of Lemma Appendix E.5

Proof. Fix $s \in \mathbb{N}$ and $\ell \in \mathcal{L}_{n,s}$. By construction, the DNN weight $w_i^{(\ell)}(\mathbf{x})$ is a measurable function of the covariate ranks within D_ℓ (together with the fixed tie-break rule), hence it depends on D_ℓ only through $\{X_j : j \in \ell\}$ and is therefore $\sigma(X_j : j \in \ell)$-measurable.
Let $h(W_i)$ be integrable. By iterated expectations and the preceding measurability,

$$\begin{aligned}\mathbb{E}\Big[w_i^{(\ell)}(\mathbf{x})\ h(W_i)\Big] &= \mathbb{E}\Big[\mathbb{E}\Big[w_i^{(\ell)}(\mathbf{x})\ h(W_i)\ \Big|\ \sigma(X_j : j \in \ell)\Big]\Big] \\ &= \mathbb{E}\Big[w_i^{(\ell)}(\mathbf{x})\ \mathbb{E}[h(W_i)\ |\ \sigma(X_j : j \in \ell)]\Big]. \end{aligned} \tag{E.65}$$

Because the sample is row-wise i.i.d., W_i is independent of $\{X_j : j \in \ell,\ j \neq i\}$ and thus

$$\mathbb{E}[h(W_i)\ |\ \sigma(X_j : j \in \ell)] = \mathbb{E}[h(W) \mid X = X_i] \quad \text{a.s.} \tag{E.66}$$

Substituting into the previous display yields

$$\mathbb{E}\Big[w_i^{(\ell)}(\mathbf{x})\ h(W_i)\Big] = \mathbb{E}\Big[w_i^{(\ell)}(\mathbf{x})\ \mathbb{E}[h(W) \mid X = X_i]\Big]. \tag{E.67}$$

For $h(W_i) = \psi_j(W_i, a, \tilde{\eta})$ and $m_j(u; a, \tilde{\eta}) \coloneqq \mathbb{E}[\psi_j(W, a, \tilde{\eta})\ |\ X = u]$, we obtain

$$\mathbb{E}\Big[w_i^{(\ell)}(\mathbf{x})\ \psi_j(W_i, a, \tilde{\eta})\Big] = \mathbb{E}\Big[w_i^{(\ell)}(\mathbf{x})\ m_j(X_i; a, \tilde{\eta})\Big]. \tag{E.68}$$

□

Lemma Appendix E.6 (Orthogonal Linearization of the Moment)**.** (→ Proof, p. 65)
Under Condition 2.5, uniformly for $(a, \tilde{\eta})$ in a local neighborhood O_j of $(\alpha_{0,j}(\mathbf{x}), \eta_{0,j})$,

$$m_j(\mathbf{x}; a, \tilde{\eta}) = \Gamma_j(\mathbf{x})\ \big(a - \alpha_{0,j}(\mathbf{x})\big) + R_j(\mathbf{x}, a, \tilde{\eta}), \qquad \big|R_j(\mathbf{x}, a, \tilde{\eta})\big| \leqslant L_v \left(\big|a - \alpha_{0,j}\big|^2 + \|\tilde{\eta} - \eta_{0,j}\|_\eta^2\right). \tag{E.69}$$

Proof of Lemma Appendix E.6

Proof. Fix an arbitrary pair $(a, \tilde{\eta}) \in O_j$ and let $\Delta_\alpha = a - \alpha_{0,j}(\mathbf{x})$ and $\Delta_\eta = \tilde{\eta} - \eta_{0,j}$. For notational simplicity let $\Delta = (\Delta_\alpha, \Delta_\eta)$. Then, by Condition 2.5 (iv), we can use a Taylor-style expansion in the following way to make use of the Neyman-orthogonality pointed out in Remark 2.2.

$$\begin{aligned} & m_j(\mathbf{x};\ a, \tilde{\eta}) \\ &= \underbrace{m_j(\mathbf{x};\ \alpha_{0,j}(\mathbf{x}), \eta_{0,j})}_{=0} + Dm_j(\mathbf{x};\ \alpha_{0,j}(\mathbf{x}), \eta_{0,j})[\Delta] + R_j(\mathbf{x}, a, \tilde{\eta}) \\ &= \partial_\alpha m_j(\mathbf{x};\ \alpha_{0,j}(\mathbf{x}), \eta_{0,j})\,\Delta_\alpha + \underbrace{\partial_\eta m_j(\mathbf{x};\ \alpha_{0,j}(\mathbf{x}), \eta_{0,j})}_{=0}[\Delta_\eta] + R_j(\mathbf{x}, a, \tilde{\eta}) \\ &= \Gamma_j(\mathbf{x})\,(a - \alpha_{0,j}(\mathbf{x})) + R_j(\mathbf{x}, a, \tilde{\eta}) \end{aligned} \tag{E.70}$$

Above, we used Condition 2.5 (vi) to eliminate the constant term in the expansion, Remark 2.2 to eliminate the derivative with respect to η, and plugged in for the derivative with respect to α using Condition 2.5 (v). Thus, it remains to be shown that the remainder term is suitably bounded, which we will achieve using the integral form of the remainder and Condition 2.5 (iv). Observe the following.

$$\begin{aligned} R_j(\mathbf{x}, a, \tilde{\eta}) &= m_j(\mathbf{x};\ a, \tilde{\eta}) - Dm_j(\mathbf{x};\ \alpha_{0,j}(\mathbf{x}), \eta_{0,j})[\Delta] \\ &= m_j(\mathbf{x};\ a, \tilde{\eta}) - \underbrace{m_j(\mathbf{x};\ \alpha_{0,j}(\mathbf{x}), \eta_{0,j})}_{0} - Dm_j(\mathbf{x};\ \alpha_{0,j}(\mathbf{x}), \eta_{0,j})[\Delta] \\ &= \int_0^1 Dm_j(\mathbf{x};\ \alpha_{0,j}(\mathbf{x}) + t\Delta_\alpha, \eta_{0,j} + t\Delta_\eta)[\Delta]\ \mathrm{d}t - Dm_j(\mathbf{x};\ \alpha_{0,j}(\mathbf{x}), \eta_{0,j})[\Delta] \\ &= \int_0^1 \left[Dm_j(\mathbf{x};\ \alpha_{0,j}(\mathbf{x}) + t\Delta_\alpha, \eta_{0,j} + t\Delta_\eta) - Dm_j(\mathbf{x};\ \alpha_{0,j}(\mathbf{x}), \eta_{0,j})\right][\Delta]\ \mathrm{d}t \end{aligned} \tag{E.71}$$

Taking norms, we can thus deduce the following.

$$\begin{aligned}\left|R_j\left(\mathbf{x}, a, \tilde{\eta}\right)\right| &= \left|\int_0^1 \left[Dm_j\left(\mathbf{x};\ \alpha_{0,j}\left(\mathbf{x}\right) + t\Delta_\alpha, \eta_{0,j} + t\Delta_\eta\right) - Dm_j\left(\mathbf{x};\ \alpha_{0,j}\left(\mathbf{x}\right), \eta_{0,j}\right)\right][\Delta]\ \mathrm{d}t\right| \\ &\leqslant \int_0^1 \left|\left[Dm_j\left(\mathbf{x};\ \alpha_{0,j}\left(\mathbf{x}\right) + t\Delta_\alpha, \eta_{0,j} + t\Delta_\eta\right) - Dm_j\left(\mathbf{x};\ \alpha_{0,j}\left(\mathbf{x}\right), \eta_{0,j}\right)\right][\Delta]\right| \mathrm{d}t \\ &\leqslant \int_0^1 \left\| Dm_j\left(\mathbf{x};\ \alpha_{0,j}\left(\mathbf{x}\right) + t\Delta_\alpha, \eta_{0,j} + t\Delta_\eta\right) - Dm_j\left(\mathbf{x};\ \alpha_{0,j}\left(\mathbf{x}\right), \eta_{0,j}\right)\right\|_{\mathrm{op}} \|\Delta\| \mathrm{d}t\end{aligned} \tag{E.72}$$

Using now the locally uniform Lipschitz property from Condition 2.5 (iv), we can construct the following bound on the operator norm involved in this expression.

$$\left\| Dm_j\left(\mathbf{x};\ \alpha_{0,j}\left(\mathbf{x}\right) + t\Delta_\alpha, \eta_{0,j} + t\Delta_\eta\right) - Dm_j\left(\mathbf{x};\ \alpha_{0,j}\left(\mathbf{x}\right), \eta_{0,j}\right)\right\|_{\mathrm{op}} \leqslant L_v t \|\Delta\| \tag{E.73}$$

Plugging back into the integral, we thus obtain the following bound where we unpack $\|\Delta\| = |\Delta_\alpha| + \left\|\Delta_\eta\right\|_\eta$.

$$\left|R_j\left(\mathbf{x}, a, \tilde{\eta}\right)\right| \leqslant \int_0^1 L_v t \|\Delta\|^2 \mathrm{d}t = \frac{L_v}{2}\|\Delta\|^2 \leqslant L_v\left(|\Delta_\alpha|^2 + \left\|\Delta_\eta\right\|_\eta^2\right) \tag{E.74}$$

As Condition 2.5 (iv) stipulates the Lipschitz continuity of the derivative to hold uniform over a local neighborhood, this bound on the remainder term is itself also uniform over said neighborhood. Thus, we obtain the following expression for $m_j\left(\mathbf{x};\ a, \tilde{\eta}\right)$ where the latter bound holds uniformly across O_j.

$$m_j\left(\mathbf{x};\ a, \tilde{\eta}\right) = \Gamma_j\left(\mathbf{x}\right)\left(a - \alpha_{0,j}\left(\mathbf{x}\right)\right) + R_j\left(\mathbf{x}, a, \tilde{\eta}\right) \quad \text{where} \quad \left|R_j\left(\mathbf{x}, a, \tilde{\eta}\right)\right| \leqslant L_v\left(|\Delta_\alpha|^2 + \left\|\Delta_\eta\right\|_\eta^2\right) \tag{E.75}$$

□

Lemma Appendix E.7 (Localized Mean Bound)**.** (→ Proof, p. 66)
Assume Condition 2.5. For any $j \in [p_\alpha]$, $a \in \mathcal{A}_j(\mathbf{x})$, and $\tilde{\eta} \in \mathcal{H}_j$,

$$\left|\mathbb{E}[\psi_j(W; a, \tilde{\eta}) \ \mid\ X = \mathbf{x}] - \mathbb{E}[\psi_j(W; \alpha_{0,j}(\mathbf{x}), \eta_{0,j}) \ \mid\ X = \mathbf{x}]\right| \leqslant L_\alpha\,|a - \alpha_{0,j}(\mathbf{x})| \ +\ L_\eta\,\|\tilde{\eta} - \eta_{0,j}\|_\eta. \tag{E.76}$$

Equivalently, using the score difference Equation (A.6),

$$\left|\mathbb{E}[\Delta\psi_j(W;\ a, \tilde{\eta}) \ \mid\ X = \mathbf{x}]\right| \leqslant L_\alpha\,|a - \alpha_{0,j}(\mathbf{x})| \ +\ L_\eta\,\|\tilde{\eta} - \eta_{0,j}\|_\eta. \tag{E.77}$$

Proof of Lemma Appendix E.7
Proof. Fix $j \in [p_\alpha]$, $a \in \mathcal{A}_j(\mathbf{x})$, and $\tilde{\eta} \in \mathcal{H}_j$. By definition of the conditional moment map m_j (see Equation (2.25)) and of the moment difference Δm_j in Equation (A.8), the left-hand side of Equations (E.76)(E.77) equals

$$\left|\mathbb{E}[\psi_j(W; a, \tilde{\eta}) \mid X = \mathbf{x}] - \mathbb{E}[\psi_j(W; \alpha_{0,j}(\mathbf{x}), \eta_{0,j}) \mid X = \mathbf{x}]\right| = \left|\Delta m_j(\mathbf{x};\ a, \tilde{\eta})\right|. \tag{E.78}$$

Apply add-and-subtract with the intermediate pair $(\alpha_{0,j}(\mathbf{x}), \tilde{\eta})$ and use the triangle inequality:

$$\begin{aligned}\left|\Delta m_j(\mathbf{x};\ a, \tilde{\eta})\right| &= \left|\left(m_j(\mathbf{x};\ a, \tilde{\eta}) - m_j(\mathbf{x};\ \alpha_{0,j}(\mathbf{x}), \tilde{\eta})\right) + \left(m_j(\mathbf{x};\ \alpha_{0,j}(\mathbf{x}), \tilde{\eta}) - m_j(\mathbf{x};\ \alpha_{0,j}(\mathbf{x}), \eta_{0,j})\right)\right| \\ &\leqslant \left|m_j(\mathbf{x};\ a, \tilde{\eta}) - m_j(\mathbf{x};\ \alpha_{0,j}(\mathbf{x}), \tilde{\eta})\right| + \left|m_j(\mathbf{x};\ \alpha_{0,j}(\mathbf{x}), \tilde{\eta}) - m_j(\mathbf{x};\ \alpha_{0,j}(\mathbf{x}), \eta_{0,j})\right|.\end{aligned} \tag{E.79}$$

By the parameter-Lipschitz conditions in Condition 2.5specifically (i) in a and (ii) in ηeach term in Equation (E.79) is bounded uniformly (in particular at $u = \mathbf{x}$) as

$$\left|m_j(\mathbf{x};\ a, \tilde{\eta}) - m_j(\mathbf{x};\ \alpha_{0,j}(\mathbf{x}), \tilde{\eta})\right| \leqslant L_\alpha\,|a - \alpha_{0,j}(\mathbf{x})|, \tag{E.80}$$

$$\left|m_j(\mathbf{x};\ \alpha_{0,j}(\mathbf{x}), \tilde{\eta}) - m_j(\mathbf{x};\ \alpha_{0,j}(\mathbf{x}), \eta_{0,j})\right| \leqslant L_\eta\,\|\tilde{\eta} - \eta_{0,j}\|_\eta. \tag{E.81}$$

Combining Equations (E.78)(E.81) yields

$$\left|\mathbb{E}[\psi_j(W;a,\tilde{\eta}) \mid X=\mathbf{x}] - \mathbb{E}[\psi_j(W;\alpha_{0,j}(\mathbf{x}),\eta_{0,j}) \mid X=\mathbf{x}]\right| \leqslant L_\alpha \,|a-\alpha_{0,j}(\mathbf{x})| \;+\; L_\eta \,\|\tilde{\eta}-\eta_{0,j}\|_\eta, \tag{E.82}$$

which is exactly Equation (E.76). Equivalently, using the score-difference notation Equation (A.6), the same bound reads

$$\left|\mathbb{E}[\Delta\psi_j(W;\; a,\tilde{\eta}) \mid X=\mathbf{x}]\right| \leqslant L_\alpha \,|a-\alpha_{0,j}(\mathbf{x})| \;+\; L_\eta \,\|\tilde{\eta}-\eta_{0,j}\|_\eta, \tag{E.83}$$

which is Equation (E.77). This completes the proof. □

Lemma Appendix E.8 (Bad-Event Envelope Bound)**.** (→ Proof, p. 67)

Let $h : \mathcal{W} \to \mathbb{R}$ be any measurable function with $\left|h(W)\right| \leqslant F(W)$ a.s and $\mathbb{E}\left[\left|F(W)\right|^{\mathfrak{q}}\right] < \infty$ for some $\mathfrak{q} > 4$.

$$\mathbb{E}\left[\left|\binom{n}{s}^{-1} \sum_{\ell\in\mathcal{L}_{n,s}} \mathbb{1}\{\,A(D_\ell)^c\,\} \sum_{i\in[ns]} \mathbb{1}\{\,i\in\ell\,\}\,\kappa(x;W_i,D_\ell)\,h(W_i)\right|\right] \leqslant C_{1/\mathfrak{q}}\,(1-\pi_x)^s. \tag{E.84}$$

Consequently, for any $\epsilon > 0$,

$$\mathbb{P}\left(\binom{n}{s}^{-1}\left|\sum_{\ell\in\mathcal{L}_{n,s}} \mathbb{1}\{\,A(D_\ell)^c\,\} \sum_{i\in[ns]} \mathbb{1}\{\,i\in\ell\,\}\,\kappa(x;W_i,D_\ell)\,h(W_i)\right| > \epsilon\right) \leqslant \frac{C_{1/\mathfrak{q}}}{\epsilon}\,(1-\pi_x)^s. \tag{E.85}$$

The same bounds hold if $h(W_i)$ is replaced by $h(W_i) - \mathbb{E}[h(W) \mid X = X_i]$.

Proof of Lemma Appendix E.8

Proof.

$$\begin{aligned}
&\left|\binom{n}{s}^{-1} \sum_{\ell\in\mathcal{L}_{n,s}} \left\{\mathbb{1}\{\,A\,(D_\ell)^c\,\}\left[\sum_{i=1}^n \mathbb{1}\{\,i\in\ell\,\}\kappa\,(x;W_i,D_\ell)\,h(W_i)\right]\right\}\right| \\
&\leqslant \binom{n}{s}^{-1} \sum_{\ell\in\mathcal{L}_{n,s}} \left\{\mathbb{1}\{\,A\,(D_\ell)^c\,\}\left[\sum_{i=1}^n \mathbb{1}\{\,i\in\ell\,\}\kappa\,(x;W_i,D_\ell)\left|h(W_i)\right|\right]\right\} \\
&\leqslant \binom{n}{s}^{-1} \sum_{\ell\in\mathcal{L}_{n,s}} \left\{\mathbb{1}\{\,A\,(D_\ell)^c\,\}\left[\sum_{i=1}^n \mathbb{1}\{\,i\in\ell\,\}\kappa\,(x;W_i,D_\ell)\left|F\,(W_i)\right|\right]\right\}
\end{aligned} \tag{E.86}$$

Taking expectations, we find the following.

$$\begin{aligned}
&\mathbb{E}\left[\left|\binom{n}{s}^{-1} \sum_{\ell\in\mathcal{L}_{n,s}} \left\{\mathbb{1}\{\,A\,(D_\ell)^c\,\}\left[\sum_{i=1}^n \mathbb{1}\{\,i\in\ell\,\}\kappa\,(x;W_i,D_\ell)\,h(W_i)\right]\right\}\right|\right] \\
&\leqslant \mathbb{E}\left[\binom{n}{s}^{-1} \sum_{\ell\in\mathcal{L}_{n,s}} \left\{\mathbb{1}\{\,A\,(D_\ell)^c\,\}\left[\sum_{i=1}^n \mathbb{1}\{\,i\in\ell\,\}\kappa\,(x;W_i,D_\ell)\left|F\,(W_i)\right|\right]\right\}\right] \\
&\leqslant \mathbb{E}\left[\mathbb{1}\{\,A\,(D_{[s]})^c\,\}\sum_{i=1}^s \kappa\,(x;W_i,D_{[s]})\left|F\,(W_i)\right|\right] \\
&= \mathbb{E}\left[\mathbb{1}\{\,A\,(D_{[s]})^c\,\}\sum_{i=1}^s \kappa\,(x;W_i,D_{[s]})\,\mathbb{E}\left[\left|F\,(W)\right|\,\middle|\,X=X_i\right]\right] \\
&\leqslant \mathbb{E}\left[\mathbb{1}\{\,A\,(D_{[s]})^c\,\}\sum_{i=1}^s \kappa\,(x;W_i,D_{[s]})\,\mathbb{E}\left[\left|F\,(W)\right|^{\mathfrak{q}}\,\middle|\,X=X_i\right]^{1/\mathfrak{q}}\right] \\
&\leqslant C_1^{1/\mathfrak{q}}\mathbb{E}\left[\mathbb{1}\{\,A\,(D_{[s]})^c\,\}\sum_{i=1}^s \kappa\,(x;W_i,D_{[s]})\right] = C_1^{1/\mathfrak{q}}\mathbb{P}\big(A\,(D_{[s]})^c\big)
\end{aligned} \tag{E.87}$$

Using now Lemma Appendix E.1, we obtain the desired bound by plugging in for $\mathbb{P}\big(A\,\big(D_{[s]}\big)^c\big)$.

$$\mathbb{E}\left[\left|\binom{n}{s}^{-1}\sum_{\ell\in\mathcal{L}_{n,s}}\mathbb{1}\{\,A(D_\ell)^c\,\}\sum_{i\in[ns]}\mathbb{1}\{\,i\in\ell\,\}\,\kappa(x;W_i,D_\ell)\,h(W_i)\right|\right]\leqslant C_{1/\mathfrak{q}}\,(1-\pi_x)^s. \tag{E.88}$$

Equation (E.85) follows from an application of Markov's inequality. □

Lemma Appendix E.9 (Drift Term under Lipschitz in u)**.** (→ Proof, p. 68)
Suppose Condition 2.5 (iii) holds for $m_j(u;a,\eta)=\mathbb{E}[\psi_j(W;a,\eta)\mid X=u]$. Then, with $f_j(w)\coloneqq\psi_j\left(w;\alpha_{0,j}(\mathbf{x}),\eta_{0,j}\right)$,

$$\frac{1}{\sqrt{n_{\text{eff}}\,(\mathbf{x})}}\left|\,\mathbb{D}_{n,w}\left[f_j\right]\right|=\left|\sum_{i\in[n]}w_i\,\Big\{m_j(X_i;\alpha_{0,j}(\mathbf{x}),\eta_{0,j})-m_j(\mathbf{x};\alpha_{0,j}(\mathbf{x}),\eta_{0,j})\Big\}\right|\leqslant L_u\sum_{i\in[n]}w_i\,\|\,X_i-\mathbf{x}\,\|_2. \tag{E.89}$$

Moreover, for any $\delta\in(0,1)$ define

$$r_+(s,\delta)\coloneqq\Big(\frac{2\log(1/\delta)}{f_-\;v_{k_x}\;s}\Big)^{1/k_x}, \tag{E.90}$$

and let $r_{\mathbf{x}}>0$ denote the fixed radius used in the trimming event $G=\{X\in\mathcal{B}(\mathbf{x},r_{\mathbf{x}})\}$. By the DNN mass-radius control (cf. Proposition 3.2), with probability $1-o(1)$ the weight mass inside $\mathcal{B}(\mathbf{x},r_+(s,\delta))$ is at least $1-\delta^2$. Consequently, the drift satisfies

$$\frac{1}{\sqrt{n_{\text{eff}}\,(\mathbf{x})}}\left|\,\mathbb{D}_{n,w}\left[f_j\right]\right|=\left|\sum_{i\in[n]}w_i\,\left\{m_j(X_i;\alpha_{0,j},\eta_{0,j})-m_j(\mathbf{x};\alpha_{0,j},\eta_{0,j})\right\}\right|=o_{\mathbb{P}}\big(\,r_+(s,\delta)+r_{\mathbf{x}}\,\delta^2\,\big)=o_{\mathbb{P}}(\,1\,), \tag{E.91}$$

and, in the CLT-scaled form,

$$\left|\,\mathbb{D}_{n,w}\left[f_j\right]\right|=o_{\mathbb{P}}\Big(\,\sqrt{n_{\text{eff}}\,(\mathbf{x})}\;r_+(s,\delta)+\sqrt{n_{\text{eff}}\,(\mathbf{x})}\;r_{\mathbf{x}}\,\delta^2\,\Big). \tag{E.92}$$

In particular, if $n_{\text{eff}}\,(\mathbf{x})\asymp n/s$, then for fixed $\delta\in(0,1)$,

$$\sqrt{n_{\text{eff}}\,(\mathbf{x})}\;r_+(s,\delta)\;\asymp\;\sqrt{n/s}\,\left(\log(1/\delta)/s\right)^{1/k_x}\;\longrightarrow\;0, \tag{E.93}$$

and choosing a sequence $\delta=\delta_s\downarrow 0$ with $\sqrt{n_{\text{eff}}\,(\mathbf{x})}\,\delta_s^2\to 0$ (e.g. $\delta_s^2=(n/s)^{-1/2}(\log s)^{-2}$) yields

$$\left|\,\mathbb{D}_{n,w}\left[f_j\right]\right|=o_{\mathbb{P}}(\,1\,). \tag{E.94}$$

Proof of Lemma Appendix E.9
Proof. By Condition 2.5 (iii), for all u,

$$\left|\,m_j\left(u;\alpha_{0,j}(\mathbf{x}),\eta_{0,j}\right)-m_j\left(\mathbf{x};\alpha_{0,j}(\mathbf{x}),\eta_{0,j}\right)\right|\leqslant L_u\,\|\,u-\mathbf{x}\,\|_2. \tag{E.95}$$

Let $f_j(w)\coloneqq\psi_j\left(w;\alpha_{0,j}(\mathbf{x}),\eta_{0,j}\right)$. Using the definition of $\mathbb{D}_{n,w}$ in Equation (A.4) with $f=f_j$ and the shorthand $m_j(u)\equiv m_j\left(u;\alpha_{0,j}(\mathbf{x}),\eta_{0,j}\right)$,

$$\frac{1}{\sqrt{n_{\text{eff}}\,(\mathbf{x})}}\,\mathbb{D}_{n,w}\left[f_j\right]=\sum_{i\in[n]}w_i\,\Big\{m_j(X_i)-m_j(\mathbf{x})\Big\}. \tag{E.96}$$

Applying Equation (E.95) inside the sum gives the deterministic bound

$$\frac{1}{\sqrt{n_{\text{eff}}\,(\mathbf{x})}}\left|\,\mathbb{D}_{n,w}\left[f_j\right]\right|\leqslant L_u\sum_{i\in[n]}w_i\,\|\,X_i-\mathbf{x}\,\|_2. \tag{E.97}$$

Shrinking Radius and Mass Split.
Fix $\delta \in (0,1)$ and set

$$r_+(s,\delta) \coloneqq \left(\frac{2\log(1/\delta)}{f_-\, v_{k_x}\, s}\right)^{1/k_x}, \qquad \mathcal{B}(\mathbf{x},r) \coloneqq \{\, u : \|u-\mathbf{x}\|_2 \leqslant r \,\}. \tag{E.98}$$

Let $r_{\mathbf{x}} > 0$ denote the fixed trimming radius used to define $G \coloneqq \{\, X \in \mathcal{B}(\mathbf{x}, r_{\mathbf{x}}) \,\}$. Decompose the weighted sum in Equation (E.97) by inside/outside the shrinking ball:

$$\sum_{i\in[n]} w_i \,\|X_i-\mathbf{x}\|_2 = \underbrace{\sum_i w_i \,\|X_i-\mathbf{x}\|_2 \,\mathbb{1}\{\, X_i \in \mathcal{B}(\mathbf{x},r_+)\,\}}_{\leqslant\, r_+(s,\delta)} + \underbrace{\sum_i w_i \,\|X_i-\mathbf{x}\|_2 \,\mathbb{1}\{\, X_i \notin \mathcal{B}(\mathbf{x},r_+)\,\}}_{\leqslant\, r_{\mathbf{x}}\,\sum_i w_i\,\mathbb{1}\{\,X_i\notin\mathcal{B}(\mathbf{x},r_+)\,\}}. \tag{E.99}$$

By the DNN mass-radius control (see Proposition 3.2), the outside weight satisfies

$$\mathbb{E}\left[\sum_{i\in[n]} w_i \;\mathbb{1}\{\, X_i \notin \mathcal{B}(\mathbf{x}, r_+(s,\delta))\,\}\right] \leqslant \delta^2, \tag{E.100}$$

and in fact the same bound holds with probability $1-o(1)$.

Unscaled probabilistic bound.
Combining Equations (E.97)(E.99) and taking expectations, we obtain

$$\mathbb{E}\left[\frac{1}{\sqrt{n_{\text{eff}}\,(\mathbf{x})}}\left|\mathbb{D}_{n,w}\,[f_j]\right|\right] \leqslant L_u\left\{r_+(s,\delta) + r_{\mathbf{x}}\,\mathbb{E}[\text{outside mass}]\right\} \leqslant L_u\left\{r_+(s,\delta) + r_{\mathbf{x}}\,\delta^2\right\}. \tag{E.101}$$

Hence, by Markovs inequality, for any $\varepsilon > 0$,

$$\mathbb{P}\left(\frac{1}{\sqrt{n_{\text{eff}}\,(\mathbf{x})}}\left|\mathbb{D}_{n,w}\,[f_j]\right| > \varepsilon\right) \leqslant \frac{L_u}{\varepsilon}\left\{r_+(s,\delta) + r_{\mathbf{x}}\,\delta^2\right\}. \tag{E.102}$$

Since $r_+(s,\delta) = (\log(1/\delta)/s)^{1/k_x} \to 0$ as $s\to\infty$ for any fixed δ, and we may choose $\delta = \delta_s \downarrow 0$ arbitrarily slowly, the right-hand side of Equation (E.102) vanishes; thus

$$\frac{1}{\sqrt{n_{\text{eff}}\,(\mathbf{x})}}\left|\mathbb{D}_{n,w}\,[f_j]\right| = o_{\mathbb{P}}(\,1\,). \tag{E.103}$$

CLT-scale Bound.
Multiplying Equation (E.99) by $\sqrt{n_{\text{eff}}\,(\mathbf{x})}$ and repeating the same expectation-Markov step yields, for any $\varepsilon > 0$,

$$\mathbb{P}\big(\left|\mathbb{D}_{n,w}\,[f_j]\right| > \varepsilon\big) \leqslant \frac{L_u}{\varepsilon}\left\{\sqrt{n_{\text{eff}}\,(\mathbf{x})}\; r_+(s,\delta) + \sqrt{n_{\text{eff}}\,(\mathbf{x})}\; r_{\mathbf{x}}\,\delta^2\right\}. \tag{E.104}$$

Under the DNN geometry $n_{\text{eff}}\,(\mathbf{x}) \asymp n/s$,

$$\sqrt{n_{\text{eff}}\,(\mathbf{x})}\; r_+(s,\delta) \;\asymp\; \sqrt{n/s}\left(\frac{\log(1/\delta)}{s}\right)^{1/k_x} \longrightarrow 0 \quad \text{whenever} \quad \sqrt{n/s}\left(\frac{\log(1/\delta)}{s}\right)^{1/k_x} \to 0, \tag{E.105}$$

and choosing $\delta = \delta_s \downarrow 0$ so that $\sqrt{n_{\text{eff}}\,(\mathbf{x})}\;\delta_s^2 \to 0$ (e.g. $\delta_s^2 = (n/s)^{-1/2}(\log s)^{-2}$) makes the second term negligible. Therefore

$$\left|\mathbb{D}_{n,w}\,[f_j]\right| = o_{\mathbb{P}}(\,1\,). \tag{E.106}$$

Combining Equation (E.103) and Equation (E.106) completes the proof. □

Lemma Appendix E.10 (Localized stability for score differences)**.** (→ Proof, p. 70)
Assume Conditions 2.4 and 2.6. Let $s = s_n \to \infty$ with $s/n \to 0$. Fix $j \in [p_\alpha]$ and the local set $G = \{X \in \mathcal{B}_{\mathbf{x}}\}$. For $a \in \mathcal{A}_j(\mathbf{x})$ and $\tilde{\eta} \in \mathcal{H}_j$ with $|a - \alpha_{0,j}(\mathbf{x})| \leqslant C\rho_n$ and $\|\tilde{\eta} - \eta_{0,j}\|_\eta \leqslant Cr_{n,j}$,

$$\sup_{|a-\alpha_{0,j}(\mathbf{x})|\leqslant C\rho_n \tilde{\eta}\in\mathcal{H}_j} \left| \mathbb{G}_{n,w}\big[\{\psi_j(\cdot; a, \tilde{\eta}) - \psi_j(\cdot; \alpha_{0,j}(\mathbf{x}), \eta_{0,j})\}\ \mathbb{1}\{G\}\big] \right| = o_{\mathbb{P}}\big(b_n^{-1}\big), \tag{E.107}$$

uniformly over $j \in [p_\alpha]$. Equivalently, by Equation (A.5),

$$\sup_{|a-\alpha_{0,j}(\mathbf{x})|\leqslant C\rho_n \tilde{\eta}\in\mathcal{H}_j} \left| \mathbb{T}_{n,w}\big[\Delta\psi_j(\cdot; a, \tilde{\eta})\big] - \sqrt{n_{\text{eff}}(\mathbf{x})}\ \mathbb{E}[\Delta\psi_j(W; a, \tilde{\eta}) \mid X = \mathbf{x}] \right| = o_{\mathbb{P}}\big(b_n^{-1}\big). \tag{E.108}$$

Proof of Lemma Appendix E.10
Proof.

Step 1: Localized class and gating.
Fix $j \in [p_\alpha]$ and let $G := \{X \in \mathcal{B}_{\mathbf{x}}\}$. Consider the localized score-difference class

$$\mathcal{F}_{n,j} := \Big\{ f_{a,\tilde{\eta}}(w) := \big(\psi_j(w; a, \tilde{\eta}) - \psi_j(w; \alpha_{0,j}(\mathbf{x}), \eta_{0,j})\big)\ \mathbb{1}\{G\} \ :\ |a - \alpha_{0,j}(\mathbf{x})| \leqslant C\rho_n, \|\tilde{\eta} - \eta_{0,j}\|_\eta \leqslant Cr_{n,j} \Big\}. \tag{E.109}$$

By Condition 2.4 the class Equation (E.109) admits envelope F and the local Lipschitz bounds in $(a, \tilde{\eta})$; by Condition 2.6 the weight negligibility and replace-one modulus hold at the $\sqrt{n_{\text{eff}}(\mathbf{x})}$ scale.

Step 2: LSE at the $\sqrt{n_{\text{eff}}(\mathbf{x})}$ Scale.
Applying Lemmas 2.2 and 2.4 to $\mathcal{F}_{n,j}$, we obtain the uniform localized bound

$$\sup_{f\in\mathcal{F}_{n,j}} |\mathbb{G}_{n,w}[f]| = o_{\mathbb{P}}\big(b_n^{-1}\big), \tag{E.110}$$

with the stochastic $o(\cdot)$ uniform over $j \in [p_\alpha]$.

Step 3: Plug-in Admissibility and Conclusion.
By Condition 2.6 we have $|\tilde{\alpha}_j(\mathbf{x}) - \alpha_{0,j}(\mathbf{x})| \leqslant C\rho_n$ and $\|\hat{\eta}_j - \eta_{0,j}\|_\eta \leqslant Cr_{n,j}$ with probability $1 - o(1)$, hence

$$f_{\tilde{\alpha}_j(\mathbf{x}),\,\hat{\eta}_j}(w) := \big(\psi_j(w; \tilde{\alpha}_j(\mathbf{x}), \hat{\eta}_j) - \psi_j(w; \alpha_{0,j}(\mathbf{x}), \eta_{0,j})\big)\ \mathbb{1}\{G\} \in \mathcal{F}_{n,j} \tag{E.111}$$

on that event. Combining Equations (E.110)(E.111) yields

$$\sup_{|a-\alpha_{0,j}(\mathbf{x})|\leqslant C\rho_n \|\tilde{\eta}-\eta_{0,j}\|_\eta\leqslant Cr_{n,j}} \left| \mathbb{G}_{n,w}\big[\{\psi_j(\cdot; a, \tilde{\eta}) - \psi_j(\cdot; \alpha_{0,j}(\mathbf{x}), \eta_{0,j})\}\ \mathbb{1}\{G\}\big] \right| = o_{\mathbb{P}}\big(b_n^{-1}\big), \tag{E.112}$$

uniformly over $j \in [p_\alpha]$, which is Equation (E.107).

Step 4: Equivalent $\mathbb{T}_{n,w}$ form.
Using the weighted operator identity Equation (A.5) together with Equation (A.2), we have for any measurable f,

$$\mathbb{G}_{n,w}[f] = \mathbb{T}_{n,w}[f] - \sqrt{n_{\text{eff}}(\mathbf{x})}\ \mathbb{E}_{X=\mathbf{x}}[f(W)]. \tag{E.113}$$

Applying this with $f = \Delta\psi_j(\cdot; a, \tilde{\eta})\ \mathbb{1}\{G\}$ and taking the same supremum as in Equation (E.112) yields Equation (E.108). This completes the proof. □

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